PRODUCT DESIGN NEA

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Design Briefs:

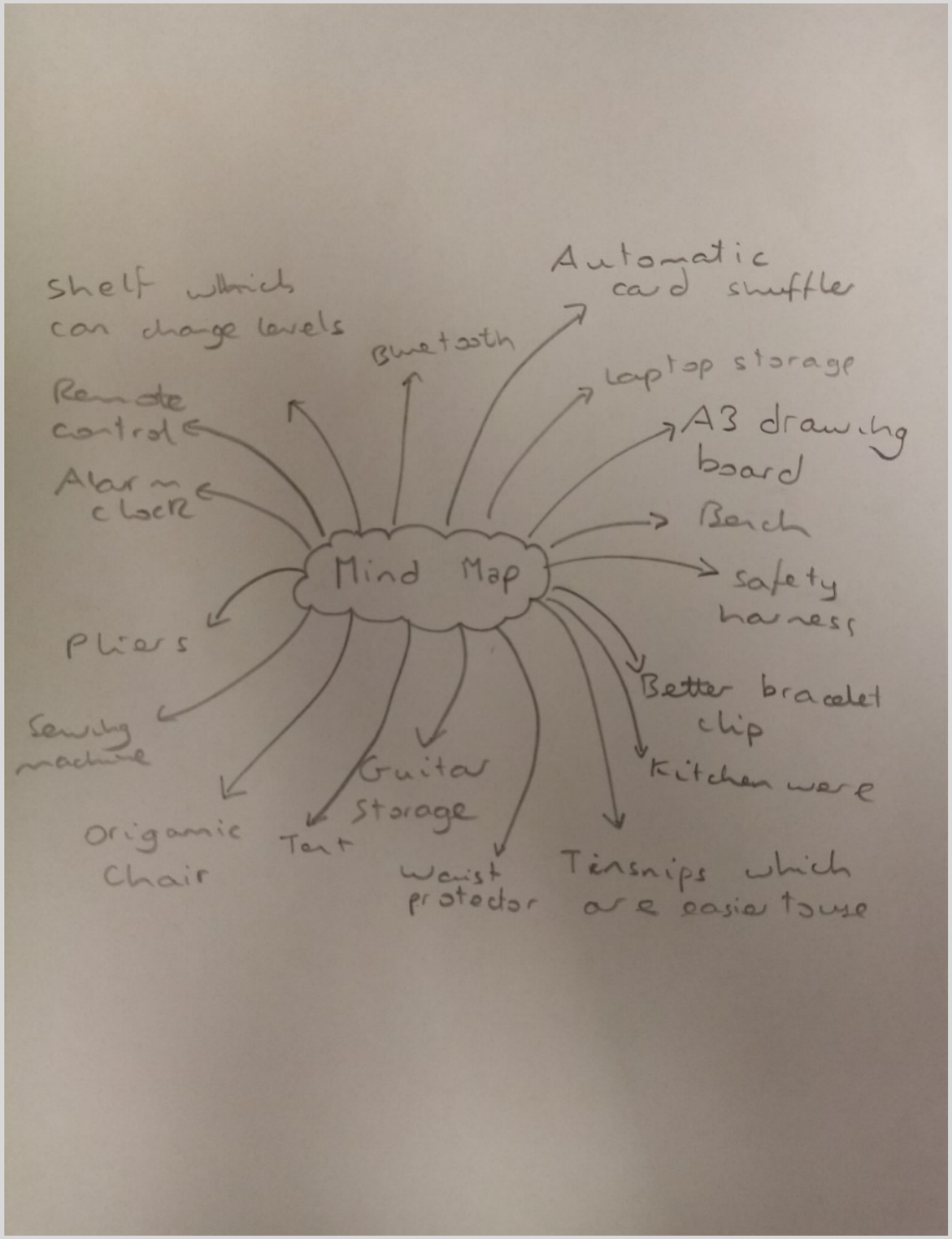
Brief 1:
A shelf whose shelf levels can change in order to facilitate the reach of higher up shelves to people with disabilities or who are vertically challenged. Both materials used, in order to create a sustainable product, and mechanisms that can be used to change the levels of shelves need to be explored in order to create the most efficient and easy to use product possible.

Brief 2:
An A3 Drawing board that can be stored inside a backpack.
The materials of which need to be carefully considered as they need to be durable enough that they will not break whilst they are shaken inside the backpack. The ergonomics of the design, both physical and visual, will need to be carefully explored and developed in order to facilitate and produce a product that becomes very straightforward for the consumer to use.

Brief 3:
An origami chair that can be easily folded away and stored whilst maintaining its strength when sat upon and ensuring that any user can easily fold it open and closed. Careful attention should be given to the choice of materials used and a decision must be taken between bending the material or using hinges on it to increase its durability.

Brief 4:
To improve the design of the existing tinsnips in order to facilitate their use by changing the design of the handle by carefully considering its ergonomics and by paying close attention to how they are used by consumers in order to work out issues within its design and thus improve on them.

Brief 5:
To create an automatic card shuffler in order to facilitate the dealing of cards within card-based games settings and to increase the speed that games can be played at by greatly reducing the unnecessary time spent on shuffling cards.



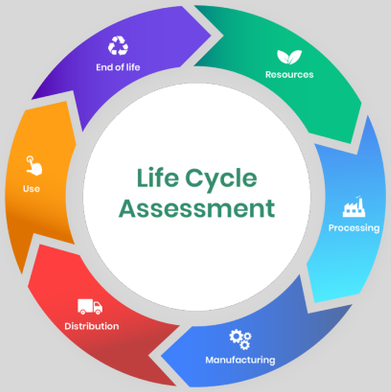
Brief 1:

A shelf whose shelf levels can change in order to facilitate the reach of higher up shelves to people with disabilities or who are vertically challenged. Both materials used, in order to create a sustainable product, and mechanisms that can be used to change the levels of shelves need to be explored in order to create the most efficient and easy to use product possible.

Reaching high up places is a big problem for many people and this often includes reaching the top shelves. To fix this problem, I would suggest the creation of shelving which would allow a user to easily change the height of the top shelf in order to allow customers to reach It.

Careful attention should be given especially to the mechanism used, as it should be safe for a user to use, making sure that there are no finger-traps as it is likely for it to be exposed to the consumer; an issue which is clearly visible in Product 1. The material choice should also be considered as it should be made out of sustainable materials whose full lifecycle should be analysed in order to create the most sustainable product possible. Ergonomics are not as important within this product; however, they should still be considered.

Problems we'll need to face:
Ease of use?
Balance?
Material lifecycles?
Mechanism?



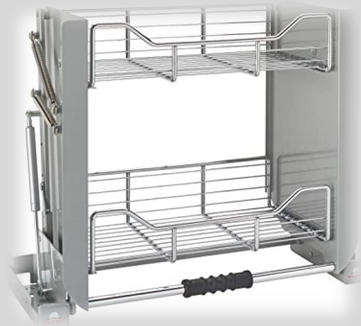
Balance:
If the shelf pivots, will there be a need to put masses underneath it to stop it from flipping if objects are put on top of it.

Mechanisms:
What mechanism works best for the user?
Consider its cost?

Ergonomics:
Consider handle size and shape?
Look at statistics for hand dimensions?
If using different mechanism, consider ease of use.
Product must clearly show how it is intended to be used.

Evaluation:
This product holds quite a lot of possibility and innovation, and it could definitely help a lot of people. Furthermore, I could look at many aspects of product design through this product ranging from the life cycle assessment to ergonomics. However, I do not think that this product would keep me interested in it for that long as I feel that the range of skills required to produce this are not enough for me to develop and struggle.

Existing products:
I have looked at both moving and normal shelves to both get an idea of mechanisms which can be used and the materials and look I'd like to go for.



Product 1:
This shelf has a mechanism on the side which relies on a lever system as seen on its side together with hydraulic pistons in order to allow the user to pull it down more easily.

Product 2:
Although this shelf does not move, it instead shows a similar style to what I would like to go to.
However, further research of products might steer me away from this design.

Product 3:
This is another shelf which can be easily pulled down by a user. It shows another method to pull down a shelf. However, it does have a handle or any easier to access method to grab this meaning that people with hand mobility problems might struggle with it.

Brief 2:

An A3 Drawing board that can be stored inside a backpack. The materials of which need to be carefully considered as they need to be durable enough that they will not break whilst they are shaken inside the backpack. The ergonomics of the design, both physical and visual, will need to be carefully explored and developed in order to facilitate and produce a product that becomes very straightforward for the consumer to use.

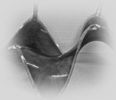
Many people have wanted to have an A3 drawing board at hand when they travel around and want to work. However, they are often large and clumsy to carry around as you need a whole other case to carry them in. However, most people carry a moderately sized backpack around with them, and to add 1 more roughly book-sized item should not be that hard. Therefore, both the dimensions and weight should be taken into consideration in order to make the board much more viable to use.

Many different methods to fold the board should therefore be considered. Including the use of smart materials or purely mechanical methods. And careful attention should be given to physical and visual ergonomics in order to allow consumers to more easily adapt and learn how to use the product without a lot of effort being put in it. For example, key parts which are interacted with should be made out of different materials and be a different colour.

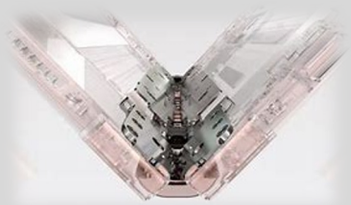
Problems which will need to be tackled:
How to create a flat surface?
How to make it bend or fit inside a backpack?
What materials?
How to fasten paper?
Weight?

How to hold paper?:
Paper clamp?
Magnets?
Slide into a triangular slot?

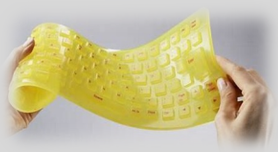
Smart Materials:
Electroactive Polymers could be used to open and close a flat sheet of material.



Samsung Galaxy Fold:
This is a famous mechanism which allows the screen to fold without bending it too much but also lets it stay flat when left opened.

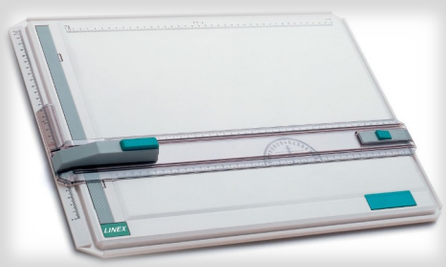


Flexible materials:
Silicone?
PP?
Wood Venner?
Rubber?



Existing products:
There are not any foldable A3 boards which currently exist. Instead, I'll take a look at existing A3 boards to inspire me.

Product 1:
This is a wooden A3 stand which can change levels in order to make drawing easier. However, it does not have any guide for a ruler or a way to securely fasten paper to it.



Product 2:
An average A3 drawing board. Doesn't have anything special about it but it still gives a good idea of what to aim for.

Product 3:
This drawing board has both the levelling feature of the first, and the ruler and triangle of the second. Furthermore, the white backdrop might make it nicer to draw on.

Evaluation:
Overall, this product holds much potential in my opinion. Lots of aspects could be explored in order to create both the system to fold and store the drawing board; and also, lots of investigations within the materials which could be used should be done in order to create the most lightweight and ergonomic product possible. I have also already received lots of interest from possible costumers who would like a product like this.

Brief 3:

An origami chair that can be easily folded away and stored whilst maintaining its strength when sat upon and ensuring that any user can easily fold it open and closed. Careful attention should be given to the choice of materials used and a decision must be taken between bending the material or using hinges on it to increase its durability.

Many people need to have foldable chairs which can normally be stored away whilst taking up little space, however, what I have found is that storable chairs often look bad and are really uncomfortable to sit on. Therefore, This product would need to fit three criteria: It needs to be comfortable to sit on, it has to look good and stylish, and lastly, it needs to be able to be folded away in a way that makes it take up minimal space.

The hardest challenge which I will need to face as its designer is the folding away method. I will most likely have to look at a way to collapse the chair and I will also need it to be able to support the weight of a person when opened, which is another issue. The ideal method would be that when it's opened, the weight of the user would cause it to stay open in some way.

Problems:
Material?
Mechanism?
Ergonomics?
Ease of use?

Material:
Strong yet very flexible?
Comfortable (although this relies more on design)?
Durable?
Ideally waterproof?

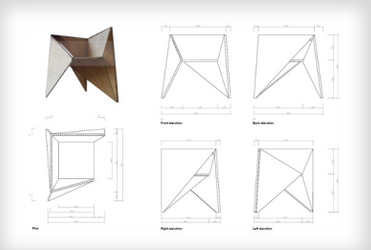
Ease of use:
Given how a user would need to fold it away, the method to fold it away should be easy to understand and use.

Mechanism:
As this is an "origami chair", there will definitely be a need to explore origami and how it works in order to develop a design which allows a user to fold it away easily.

Evaluation:
This product will probably be quite challenging to produce as to fulfil all three criteria in a single product together is quite hard. However, if I am able to, it should result in a very good product. Nonetheless, I have found little interest from consumers in this product, mostly because it has been seen as more of a gimmick than an actual preferred product to the standard folding chair.

Existing products:
There are a few products which market as origami chairs, however not all of them can be folded away.

Product 1:
This is a stylish chair which looks good but can also be folded away as it flattens laterally. However, it does not look like it would be very comfortable to sit on due to all the cracks.



Product 2 and 3:
These are both 2 products which both fold away and then can be folded together in order to create the actual chair. They both look quite stylish; however, they also look like they wouldn't be that comfortable to sit on, especially the first one. Another large issue which can be seen with the second design is that when unfolded, it takes up a very large amount of space.

Product 4:
This product does not fold away; however, it looks very stylish and much more comfortable to sit on than the other designs.

Brief 4:

To improve the design of the existing tinsnips in order to facilitate their use by changing the design of the handle by carefully considering its ergonomics and by paying close attention to how they are used by consumers in order to work out issues within its design and thus improve on them.

The current design of the tinsnips is extremely flawed. The handle is far too small and too uncomfortable to hold and the blade dulls too easily. Another problem is that the bearing on it is too weak and has too much friction which makes the tin snips even harder to use.

Therefore, there will be a need to create a more ergonomic and confortable handle, and also a way to either use better materials for the blade or a mechanism to quickly remove and swap out blades.

Current design:
As you can see from the current design, the handle is very thin, very hard, and does not even consider the ergonomics of a person's hand. This means that when you try and grip it and squeeze, you cannot get a good grip and you cannot squeeze with a lot of strength.

The blade is also made out of very low-quality steel which means that it blunts extremely easily resulting in it needed to be resharpened very frequently.

Evaluation:
Although this product is something which would be very helpful to create. I do not think that it would allow me to use my creativity as much as I'd like to as I would only be changing an existing one and not innovating or creating an entirely new product.

Brief 5:

To create an automatic card shuffler in order to facilitate the dealing of cards within card-based games settings and to increase the speed that games can be played at by greatly reducing the unnecessary time spent on shuffling cards.

This is a product which would be very helpful to have for when one is playing almost any card game. It would have to be quick and must reliably randomise the cards. To do this, it could use a computer program to randomise how the deck is shuffled in order to shuffle it well.

DIfferent methods to shufle the cards should be looked at. Especially by looking at existing products:

Both of these products use similar methods to shuffle the cards, except that one shuffles the whole deck inside the machine, whilst the other shuffles by choosing a card from either side of the split deck.

Evaluation:
Overall, this product holds next to no real potential. There isn't much development or exploration which can be made with it. And almost no innovation on existing products. At most, I could look into materials, but even that does not require much skill to do and so has very little creativeness linked to it.

Peer Assessment And Research plan:

Brief 1:	Find the general height and wingspan of people. And research the average weight of objects on shelf. Think about the target demographic and about having something special to make people want to buy these shelves.
Brief 2:	Make it usable in both A3 and A4. Making completely flat would be hard. Test how easy it is to write on different materials (ask multiple people to remove bias). Figure out how to make a stand that works in both A4 and A3 mode. Will need a foldable ruler. Try and target it at a certain type of demographic that use drawing boards such as architects and engineers.
Brief 3:	Figure out how to fold and stay flat when folded so easy to store. Research Target market and who wants the chair and why they'd buy it. If flat-packed, try and think what shape it will be stored. Think about sustainability
Brief 4:	Find mechanisms to change the blade. Research Ergonomics so easier to grip and create pressure. Try and think pf a specific demographic.
Brief 5:	Make it portable? People who use cards who want to buy this probably already know how to shuffle cards and so will not be interested in it so investigate the possible target markets. Focus on sustainability.

Evaluation of Briefs:

For my project, I have decided to do the second brief. I decided to do this brief after careful consideration of all the briefs. The first one to go was Brief 5, as I said in my evaluation because it did not hold any potential to its production. Then, Brief 4 was unfortunately scrapped; this was because even though it was a fairly interesting and useful product, I did not think that it held enough potential as it could not test my skills enough nor allow me to develop new knowledge or multiple ideas. Then I had to decide what next had to go. I found this quite hard as the remaining briefs all really interested me; nonetheless, I had been told by a fellow classmate that he found the Origami Chair gimmicky, and that he doubted its actual usefulness, so after some evaluation, I also scrapped this idea. To decide between the last 2 took quite a bit of time. Although the first one seems more useful to a general market, I found that it does not allow me to develop and challenge myself as much as I'd like, as a shelf is at its core a simple product. Therefore, I decided to go with Brief 2, as I think that it will allow me to utilize my creativeness and develop my skills in the process in order to produce an innovative and interesting design of a product.

Research Plan:

Colour Theory:

I will need to research colour theory in order to both decide what coulour pallete should be used for my product, and whether to highlit key points on the board, such as handles, in different colours so that they stand out more.

Ergonomics:

The key part of ergonomics which I will need to research is dimensions and how users interact with the product so that I can work out the different shapes and textures given to the product. I will also need to look at the visual ergonomics of the product and how they link to colour theory.

Materials:

There are 2 different things which I will need to consider in my materials research. Firstly, I will need to look at what materials the product itself is made out of, as different materials have different feels on their hands, and thus different tactile responses. Secondly, I will need to research smart materials and different materials which could be used for the surface of the drawing board.

Mechanisms:

The mechanisms used to bend the frame (if a mechanical soultion is used) will probably entice the bulk of the research. There will need to be a thorough investigation of existing methods used to fold surfaces in half in order to develop a useful and reliable mechanism, so both pros and cons of existing methods needs to be explored.

Costumer's profile and needs:

I will need to discuss with the consumerin order to work out the various different aspects and requirements of the product which the user will need, such as size, weight, material and coulour. I will also need to examine the user while he uses similar products in order to analyse and work out how they interact and use it.

Why is there a need for my product?

The main issue:
The main issue with the current A3 drawing boards is that they lack portability for the average user. Although you can different solutions to this problem, all of them have their flaws.

Reducing waste:
Currently, the only solution for people who want to bring an A3 drawing board around with them is to either buy a separate A3 drawing board case or to buy a separate A3 drawing board that has a handle attached to it. However, these two solutions both greatly increase the waste, thus going against the sixth Hannover Principle. This means that there is a flaw in the design, as we cannot create products to fix another product's issue, without in turn creating more waste. Instead, we should be fixing the issue in the original product.

The main issue with the case is the materials which it is made out of. Most seem to be made out of plastics such as PP, which are not very resistant, and so results in a product that has been planned to ruin over time, meaning that for the lifecycle of a single drawing board, multiple cases must be used.
However, in the other design, the need to add a handle, although solving the original issue, results in the surface being exposed to the environment. In the average commute, this can result in the surface becoming scratched, scuffed, and in turn ruined, meaning that a whole new board needs to be bought, thus creating more waste.

My product should be able to reduce the waste greately, as by being stored within a backpack, its lifetime can greately increse. This means that the product will need to be rebought a lot less times which in turn results in less waste being produced.

Looking at dimensions:
To prove that it is unrealistic for an A3 drawing board to fit within a normal backpack, I looked at an average A3 Drawing Board from Linex and compared it to the sizes of different backpacks which I could find (the dimensions can be seen in the specification [however, the diagonal dimension is 648mm]).

Specification

Colour	Grey
Warranty Duration	1
Country of Origin	Germany
Pack Unit Of Sale Barcode	5701221470706
Brand	Linex
Height mm	510
Weight kg	1.12
Product Unit	1
Width mm	400
Pack Contains	10
Depth mm	18
OEM	100411028

To compare the dimensions, I went on Amazon and looked at the Best Sellers Backpacks to see how the dimensions of the A3 drawing board compares to it.

Rank	Name & Link	Dimensions	Does it fit?
1.	Sosoon Laptop Backpack	Up to 250x360mm	No
2.	MATEIN Travel Laptop Backpack	500mm in Diagonal	No
3.	WENIG Anti-Theft Laptop Backpack	500mm in Diagonal	No
4.	LIWAG Extra-Large Travel Backpack	520x400mm	Yes
5.	Della Gao Extra Large Backpack	500x320mm	No
6.	AMBOR Laptop Backpack	495x318mm	No
7.	Lekesky Backpack	420x280mm	No
8.	KROSER Laptop Backpack	429x283mm	No
9.	Tzowla Travel Laptop Backpack	500mm in Diagonal	No
10.	QINOL Travel Laptop Backpack	460x360mm	No
11.	VSNOON Backpack	432x300mm	No
12.	SOSOON Extra-Large Backpack	488x373mm	No
13.	Della Gao Lightweight School Backpack	520x340mm	No
14.	Lekesky Laptop Backpack	440x300mm	No
15.	XQXA Anti-Theft Backpack	457x279mm	No

Client Research:

Primary Client Profile:

Name:

- Edward Reamer

Description:

- Age: 19
- White male

Occupation:

- University Student studying Computer Science at King's College London.

Information about the product from the Interview:

- Uses a Drawing Board recreationally about 3 to 4 times a week.
- Normally draws models, or artistic drawings for fun.
- As a student, he requires the drawing board to be portable as he likes to study in the Maughan Library but that's far from his house, so he has to be able to bring it in-between them.
- He Is worried about the cost that the product might end up at and its environmental impact.
- When talking about materials, he liked the look of the cherry wood in the material research after I showed him the different materials. He also said that he liked the smoothness of acrylic but is worried about its durability and scratch-ability.

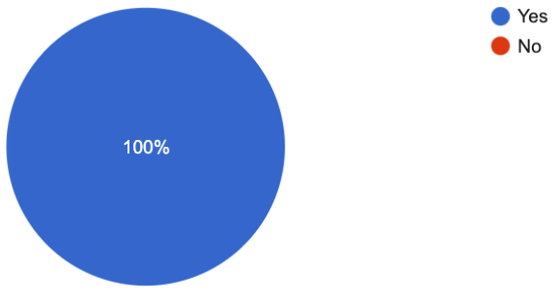
Evaluation of the Primary Client Profile:

Looking at the responses from my client, I have come to a few conclusions: Firstly, he prefers wood over most other materials, as he does not like how plastic ages badly and makes ink look ugly on it; whilst he prefers the look of a polished wood surface, such as the Cherry wood (which can be seen withing the material research). His thought of using a vinyl layer is interesting, as it could be used to create the bending surface of my drawing board, however, the problem with it is that it's quite heavy, and also is not that flexible compared to other materials, so it would result in the sheet being much larger then is needed. The size of his bag isn't the biggest, but it is big enough to fit a slightly larger A4 sheet, so it would definitely fit the planned dimensions for my bag. The sustainability and environmental impact of the materials is a concern to him, so it will need to be considered. This means that it would be good for me to avoid plastics and other materials which have a long-lasting impact on the planet after the lifetime of the product has ended. I'll also need to try and avoid the use of glues and other fixing methods which result in the product being harder to disassemble and thus recycle its key components. Lastly, the cutting matt is a good idea, however, having researched it in the materials research 2, I do not think that it is a good idea to use it, as it was not very flexible and would shatter easily,

Response to questionnaire by the client:

Are you interested in acquiring a Foldable A3 drawing Board?

1 response



Are there any materials which you'd like the Drawing Board to be made out of?

1 response

Plywood and some sort of vinyl layer - much like a cutting Matt. That pr a highly polished wooden surface

When using a drawing board, what do you normally find annoying or frustrating about it which could be improved on in this product?

1 response

Plastic. It ages badly. Cuts, ink look ugly on it. Wood absorbs better

How much space is normally available in your backpack which could be filled by the drawing board?

1 response

A laptop size - something around 25x35 cm

Is there anything else which hasn't been asked in this form which you think should be considered?

1 response

Sustainability of materials, manufacturing process.

Client Research 2:

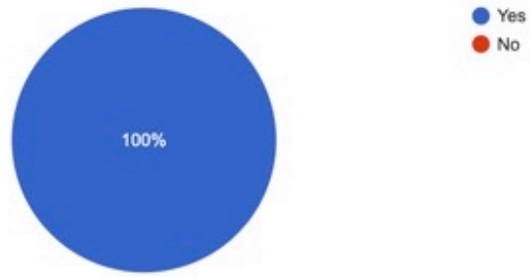
Secondary Client Profile:
Name:
Duncan Stewart
Description:
Male.
56.
White
Religion : Jedi with Buddhist tendencies
Occupation:
Carpenter/ furniture maker/teacher/musician
Information about the product:
Materials. Maybe leather for drawing comfortably.
Rigid framework.
Possible polycarbonate inner.
Storage for Rotring fine liners X 2
Foldable ruler with included scale? 1:20 1:50 1:200?

Evaluation of the Secondary Client Profile:
Looking at the responses of my client in both the Profile Questions and in the public questionnaire, I have come to a few conclusions: firstly, leather as the drawing board seems to be a very popular material for the drawing surface, so I will need to research different types of leather in order to find out which is the best one to use. The foldable ruler will be important to have, but, as non-brittle materials are required, acrylic and similar materials should be avoided, unless the ruler can be protected during storage in some way. The use of "clever ergonomic styling" will be important to me as I want to create a very attractive product, so lots of research and modelling will go into this. The use of a "magnetic latching system" is also important, as magnets could prove to be very useful in my design, as they can offer space-saving solutions to many problems, such as how to attach the ruler easily to the frame. I will need to consider how I'd want to include an accessory holder in my design, and whether it should be on the outside of the body or stored inside the frame. The use of polycarbonates will need to be researched, as they are a viable option to this at the place of using woods. To store the Rotring Fine liners, I'll need to look at possible methods of storage, ranging from the accessory holder to a potential outside holder made of fabric or similar materials, as their dimensions might be so that if stored inside the frame, it might result in a very large extrusion in one of the corners, so in the design, there be some consideration of a method to attach them outside. Lastly, one should consider how many features could be included in the ruler so that it reduces the number of extra pieces which are needed.

Response to questionnaire by the client:

Are you interested in acquiring a Foldable A3 drawing Board?

1 response



Are there any materials which you'd like the Drawing Board to be made out of?

1 response

Not too brittle. Must be flexible. And comfortable to draw on.

When using a drawing board, what do you normally find annoying or frustrating about it which could be improved on in this product?

1 response

Size! Too big to transport. Cumbersome. Material is too brittle. Not rugged enough. Would prefer them to be more user friendly ie clever ergonomic styling for comfort. Accessory holder with magnetic latching system maybe.

How much space is normally available in your backpack which could be filled by the drawing board?

1 response

Roughly A4 size but would be able to carry slightly bigger if it was worth the effort.

Is there anything else which hasn't been asked in this form which you think should be considered?

1 response

I would love to have a system which is useful multi functional. Looks cool is easy to use and transport. The materials I would suggest need to be for me slightly rubberised firstly for grip but also to feel nice to draw on. Like old leather writing desks. In fact leather is perfect.

Target Market Research:

Making and posting:

I needed to find data about my product, so to do that, I posted on multiple forums about my drawing boards. These were mostly on Reddit, such as r/Architecture, r/Drawing, r/Art, and many other smaller ones. I also posted on other forum sites, such as SkyScraperCity and Archinet which gave me a few responses. The questionnaire was made in Google Forms (Foldable A3 Drawing Board Questionnaire (google.com)):

Are you interested in acquiring a Foldable A3 drawing Board?

- Yes
- No

Are there any materials which you'd like the Drawing Board to be made out of?

Your answer

When using a drawing board, what do you normally find annoying or frustrating about it which could be improved on in this product?

Your answer

How much space is normally available in your backpack which could be filled by the drawing board?

Your answer

Is there anything else which hasn't been asked in this form which you think should be considered?

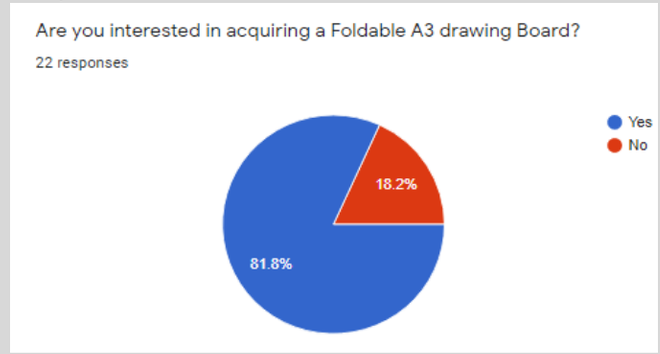
Your answer

Responses:

The questionnaire gave me a total of 22 responses, some of which were more helpful than others:.

Q1:

With 81.8% of responses being a yes for acquiring the foldable drawing board, it shows that there is interest in this product.



Q2:

The majority of the answers asked for wood to be used within my product. However, some asked for leather, there was one silicone, and a few other interesting ones (as can be seen on the right). There were only responses which asked for plastic and one for metal, meaning that I will probably not be using them. And some which suggested leather as a drawing surface, probably because it is comfortable to draw on.

Interesting responses for Q2;

"Plywood and some sort of vinyl layer - much like a cutting Matt. That pr a highly polished wooden surface."

"a balance between sturdy and hand-friendly- my drawing board is made out of wood sheets; while i may end up getting good neat sketches, my wrist legit hurts lol"

"No specific materials, as long as it is light weight but strong."

"Not too brittle. Must be flexible. And comfortable to draw on."

Q3:

For this, the majority of answers were either with the way the paper was attached, or with the ruler being annoying to use. However. there were a few outliers which were interesting as they gave a wider range of problems to consider.

Interesting responses for Q3;

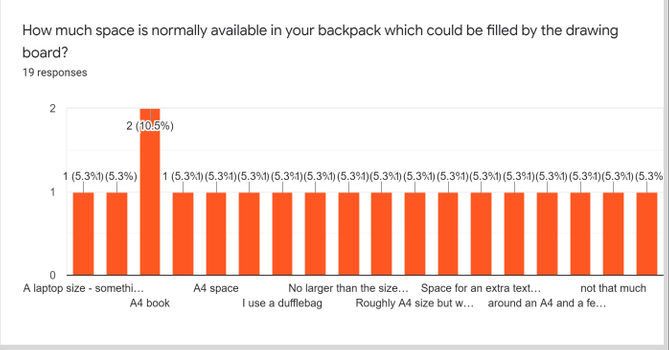
"any bumps that mess up the work so not having a perfectly flat surface. not being able to see beneath it so that i could copy / trace a drawing for example"

"Plastic. It ages badly. Cuts, ink look ugly on it. Wood absorbs better"

"at times annoying, the wood keeps splintering me or when i use is as a table top to cut stencils out- the wood peels easily leaving dents when i sketch or draw on my drawing sheet"

Q4:

As the graph on the right shows, most people have around an A4 size of available space in their backpack, meaning that my product should have around that much space available.



Q5:

The only notable responses were about cost, lots of emphasis on weight, and the material choice regarding their sustainability and manufacturing process.

Target Market Research 2:

Analyzing other people using Drawing boards:
To see how other people use drawing boards and to understand problems with them, I took part in a class where students would use drawing boards and I examined their interactions with the product, looking at 2 key things: Their kinesthetics (explained in the Ergonomics page) and their interactions and possible frustrations with the product.

The two main issues found is that lots of people were complaining about how hard the mechanism to attach the paper to the board was, especially aligning it correctly, and with the ruler. Regarding the placement of paper, the main issue which I was told was that the mechanism (as shown below), which was very hard to use, as it was very hard to grip the handle, and to pull it back, in order to put the paper underneath; furthermore, aligning the paper so that it was straight was very hard as the line which you had to align it to was very hard as the line was too small. A solution which was suggested to this problem was to put clips in all corners, as it'd both make it easier to align and would also stop the paper from crimpling.

The main issues regarding the ruler were that some students were struggling to clip them in without them falling out and that the ruler was sagging on the extremes. The most common complaint is that ruler was hard to slide in its track so an improvement to that should be made (a possibility is to use magnets instead).

The standard A3 drawing board sticks out of the school issued (and relatively large) bags by 30mm in height and 120mm on each side.

Interviewing people outside 4D Model Shop.
To look at more possible clients, and their opinions regarding my product, I went outside the 4D Model Shop near Aldgate East. Having spent a few hours outside it, I talked to a lot of people, but most did not use an A3 drawing board. However, I did find some people, and these are their responses to the following questions:

Q1 – How often do you use a drawing board?
1. "I work for an architecture firm in central london so quite a lot"
2. "'Used to in school but stopped when I left it"
3. "I tend to draw on a desk these days but I used to years ago"
4. "Daily since I'm an engineer"
5. "Only in art– I want one but they're all too costly"
6. "Every once in a while for fun"

Q2 – Would you be interested in a foldable A3 drawing board?
1. "if it was made well, it'd get it"
2. "Not right now"
3. "I don't need it but if I was looking for one yeah sure"
4. "Seems like a cool idea"
5. "Only if cheap"
6. "I don't really use my drawing board when I go about but I like the idea"

Q3 – What problems does your drawing board have?
1. "I used to have one for a few years which I brought to and from home but I've stopped doing that after I dropped it on the train and it dented and I couldn't straighten it"
2. "I remember hating the ruler thing cause it'd get stuck like always"
3. "A spot on my desk got ruined by some tea I spille don it a while ago so you waterproofed it"
4. "Mine is built really well so I don't have any complains about it"
5. "The ones at school are all dirty and crusty and old and such"
6. "Too small"

Q4 – What materials would you like your drawing board to be made out of?
1. "Defintely not Allumium since that's what dented"
2. "Something Eco-friendly and recylcable"
3. "The environment is a big thing these days so something to do with that"
4. "I like steampunk vibes - steel and wood basically"
5. "Smooth plastic and such"
6. "Wood"

Q5 – Is there anything else which I should take into account?
1. "I found the surface wasn't grippy enough so paper on it slipped a lot"
2. "I don't know the ruler I guess"
3. "Don't think so"
4. "Not sure how you'd make it foldable without having a nasty notch in it"
5. "Cost"
6. "Nah"

Evaluation of Target Market Research:
Looking at the research, I can see that there are a lot of issues with the current drawing boards. Most of these are about either with the ruler or with the way the paper is attached. However, there were some complaints about the materials used and their durability and strength. Some other things which will need to be considered are the cost and weight, as some of the responses in the questionnaire and in the interviews are worried about that. The dimensions of the drawing board will also need to be looked at, as it needs to be small enough to fit in most bags. Lastly, as there was a few mentions of eco-friendliness in my interview so the material choice should reflect that.

Materials:

Creating a flat surface:
There are 2 methods to create a flat surface. You can either have 2 pieces which have very flat edges and so create a gap which is barely noticeable. Or you can bend a material so that there is a no gaps.

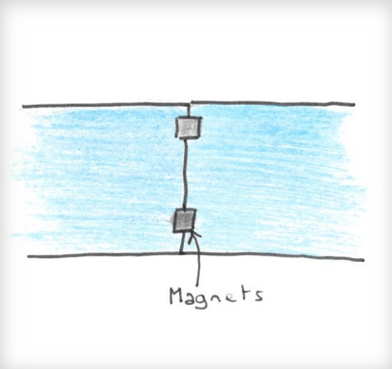
Bending a material:
Flexible materials, such as silicone or PP (or similar plastics) could be used in order to create the drawing board. However, a problem with this method, is that the board might dib in the middle without using proper support. To avoid this. The material could be put under tension much like the canvas of a painting so that it is easier to draw on.

Bending a material continued:
Further consideration of this led me to 2 conclusions. I could use a single material such as Silicone or Leather, Or I could layer 2 materials such as Canvas and PP, as by doing so, you get the advantages of the PP's smoother surface, and the strength of the canvas, without needing to worry as much about the PP being pierced through by the pencil.

The 2 pieces method:
To do this. I would need to use a material like acrylic whose edges have been cut exactly horizontally in a way that when the two sheets are pushed together, the gap between them becomes seamless. This will be very hard to do, and it might be worth trying to find a compromise between the 2 methods.

Using smart materials:
Using smart materials, I could have the board be rolled up, and then using Electroactive Polymers, activated through electrical currents, such as Solvene® EAP, could be used to have the board roll out and become rigid. However, there are a few concerns around this, such as the strength of the board, how often it needs to be recharged (impacting its usability), and finally how expensive smart memory alloys are.

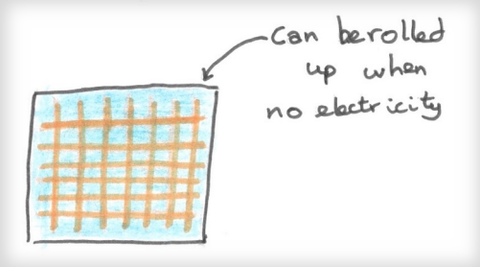
Inside a backpack:
Inside a backpack, the drawing board is likely to be hit and shaken around. So, the materials used must be durable enough that they will not break or be ruined inside a backpack. This means that materials which shatter easily should be avoided unless a method to protect them is developed.

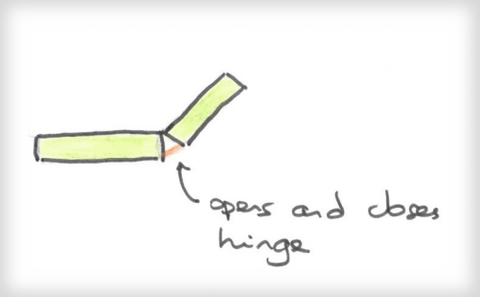


Resistant materials:
Materials which could be used to create the casing of the product range from wood, to aluminum and acrylic; as many other materials cannot be used due to their weight or other properties.

Weight:
As this is supposed to be a handheld and portable product, the materials used must be light weight. This means that heavier metals such as steel or heavier woods like Verawood should be avoided as even though they are very tough and resistant. Their weight makes them unsuitable to a portable product. However, lighter metals such as aluminum or wood like MDF could be used to make it more portable whilst maintaining its rigidity.

How Smart Materials influence the design:
There are 2 ways the EAP could be used in my design. The first method is to create a net, as shown in fig. 1, which supports a bendable material creating a surface to draw on. And the second method is to use it to move the hinge automatically.





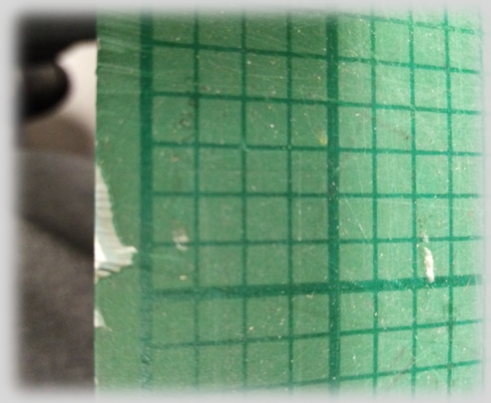
Contacting the company:
I asked the company some key information about their product. This is the voltage it runs at, a datasheet of the product, and how easy it is to use and work. The price is extremely high at £187 for 10g, so I will need to research more materials in order to find cheaper ones. I never ended up receiving a reply.

Continuation of research:
As I never received a reply from Solvene EAP; I decided to research other EAPs companies in order to find other possible materials and products which I could use in my product. This can be seen on the next pages.

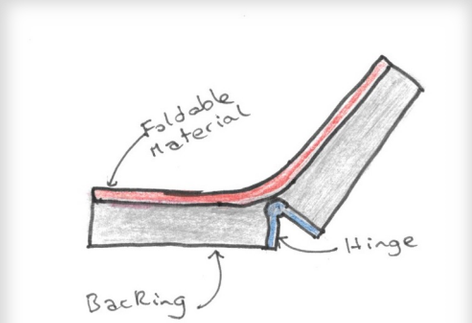
Materials 2:

Feedback from peer:
A feedback from pier has led me to research cutting mats as a possible option. This material is a composite material which has an inner layer of foam to give it flexibility, and an outer layer of a hard plastic which is scratch resistant. The outer layer protects it while the inner layer lets it bend more easily. However, a problem I found is that when the mat is bent too much, the outer layer of hard plastic shatters which means that it would not work for my product. However, composite materials should still be considered within my design.

Composite materials:
Research of composite materials has led me to consider other options such as Canvas with PP and Leather with a Wooden backing (which the idea for I got from looking at a wallet and the cards inside which creates rigidity). The use of composite materials would allow me to get the best of both worlds, with the strength and rigidity of one, and the flexibility of the other,



Looking at the wallet (as seen below) has led me to identify a problem with using leather which will need to be tested is that if the leather is folded multiple times, then it starts to get ruined and become creased.



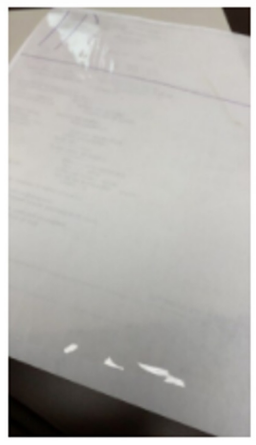
Primary Research - Comfortability of writing on and tactile responses:
Primary research needs to be conducted in order to work out which materials are the most comfortable to write on and to handle (see ergonomics page for research). To do this, I have asked multiple people to hold onto a material and try to write on it and to give a score to the different materials in the two categories from 1 to 10. Then, I have used this information to assess the different materials:

Material:	Tactile Response Scores:								Mean Tactic Response:	Drawing Scores:								Mean Score for Drawing:
Aluminum.	7	7	9	7	9	6	7	8	7.5	8	9	10	10	7	8	8	8	8.5
Silicone Rubber.	3	7	3	6	5	1	6	10	5.1	3	4	2	4	3	2	3	4	3.1
Leather.	9	7	10	6	9	9	10	10	8.8	7	6	5	6	9	9	8	7	7.1
MDF.	5	4	3	5	4	7	5	8	5.1	9	8	4	8	4	7	6	10	7.0
Acrylic.	8	8	8	8	8	7	6	7	7.5	10	7	9	7	7	8	8	9	8.1
Canvas.	8	9	10	6	7	8	7	5	7.5	2	6	1	1	1	3	2	1	2.1
PP.	7	6	7	5	6	4	8	7	6.3	10	6	9	7	3	6	9	9	7.4
Latex Rubber.	6	8	4	7	5	1	4	8	5.4	2	6	7	8	4	2	4	6	4.9
Foam.	2	2	1	1	3	1	1	1	1.5	1	3	1	1	1	1	4	1	1.6
Cherry wood.	10	9	10	9	10	9	8	9	9.3	9	9	10	10	9	10	8	9	9.3

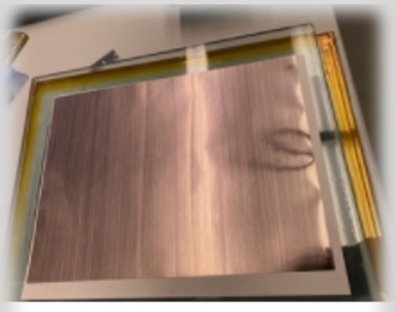
Specific Material Research – EAPs:

Beginning the research:
I began my research by looking at a NASA website (EAP material and product manufacturers (nasa.gov)) which gave me links to many different websites which sell and produce different EAPs. Although most of them were not useful. I did find one site which helped me, this was PolyK Technologies.

PolyK Technologies:
Home | PolyK Technologies (polyk-lab.com)
There are three films which I could use which PolyK produces. The first one has a sheet size of 180mm by 270mm, and the amount which it bends is considerable, so a single one could probably be used . The second one is a similar material, but it's slightly smaller, as it's sheet size is 180mm by 260mm. Lastly, the last one has a sheet size of 200mm by 270mm, making it the biggest one. They all have similar actuating potential meaning that they all bend similar amounts when power is sullied to them. Furthermore, they need 600V to activate them meaning that they would be unsafe for any consumer to use.



Piezo PVDF film, Rated for 100C, high d33 and d31
$300.00



Metallized piezo sheet PVDF film, 45 um thick, 200 nm thick sputtered Cu/Ni
$300.00

Metallized piezo sheet PVDF film, 28 um thick, sputtered Cu/Ni Electrode
$250.00

Electroactive Polymers:
Searching for websites which gives me a wide amount of information on EAPs. Doing this, I found Azom.com which gave me access to information about many different EAPs. From this, I found out about Electro-strictive Paper, which is a flat surface which when an electric current is applied to it, bends towards the direction which it has been programed to.
Piezotech® sells Electroactive Films. However, looking at the prices from PolyK Technologies, One can assume that they are very expensive. However, to be sure of this, I contacted them (as can be seen below). Lastly, through the Azom website, I found out about Liquid Crystal Elastomers, which change state depending on the voltage which is passing through them, hardening or softening by forming a crystal structure; however, they are extremely hard to find online. Therefore, I decided to look at Carbon Nanotubes as they could also be used as a liquid which hardens when electricity passes through it. This would mean that you could have a thin pouch which contains it, and when electricity passes, it will harden the surface making it writeable. However, the cost of the nanotubes is extremely high at £50 for 500 mg, meaning that it would cost too much for it to be used within my product.

Sub-Application Piezoelectric Films

Type of request* Quotation

Your message*

I am in need of a quotation for an A3 sized sheet of Piezoelectric Film.
I would also like to ask for a small sample in order to test the material if that would be possible.
Many Thanks, Rastelli.

Dear Francesco,

Thank you for your interest in our products.

Unfortunately, we do not send samples anymore. We use to offer a 5 % discount on Universities' orders to support their academic developments. If you have already published a communication (Article, Abstract, or Oral communication) citing our company and products, please send it to us, and we will be glad to increase to the discount to 10 %.

The only film size we sell (FC20 copolymer) is 12*12cm². With a thickness of 10 microns, metallization and polarization, the film is sold for 630€/u.

If you are looking for classic PVDF film, we advise you to contact Polyk https://piezopvdf.com/ who can also provide you with larger sizes, you can tell them that we sent you to them.

I remain at your disposal,

Sincerely,

Mickael Pruvost

Evaluation of EAPs:
Looking at all the different EAPs, I have concluded that they will likely not be used in my product for a few reasons. Firstly, cost is a big factor. As my costumer does not have an extremely large budget, it would be inconsiderate to use such expensive materials within my product, as to even create something with the most basic ones would cost upwards of £200. Secondly, the use of EAPS would create a layer of complexity which is unrequired within a product such as mine which should be developed to be portable and lightweight; and the use of batteries and the need to charge would hinder that process and its usability. Lastly, the delicate nature of these materials means that they are likely to not be even storable within a backpack.

Specific Material Research – Changeable Rigidity Materials:

Shape Changing Robotic Fabric:
https://newatlas.com/materials/robotic-fabric-shape-change-soft-hard-heated-cooled/
This fabric can change its rigidity depending on the temperature that it is at. It was developed at Yale by a team of scientists and works by using a range of fabrics that have different abilities, which in turn are interwoven into a piece of base fabric to create a composite which can utilise all of their abilities.

Their Field Metal-Epoxy Composite can "become as flexible as latex rubber or as stiff as hard acrylic, over 1000 times more rigid, just by heating it up and cooling down", as quoted by the lead author of the study. This means that this product could easily be used by me in my project as the drawing surface for it, or as support for the drawing surface.

Possible uses of this fabric range from high tech clothing, to tents that set themselves up and robotic parachutes. However, the most interesting thing that it is currently being tested on is Stowing wings, which is very similar to my product and how I could use it in it.

An issue with using heating and cooling, however, is that it could pose a danger to the user and might result in burns if any heating or induction coils become exposed through the use and wear of the product. Therefore, it is unlikely that I would want to use this product. And, as the video [on the stowing wings link] shows, it takes quite a bit of time for it to deploy which could impact user experience.

Searching for other materials:
As I was searching for other materials, I started reading a paper on Controllable Tuning of Material Rigidity in order to find other materials which I could use within my product. There are two types of Variable-Rigidity Materials: adaptive semi-active and active materials. Adaptive Semi-Active undergo transformations in response to an external stimuli, meaning that their mechanical properties change depending on a force which is exerted upon them; different materials which fall within this branch is SMAs, SMPs, MRFs, and other elastomers and fluids. However, active materials act as energy transducers, meaning that they convert different energies, such as mechanical energy to heat or electricity or vice-versa; most of these materials are EAPs, PZPs and MSMAs.

Another useful source which this paper gave me is a table which had different materials and the projects which they were used within. From it, I found three adaptive semi-active materials which I think that I should research more:

Electric	Electroactive gel*	Actuators	Wearable assistive device [23]
Magnetic field	MR elastomers (MRE)	Dampers	Prosthetic device [21]
Electric	Electrochemo-mechanical conducting polymers (ECMCP)*	N/A	N/A

The first one that I started looking at is Electroactive gel. The idea that I had in regard to this is to contain the gel within a thin pouch which acts as the backing for the drawing surface; then, when electricity passes through it, It hardens resulting in a hard drawing surface. However, there are many issues with this material: looking at an example from the Unisversity of Bristol; the material contracts by a mere 5.5% when 600V are applied; meaning that it is unrealistic that I could use this product. Furthermore, it is commercially unavialbe to the public and can only be synthesised in a specific lab meaning that I could not even acquire it.

The next option is MR elastomers. These have ferromagnetic particles within them which change the state of the polymer when subjugated to a magnetic field. Therefore, I could have an electro-magnet within my product which I can turn on in order to make the board become more rigid. However, the main problem with this is again the amount of current required to create a magnet strong enough to actually activate the properties of the elastomers. Furthermore, this is another material which is extremely hard to purchase to the public as almost no companies produce it internationally and the few that do only sell to large bulk buying companies. However, it might be possible for me to create a similar material by crushing a magnet into a dust and mixing it into a solution, such as melted silicone, which in turn will harden keeping the magnetic dust in place.

Lastly, Electrochemo-mechanical conducting polymers are very similar to EAPs, meaning that they change firmness when a voltage is applied. However, they require much lower operational voltages and have larger strains. The current best ones use a layering of the materials in order to increase the difference of actuation which the overall material uses. However, these are also created within a lab and so I would be unable to use them within my project as they are too hard to acquire for a student like myself.

Specific Material Research – Leather:

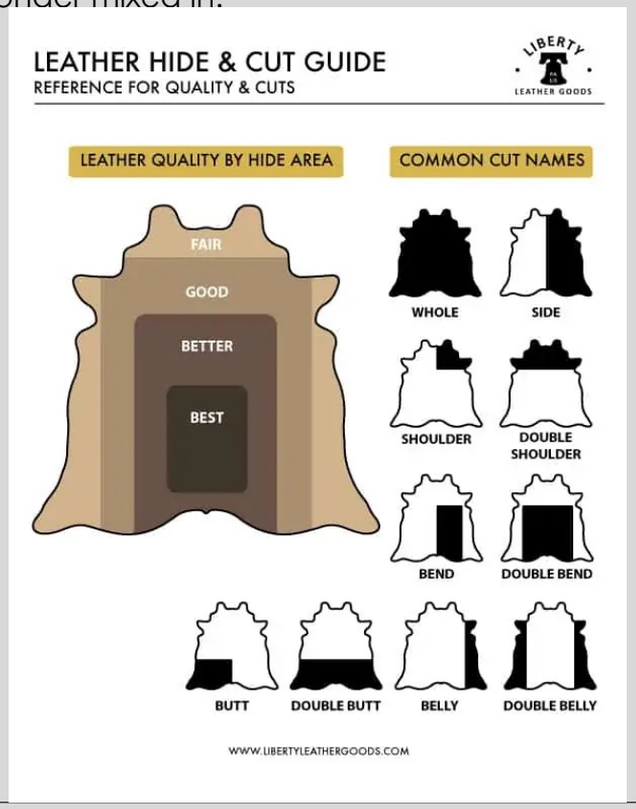
Cromford Leather - Leather, Suede and Shearling Specialists in London
Old Style Aniline Goatskin leather hide - BuyLeatherOnline
Types of Leather: All Qualities, Grades, Finishes, & Cuts (libertyleathergoods.com)
Italian Leather Supply Store: Full Grain Hides & Sheets - BuyLeatherOnline

Types of leather grades and qualities:

- Full Grain leather:
  - Uses the outer layer of the hide and hasn't been sanded or buffed to remove imperfections. It is very strong, durable, waterproof and can withstand tough use. Develops a patina (color change) over time that can be pleasing to the eye. Often used for saddlery, footwear and furniture.
  - Nappa is a subclass of this that is generally smoother and softer.
- Top Grain leather:
  - Similar to Full Grain except that it is sanded to remove imperfections. Often softer and more malleable, but less durable and removes some of its waterproof abilities.
- Genuine Leather:
  - Leather which has been treated to give it a uniform appearance. Often used for belts and similar goods.
- Split Grain Leather;
  - Cut from the lower levels of the hide. Not as dense and tight as Full grain and Top Grain.
- Bonded Leather:
  - Made up of leather scraps which are bonded together and mixed with polypropylene or latex into a sheet. Its characteristics depend on the composition and percentage of bonder mixed in.

Different cuts:

- Whole Leather cut:
  - Offers a wide range of softer and harder areas.
- Side Leather cut:
  - Ranges from softer stretchier areas to thicker stiffer areas.
- Shoulder Leather cut:
  - Firm yet malleable and flexible feel to it. Often used for tooling.
- Bend Leather cut:
  - Area from the hide to the belly. Gives you some of the best cuts of leather.
- Butt Leather cut:
  - Thickest and Firmest area of the hide. Used for heavy items such as belts.
- Belly Leather cut:
  - Much softer and malleable than other areas.



Different Animal Hides:

- Bulls:
  - Thick, heavy hides. Useful in heavy belts and boot soles. More expensive than most other ones.
- Steer:
  - Lighter than bull and more widely available. Still quite tough.
- Cows:
  - Thick and soft. Recommended for most needs.
- Heifers:
  - Soft and pliable. Recommended for shoes, boots and similar leather uses.
- Dairy cows:
  - Soft and Thin. Makes great uses for lighter leather products such as wallets, clothing and upholstery.
- Calves:
  - Very soft, thin and supple. Useful for wallets, watchbands and smaller leather accessories.
- Pig:
  - Used primarily in clothing due to breathiness and lightness of skin.
- Sheep leather:
  - Contains wool on one side. Often used in clothing.
- Goat:
  - Very soft, strong and durable. Skin is suppliable and flexible and can be used to give a more comfortable feel.

Useful Finishes:

- De-grained Leather:
  - Has the grain layer removed. Results in a much smoother and more consistent surface. Makes it less resistant to moisture,
- Stretch Leather:
  - Has a composite leather which allows it to stretch when used. Uses a processed leather surface mixed with a synthetic layer underneath.

Evaluation of Leathers:

Looking at all the different leather types, I have chosen three which I will take samples of and test before deciding on the final one. These are Anline Goatskin, Vintage Vegetable Calfskin, and Anline Nappa Lambskin. This is as they are all very flexible skins which allows them to be folded easily without ruining the leather. Furthermore, They are all very thin which means that they will be easier to attach to the backing without it resulting in a very thick product being formed which will hinder its storability within a backpack..

Mechanisms:

Different Mechanisms:
There are three different mechanisms that I can think would be useful to my design. One is the use of a normal hinge to create a much simpler system; the other is to use something like the folding mechanism found in the Galaxy Z fold as it was developed for similar uses; and lastly, I could use something similar to one used within a Huawei Patent.

Normal Hinges:
The normal hinge could be used to attach the two backings together and letting it bend inwards to close it, but when opened, it will result in a much smaller gap if the hinge sits flush within the backing.

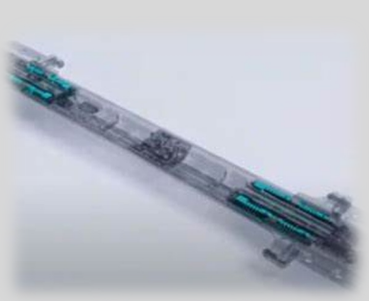
The main advantage of using this method is that it's a lot easier to make, cheaper, and more reliable. However, the material inside of it would end up being bent in half flatly which would result in it likely tearing, with no real way of avoiding that except to have a hollow backing, which would render the backing useless.

Huawei Patent:
The Huawei Patent uses a number of interconnected Hinge segments and a linear actuator to allow it to change its angle while keeping a uniform distance between both sides, thus ensuring that the screen within it is protected. However, this also results in it becoming very complicated and hard to produce.

Galaxy Z Fold:
The Galaxy Z Fold is an existing mechanism which allows me to fold a material without it tearing as much whilst maintaining a rigid backing. This system was developed for a similar use, so I feel that it could prove to be very useful to my product.

Looking at the mechanism, I can see that it uses a gear mechanism to keep the parts aligned, and as can be seen in the first image, it uses springs to make the opening and closing system smoother. I will need to test in my modelling whether I need the springs and gears for my design or not, and thus whether I can simplify it or not; as to make very small gears and springs is quite hard given the tools available to me.

Overall, this system has many more advantages that the hinge one, as it allows me to make sure that the bent material is much more protected, and it would also result in a much more rigid and strong backing.



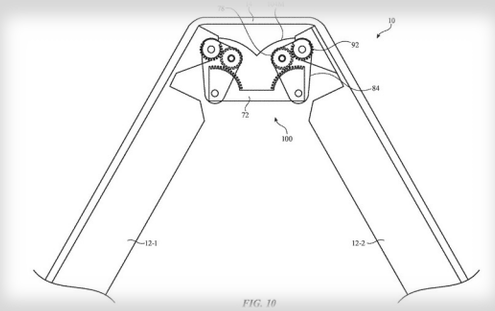
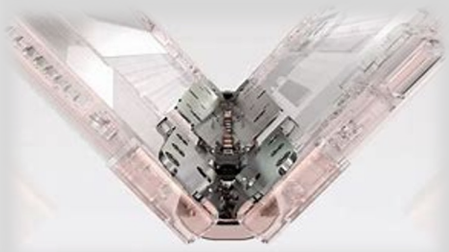


FIG. 10



Snap Bands:
I could use something similar to snap bands on order to make a backing for my board which stays flat when opened; and then you can hit it in order to roll the board up and close it. However, there are 2 concerns with this mechanism: firstly, if someone applies too much force to the drawing board while sketching, it might cause the board to close on them, And secondly, the metal might damage the drawing surface when the board is closed as it might deform it.

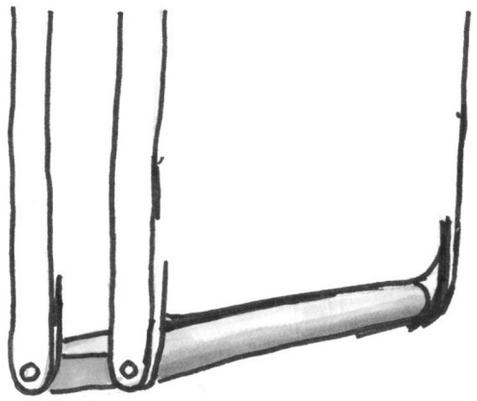
Evaluation of mechanisms:
Comparing the three mechanisms, I have come to the conclusion that I would use something similar to the Galaxy Z Fold method in order to be able to create a product which is much more reliable and easier to use, as I do not think that the Normal Hinge method would work for the materials, I want to use without something getting damaged. Furthermore, The second mechanism results in a much nicer finish and looks cooler in the process meaning that the user is more likely to buy it. Furthermore, I have not considered the Huawei Patented Mechanism due to its complexity and machinable accuracy required to fabricate it in a way that would allow it to work effectively. Lastly, to make a final decision on the snap bands I will need to test them.,

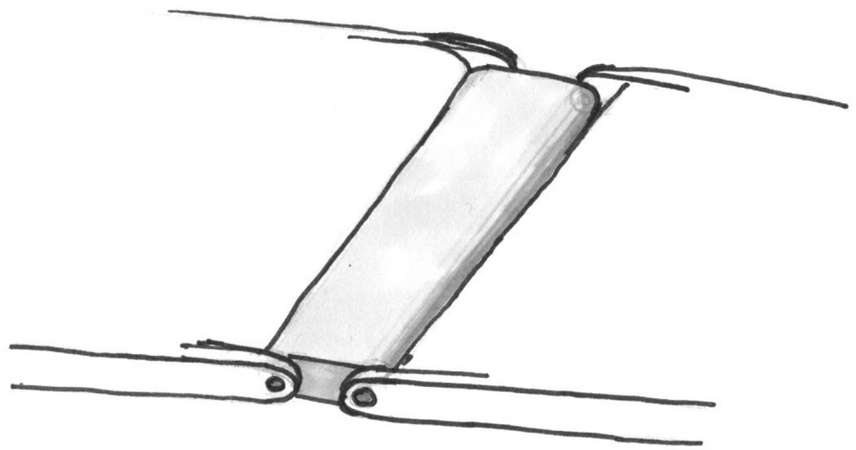
Mechanisms – Initial Sketches:

To explain some of my ideas for mechanism, I sketched some down in order to show some of the initial ideas that I had for the mechanisms.

Samsung Fold Based design:
As seen on the drawings on the right, This would have a hinge in the middle (colored gray), which holds the two pieces together. This would allow it to bend with the leather without it stretching or compressing the leather surface too much.

There will need to be modelling and development of this mechanism to test whether there will need to be gears or magnets or another system which will make the opening and closing more comfortable and reliable to use. There will also need to be research into a way to lock it when it is both open and closed.



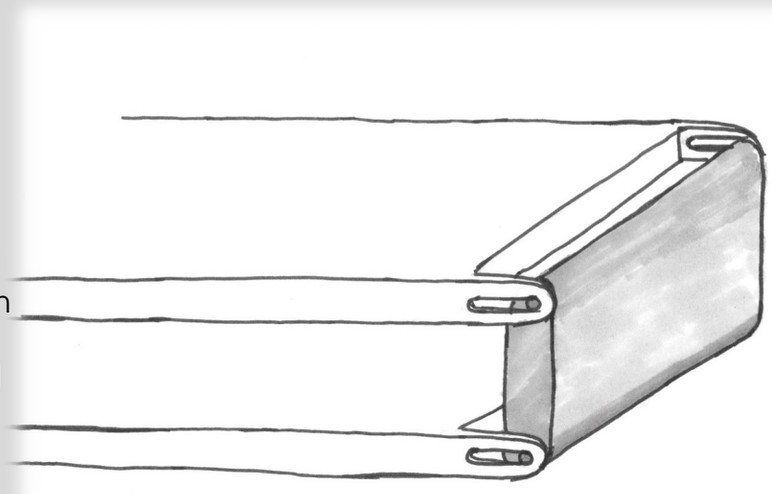


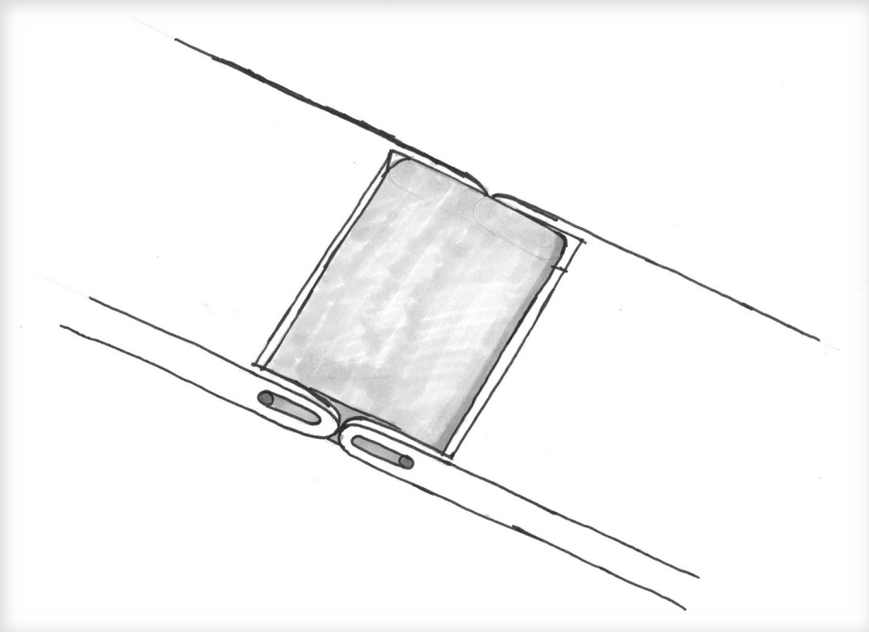
Own Mechanism:
I came up with this mechanism as a way to lock the board in place when it is opened and closed.

I will need to test weather I'd want to use magnets, a mechanical lock, or a spring system in order to lock the board in place when it is opened and closed.

I will also need to see how the leather would sit on this board and weather it would stretch when the board is open thus potentially damaging it meaning the mechanism could not be used as it might place the lather beyond it's tensile limit which would permanently deform it, thus ruining the drawing surface.

I could also make the faces flat on this meaning the gap in the board is even less noticeable.

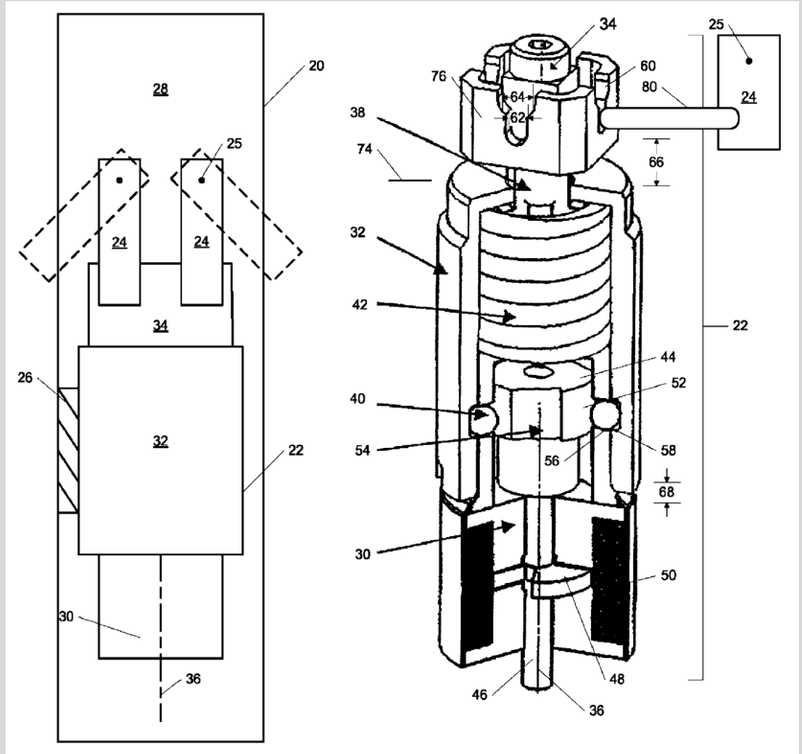




Mechanisms – Locking Mechanism:

As I was thinking and further improving the design of my product, I realized that there would be a need for a locking mechanism in my product in order to make it rigid when it is opened and closed. This means that the mechanism to lock it needs to be able to block it at 0° and 90°.

Given the restrictions of my design, it is likely that my mechanism would need to fit inside a very small circular tube, as it cannot be too thick without it making the whole board much thicker, and it should rely around affecting the pivot of the board.



The first mechanism I looked at is a Rotary actuator ball-detent locking mechanism which uses balls around a cam to block the shaft from rotating any further through the use of a spring to push on them and forcing them in the gaps within the cam.
I could use a similar method to apply force to the pivot on the board in order to lock it in its closed and opened positions. However, an issue with this would be how to unlock the mechanism without it being extremely hard and uncomfortable to use.

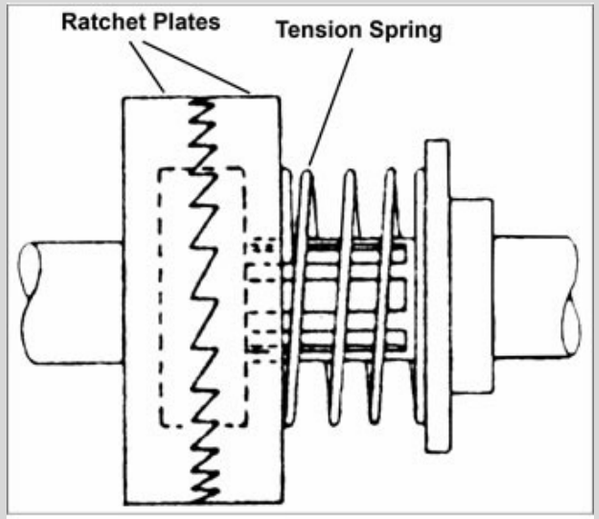
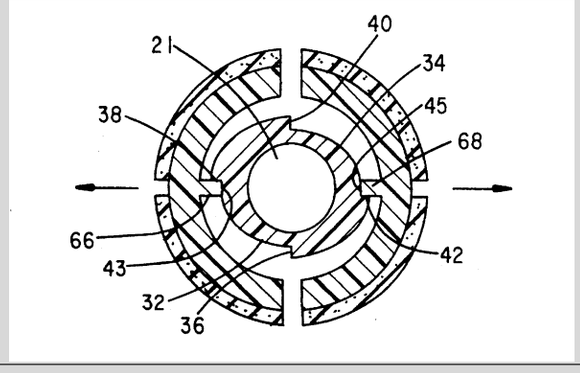


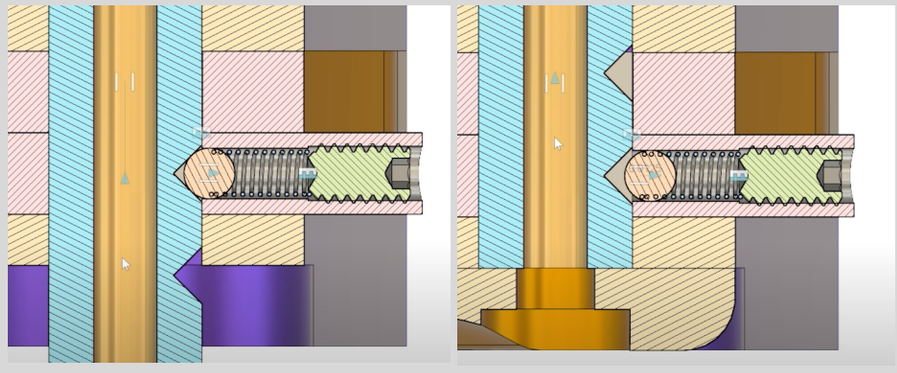
Figure 5.
One-way dog clutch: Becker Orthopedic 9001 E-Knee integrates dog clutch into its design where ratchet plates are spring loaded to separate. Surrounding electromagnetic coil works against spring to engage plates in stance.

The second mechanism I looked at is a One-way dog Clutch at a mechanism which uses two ratchet plates which are held together using a tension spring. This means that the shaft can be held at very specific angles without it moving meaning that I could design this in a way that blocks it at 90° and 0°. However, the main issue with this design is that it can only turn one way, so even though my board would be able to open. It would not be able to close without it disengaging the two plates.



The third mechanism I looked at is Locking mechanism for telescoping tubular poles which stops a cylinder within another cylinder from moving thus stopping it from moving out of the other tube. I could use something similar to not only lock my board in place, but also limit the amount which it moves to the constraints needed.

The last mechanism I looked at is steel ballicissors. Although at first they might not seem very useful, they contain a mechanism inside to lock the scissors in place so that they can be used to cut.



To do so, it uses a very small ball in front of a spring to push itself inside of an indent in order to block it from moving without the user applying enough force to move it to the other level.
I could a similar system to lock the board at 90° and 0°. However, worries regarding this system is that it will not be strong enough to make the board rigid, and it might also be too complicated to build without resulting in it being extremely thick, as the whole system needs to be very small, probably around the size of millimeters, which could in turn make it much easier to break.

Ergonomics:

Haptic Ergonomics:
(index.dvi (gla.ac.uk)) - Interesting paper on haptic ergonomics standards.

With this product, as it will be often handheld, it is important that the haptic responses of the materials to the user will be positive. This means that research should be conducted into what materials to use, especially for the casing. As the paper says, 2 key components of haptics should be considered, these are the tactile aspects of the product, such as how the material feels to the skin, or the usability and cleanliness of mechanisms are felt by the user. However, Kinesthetic responses should be considered, this has to do with how the user interacts with the product and the position of the user's limbs while interacting with it; observing and analyzing how the user interacts with similar products will allow me to get a better picture of how to design this product.

For tactile, materials such as aluminum can be considered, however, in colder temperatures they often become much more uncomfortable to hold as they also cool down really quickly due to their high thermal conductivity. However, materials such as wood, are much nicer to handle, especially when they have been polished and treated. The other aspect of tactile response is the materials which the area on which you draw on is made out of. Thiis is because different materials will feel differently underneath a pen, and while some materials such as silicone are really uncomfortable to write on, as they are quite soft and rubbery. Others like wood are much nicer as they form a hard backing for the paper to go on. The other part of tactile response is the texture of the materials. Smoother materials such as acrylluc sheets or polished wood are more likely to be looked over. But if you give texture, such as little bumps or dimples in the material, this will make it stand out meaning that the user is more likey to interact with it; therefore, key components such as handles will need to be textured in order to make it better for the user.

For Kinesthetic, the two factors which need to be considered are the placement of key parts, such as the removable rulerand the handles, and the mechanism to open it. If the mechanism is too tough, the user might be inclined to put their arms in uncomfortable positions in order to open it; however, the mechanism will need to be strong enough to hold it closed. Furthermore, the placement of the key components should be such that none of them make the user feel like they have to unnecessarily stretch in order to reach them, or that they have to put their arms and hands in cramped positions in order to use them.

Visual Ergonomics:
Visual Ergonomics | The International Ergonomics Association is a global federation of human factors/ergonomics societies, registered as a nonprofit organization in Geneva, Switzerland. (iea.cc)
Visual Ergonomics is very similar to color theory, in the way that it is the study and analysis of the interaction between humans and elements in a system with regards to the visual stimuli of the system. To create a product which uses this within it, special attention will need to be given to the visual displays and information design of the product, or in other words, the change in texture and color of specific areas in order to make them stand out so that the user is more likely to interact with them.

With regards to the change of color, Color Theory will be used to create a more pleasant and sensible color pallet. The pallet will also highly depend on the material choice of the board, as the different colors should fit well together, and, as the client interview shows, it needs to fit well with the color of the leather. Looking at existing color pallets on Design Seeds, I have decided that these are my favorites:

Evaluation of ergonomics:
Evaluating the ergonomics, I have decided that the Haptic ergonomics will be important in order to create a more comfortable and easier to use design by changing the texture and shape of different parts which you interact with; this will lead on and be connected to the research which was conducted in the material research in order to create a better tactile response and thus a more useful and enjoyable product. In terms of Visual ergonomics, the change in texture to the eye will definitely help, but in terms of the color pallet, I am thinking of using more browns and beige and blues to highlight and draw attention to parts which the user interacts with.

Dieter Rams:

Dieter rams is a famous designer who greatly inspires me and my product. I like the minimalistic and simpleness of his designs, and how form follows function. Something else of his which inspires me is his 10 principles of good design [as seen below]; which I would like to focus on in my product in order to maximize its usefulness and potential; whilst also being more environmentally friendly and aesthetic.

Dieter Rams' 10 principles of good design:

1. **is innovative**- The possibilities for progression are not, by any means, exhausted. Technological development is always offering new opportunities for original designs. But imaginative design always develops in tandem with improving technology and can never be an end in itself.
2. **makes a product useful** – A product is bought to be used. It has to satisfy not only functional, but also psychological and aesthetic criteria. Good design emphasizes the usefulness of a product whilst disregarding anything that could detract from it.
3. **is aesthetic** – The aesthetic quality of a product is integral to its usefulness because products are used every day and have an effect on people and their well-being. Only well-executed objects can be beautiful.
4. **makes a product understandable** – It clarifies the product's structure. Better still, it can make the product clearly express its function by making use of the user's intuition. At best, it is self-explanatory.
5. **is unobtrusive** – Products fulfilling a purpose are like tools. They are neither decorative objects nor works of art. Their design should therefore be both neutral and restrained, to leave room for the user's self-expression.
6. **is honest** – It does not make a product appear more innovative, powerful or valuable than it really is. It does not attempt to manipulate the consumer with promises that cannot be kept.
7. **is long-lasting** – It avoids being fashionable and therefore never appears antiquated. Unlike fashionable design, it lasts many years – even in today's throwaway society.
8. **is thorough down to the last detail** – Nothing must be arbitrary or left to chance. Care and accuracy in the design process show respect towards the consumer.
9. **is environmentally friendly** – Design makes an important contribution to the preservation of the environment. It conserves resources and minimizes physical and visual pollution throughout the lifecycle of the product.
10. **is minimal** – Less is more. Simple as possible but not simpler. Good design elevates the essential functions of a product.

Looking at his products:

The first product that I looked at was a handheld portable radio device [as seen on the right] called the TP 1 Radio / Phono from Braun. It was designed by Dieter Rams in 1959 and there is a few key things which I can take from this. Firstly, it relies on a neutral base whose interactive components are made of a different colour so that it stands out, such as the leather strap. The disc player is also in chrome in order to stand out to the user as where any disc which they'd want to play would be placed.

The second product that I looked at is the Braun Electric Shaver. This is a very minimalistic design which only has one button to turn it on, thus making it as minimal as possible. The only other part which you interact with is the front which is also coloured in black to show that. Lastly, the design has been made unobtrusive for the user, as its shape, dimensions and simple pallet would make it less intrusive for a user and more comfortable to use.

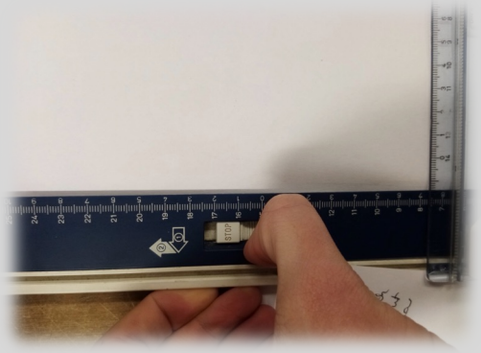
I then looked at the HL Desk Fan which. In its simplicity, just had a single button [in black] to turn it on, and to change the direction of air you could just turn the acrylic covering over the fan either up or down. This design is another very minimal and unobtrusive product for the user.

Lastly, I looked at the famous SK 4 from Braun which has a very simple design, with a neutral plastic base, wooden highlights on each side, and a clear acrylic top. I would argue that the use of a clear top is what makes this product as iconic as it is as it shows the user how it should be used without feeling obtrusive or unaesthetically pleasing.

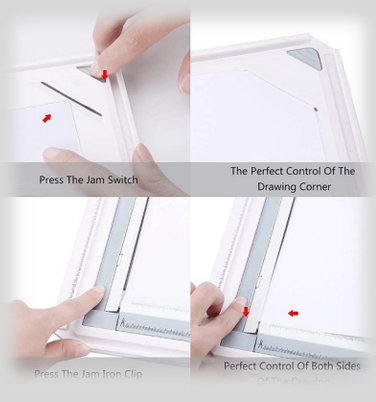
Existing Products:

School's A3 Drawing Board:
I have analyzed the school's drawing boards in order to work out flaws and strengths within them.

The first flaw I found, as shown in the images below, is the method to open the ruler in order to insert the paper inside it and clip it in place. So, to clip the paper in place, I will definitely need to use a different method, probably being inspired by other existing A3 drawing boards.

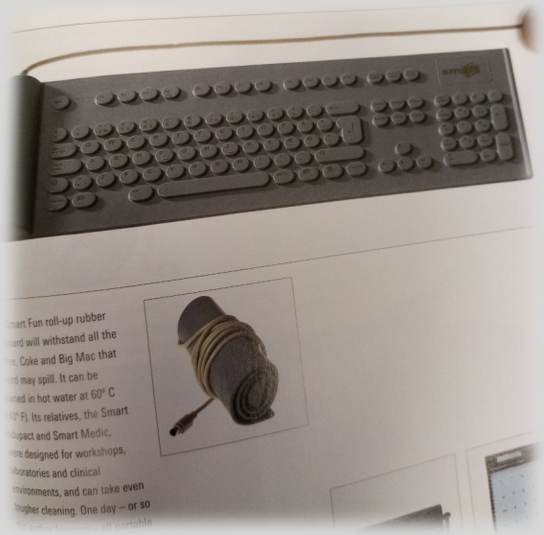


Another flaw that I found is that the ruler did not attach properly to its railing and would easily fall out. And it would also sag on the extremes. A solution to this is to use a similar system to the Hellermans A3 Drawing Board, (as seen below) which has a railing on both sides allowing it to be held firmly in place. The last issue with this is that it was very hard to align the paper straight, as both the line to align the paper and the method to attach it is very flawed. Therefore, a better system should be used within my product in order to make it a lot easier to align it; this ranges from using a better method to clip the paper (as seen below on right), to using a longer alignment line.



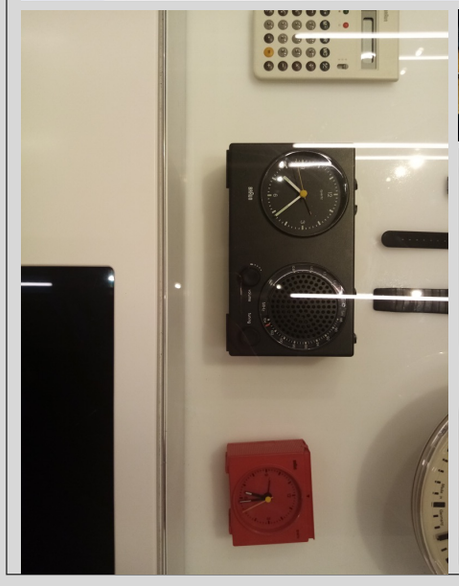
Old Leather Desks:
As my Client requested, he'd like my product to be influenced by old leather drawing desks; so, I spent a bit of time researching them. From this, I have seen a few patterns emerging: They all use similar dark pallets, with both the leather and wood. A popular choice of wood seems to Mahogany, as it polishes and oils very well, and also produces a very nice shine to it.

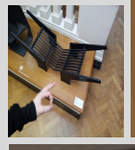
Bendable Keyboard:
This keyboard which I found within the Collapsibles Book is a keyboard that rolls up so that it is more portable and so allowing it to be used with laptops whilst travelling. Although I do not know the exact material used, I could assume that it is silicone or rubber or a similar material as it fulfils all of its properties. However, it is very unlikely that those materials will be used, as shown in my material research, as they are too soft to comfortably write on.

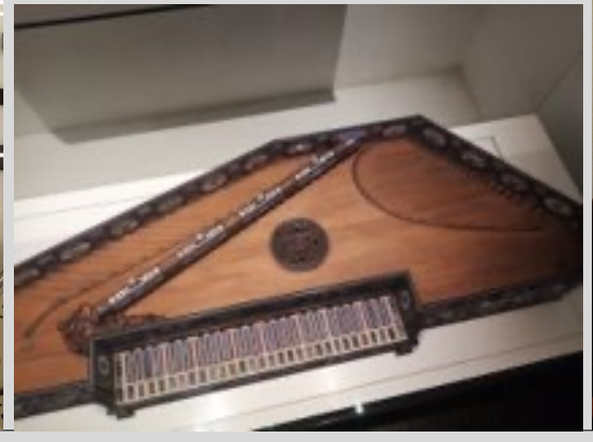


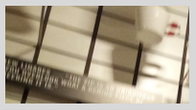
A3 Drawing Board Case:
An A3 Drawing Board Case shows a clear need for my product. This is because at the place of creating extra products to fix an existing issue with a different product (in my case how an A3 Drawing Board doesn't fit within a standard backpack), we should instead be fixing the issues within those products in order to reduce their waste and thus create more sustainable products for consumers. Therefore, not only does my product fix this issue, but it also allows the consumer to feel better about their purchase of it as they feel less guilty given its lower impacts on the planet.

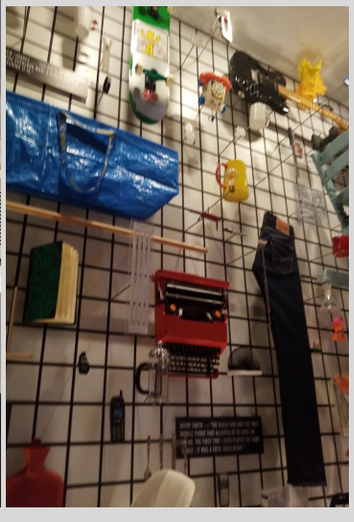
Inspirational Trip To The V&A, The Design Museum, And Japan House:











Design Brief:

A foldable A3 Drawing Board which can be stored inside a backpack. Its design must follow the 10 principles of good design meaning that it must be: minimalistic, unobtrusive, easy to use, useful, innovative, and aesthetically pleasing. To make it unobtrusive, the design must be simplistic and thus minimalistic. To make it easy to use, the mechanism must not be too complicated and if anything in it needs to be interacted with by the user, it should be easy to access and grasp or push. To make it useful, the board should remain rigid when opened but should also stay shut when it is closed. The design should aim to use materials which are eco-friendly and so can be recycled, reused, or are compostable. The design should be able to fit into most backpacks when closed, so it should aim to be around A4 sized. Lastly, the design's materials should be as long lasting as possible and should attempt to not use any fixing methods such as glue to make it easier to recycle.

Specification:

Function:
1. The A3 foldable drawing board must fold neatly and easily so that it is easier to use and will not occupy too much space in the consumer's backpack.
2. The A3 foldable drawing board must be rigid when opened so that it is easier to draw on.
3. The A3 foldable drawing board must not open by itself accidentally so that it does not open in a backpack and damage anything inside it.
4. The A3 foldable drawing board must be strong and resistant so that it's lifetime increases.
5. The A3 foldable drawing board must have all the normal functions of an A3 drawing board so that it can be used like a normal A3 drawing board.
6. The A3 foldable drawing board must be able to store some accessories as the clientele have asked for that.
7. The A3 foldable drawing board should be comfortable to use so that the user enjoys using it.

Cost:
1. The A3 foldable drawing board must be affordable to the user.
2. The A3 foldable drawing board should aim to be easy to disassemble so that broken parts can be replaced in it more easily.

Materials:
1. The A3 foldable drawing board should aim to use materials which are recyclable or reusable.
2. The A3 foldable drawing board must use materials which are comfortable for the user to interact with in both their haptic response and visual response.
3. The A3 foldable drawing board should aim to use the least number of different materials so that at its end of life, it is much easier to recycle without creating a lot of different types of waste.
4. The A3 foldable drawing board should aim to not use glues to connect pieces to it.
5. The A3 foldable drawing board must use lighter materials in order to reduce the weight of the board to make it more comfortable to use.

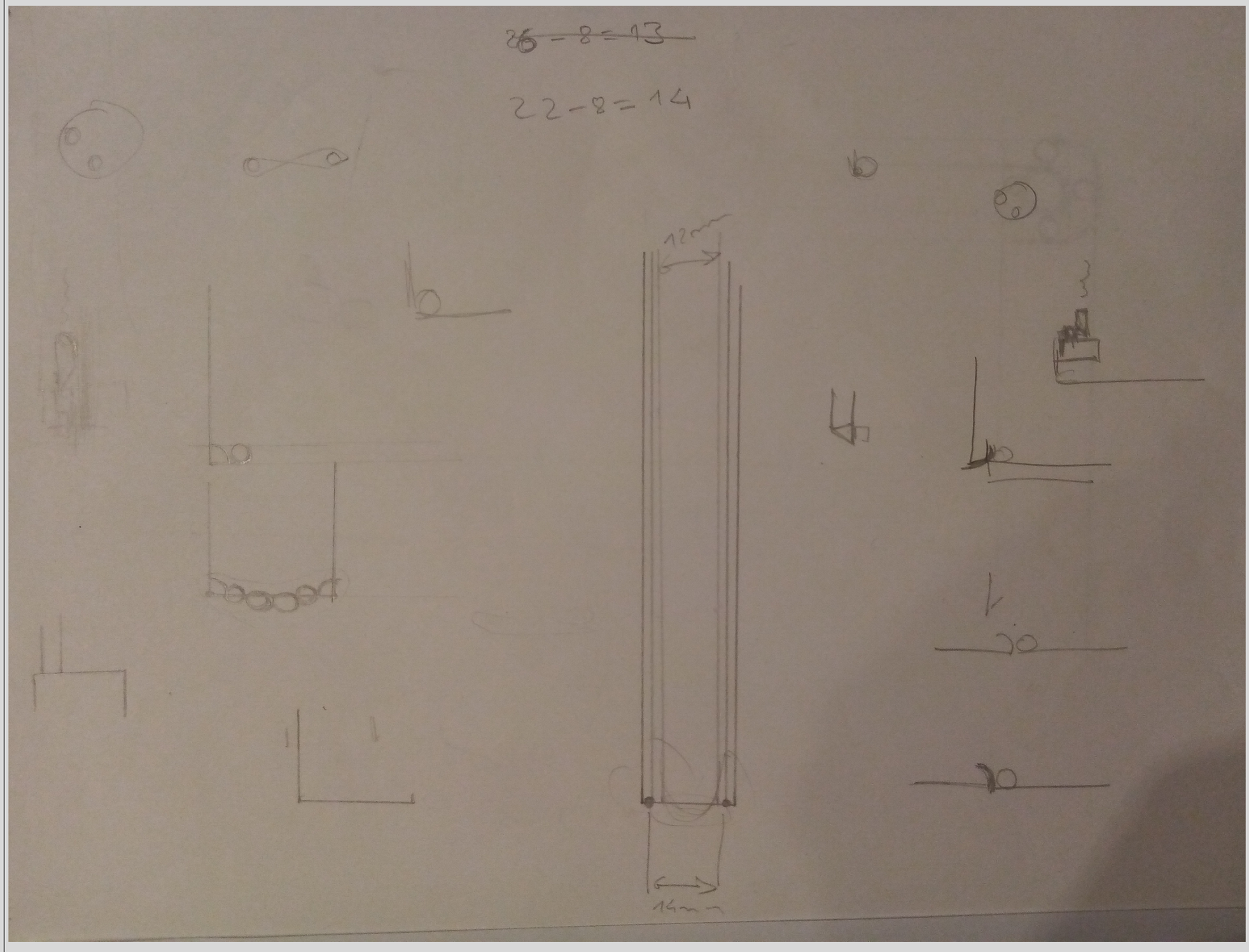
Aesthetics:
1. The A3 foldable drawing board must have a minimalistic design.
2. The A3 foldable drawing board must have a simple colour pallet.
3. The A3 foldable drawing board must have an unobtrusive design.
4. The A3 foldable drawing board must have an eye pleasing design.
5. Parts the user interact with should have contrasting colours.

Maintenance:
1. The A3 foldable drawing board must be easy to disassemble so that pieces can be replaced by the user.
2. The A3 foldable drawing board should aim to not use glues to attach pieces to it so that it is easier to replace pieces within it.

End of life:
1. The A3 foldable drawing board should aim to not use any materials which cannot be reused or recycled.
2. The A3 foldable drawing board must be easy to disassemble so that different pieces can be recycled more easily.
3. The A3 foldable drawing board should aim to not use glues to connect pieces to it so that when it is disassembled to be recycled, it is much easier to do so.

Size and weight:
1. The A3 foldable drawing board must be light enough so that a user can easily carry it without having to exert too much force or putting too much effort into it.
2. The A3 foldable drawing board must be able to fit inside a backpack neatly and easily when closed, so it must aim to be slightly larger than an A4 paper at most and must not be thicker than 3cm.

Designing - Continuing The Mechanism 1:



To begin with my design, I did some random initial sketches on a spare piece of paper to being drawing down ideas.

The first problem I tried to tackle was to how to block the board so that it woud stay open and closed. For this, I thought about using springs, having a pin locks every 90° to lock it there.

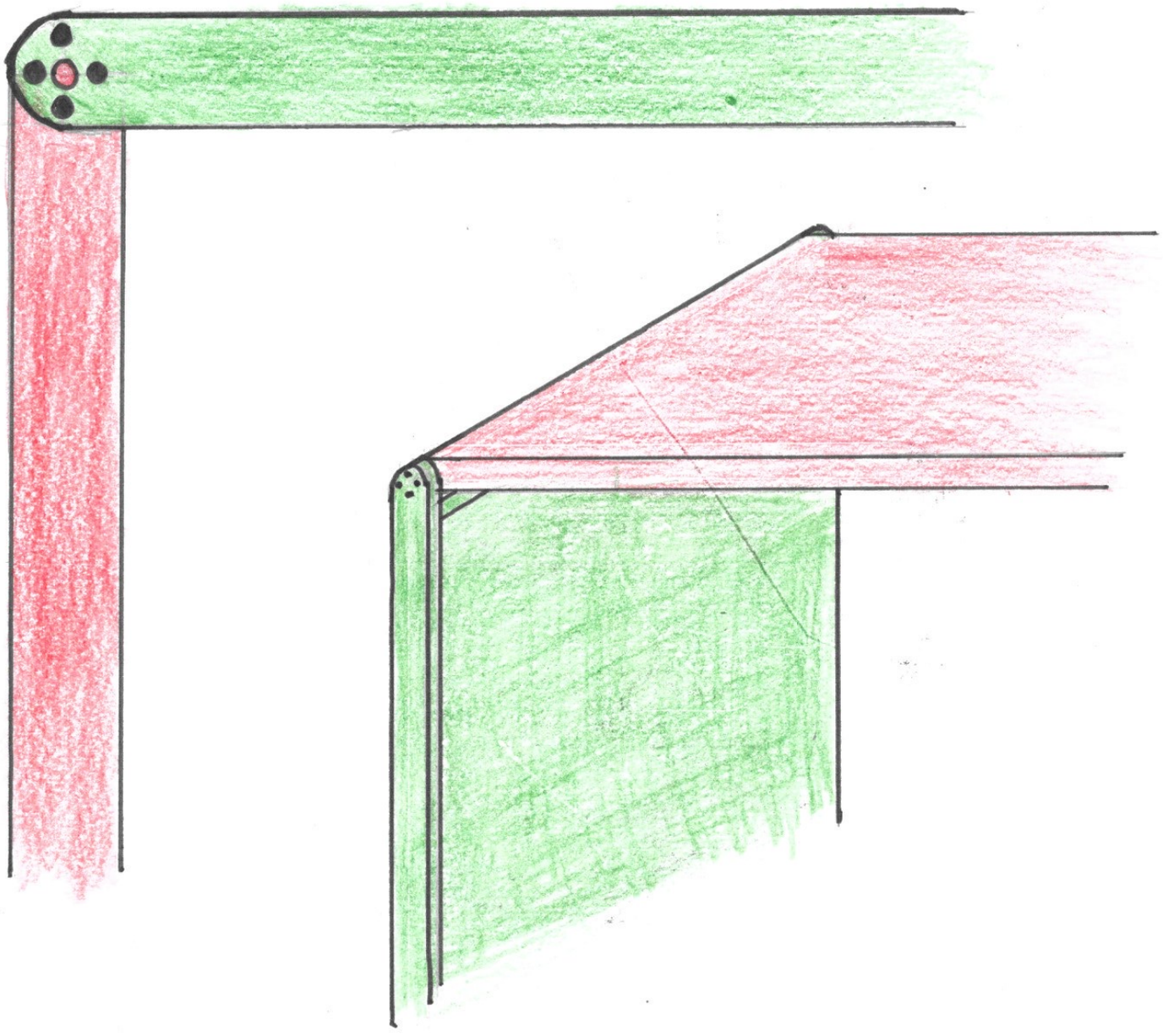
Then, I thought about a way to make the boards move together at the same time, so to work that out, I drew the middle sketch to get a rough idea of how large the distance between the pecies would have to be so that I knew the dimesions of it. From this, I knew that I'd need an even number of gears, of about 4mm thick each.

This developed into an idea to use a pulley to connect the two boards in order to simplify the pecies.

All these ideas are draw in more details in the next few slides to show them better.

Designing - Continuing The Mechanism 2:

① Using Magnets at the joint.

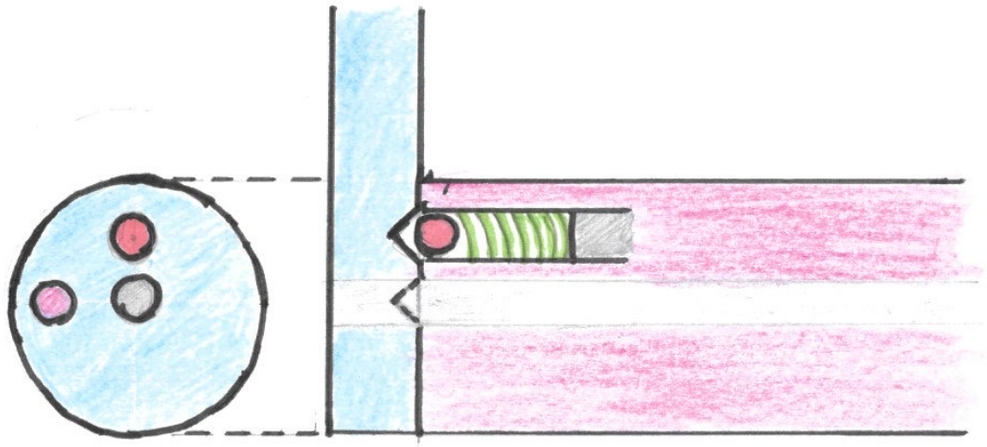


I could use 4 magnets to block it at 90° angles.

The only worry with this method is that as they will be very small, they may not be strong enough to make the board rigid.

Designing - Continuing The Mechanism 3:

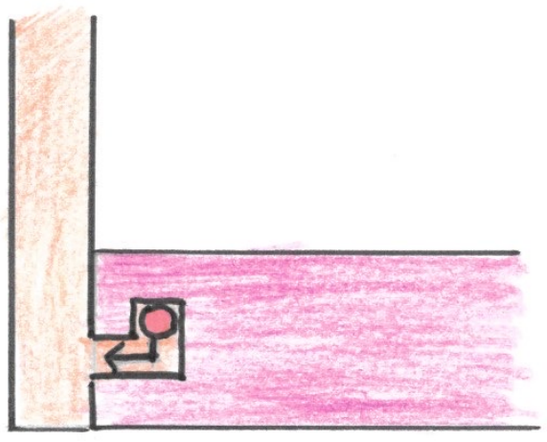
②

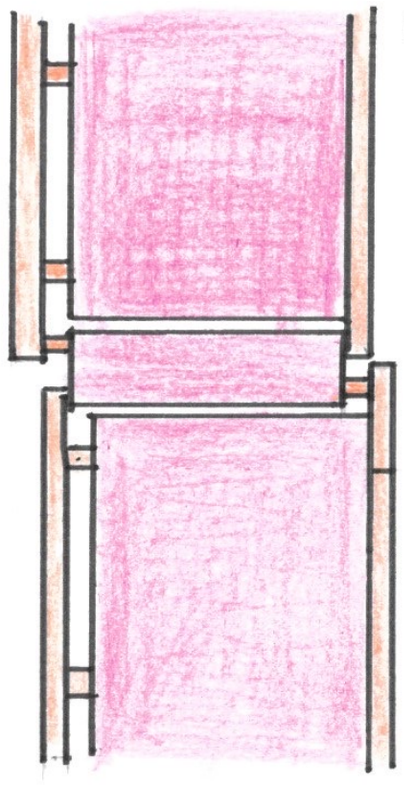


Inspired by the balliscissors, this design uses a spring to push a ball into an indent to lock it at the 90° and 0° positions (closed ad opened).

As the mechanism would be very small, it might be extremely hard to make.

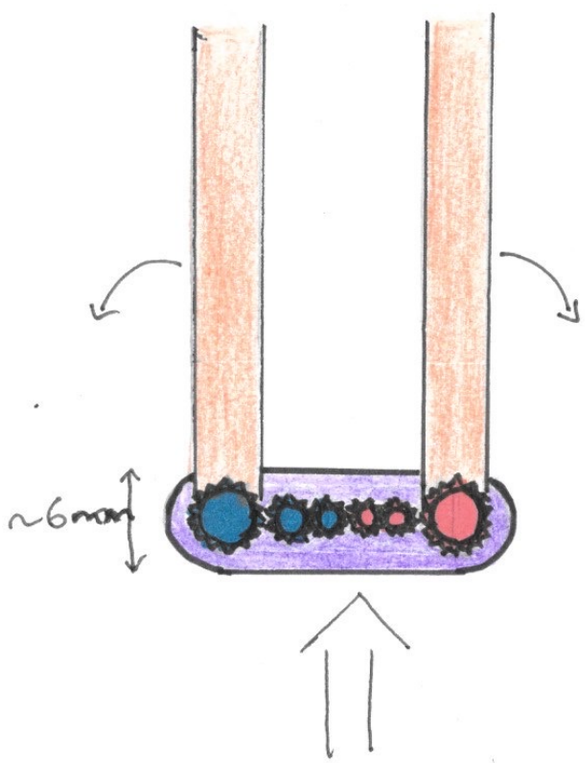
③



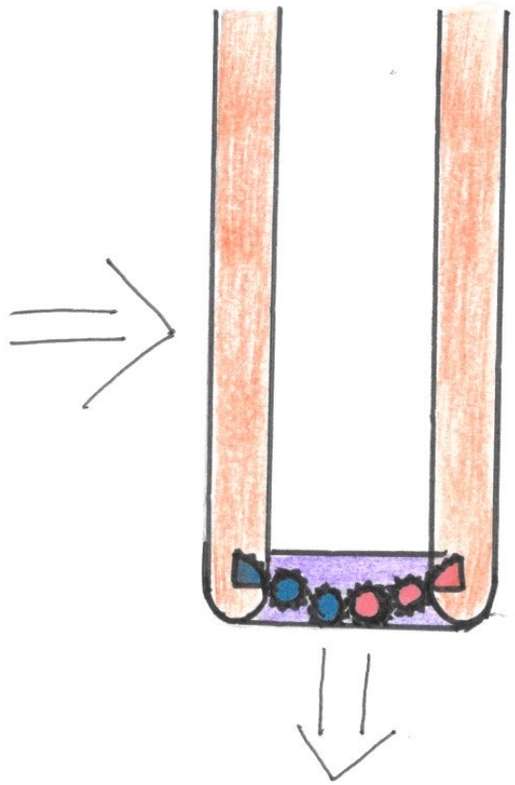


This mechanism uses a twistable lock which you turn and pull out in order to change from a lock position to a closed position.

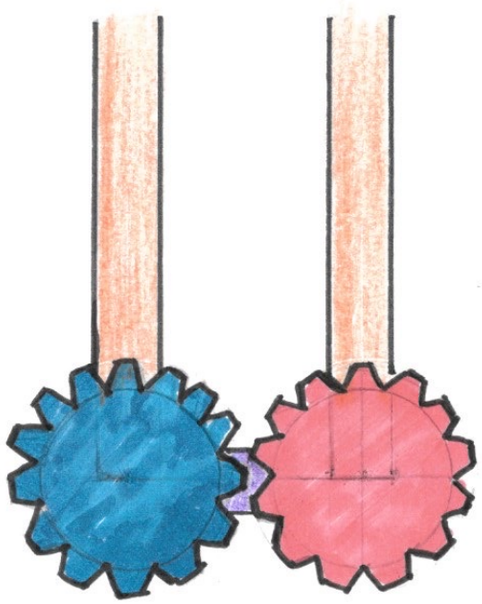
Designing - Continuing The Mechanism 4:



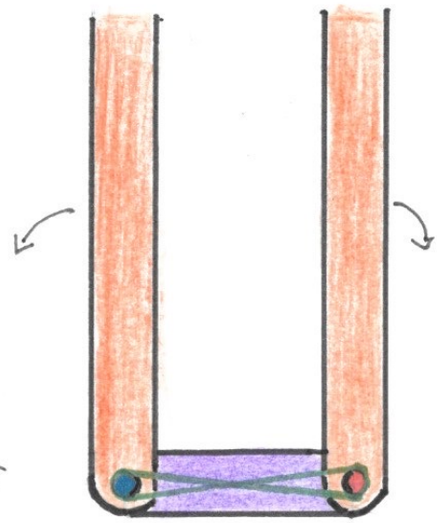
To make the mechanism more comfortable to use, I decided to put gears inbetween the two boards so that they move together. However, the gears would have to be very small at less than 6mms. so this may be hard to make.



To make it simpler, I would only require a quarter sized gear at the joints which means that I could probably make it thinner. However, there would still be a need for a very small mechanism. which is easier to break.

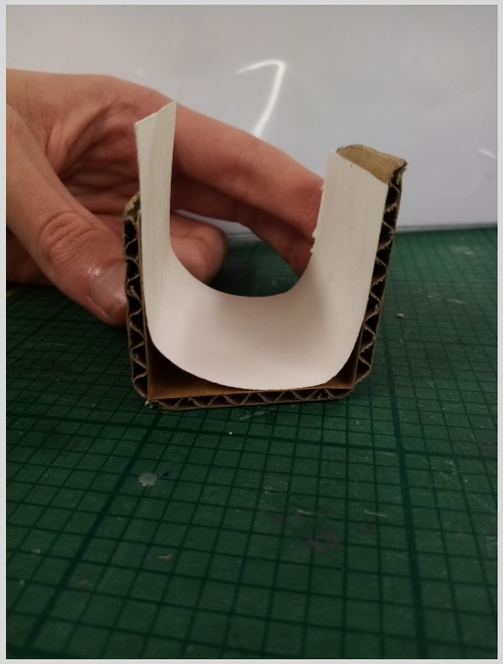


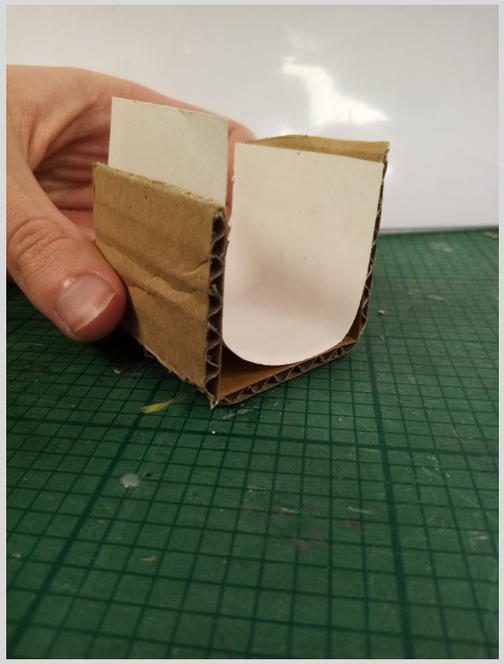
The design began with finding a way to make the two boards move together at the same time. The easiest way to do this was to use an even number of gears to interconnect them.



To make the overall system easier, I thought to use a pulley instead of gears. The pulley would be inverted to make sure that the two pieces go in opposite directions.

Modelling - Looking At The Mechanism:



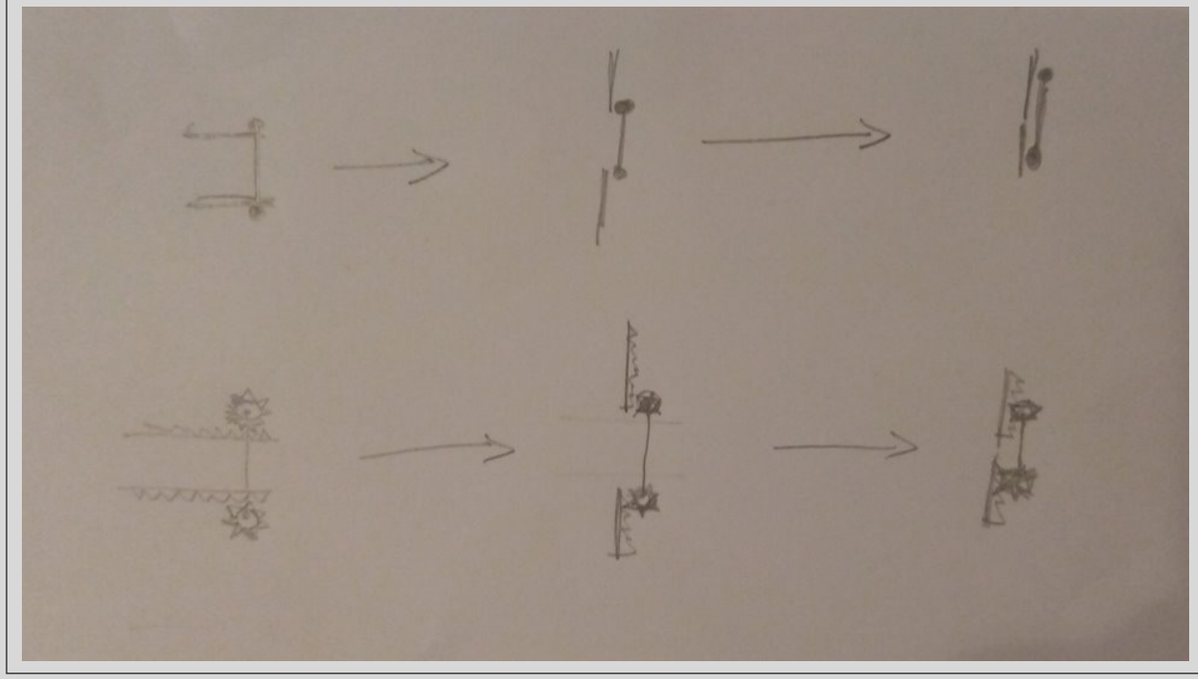


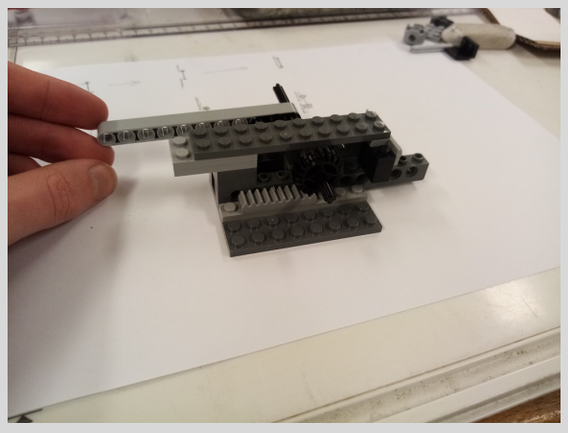
I needed to test my theory as to wether, when the board folded, it would cause the drawing surface to bend out of shape. However, what happened was that it rose above the other piece resulting in me needing to design a method which would prevent it from that happening.

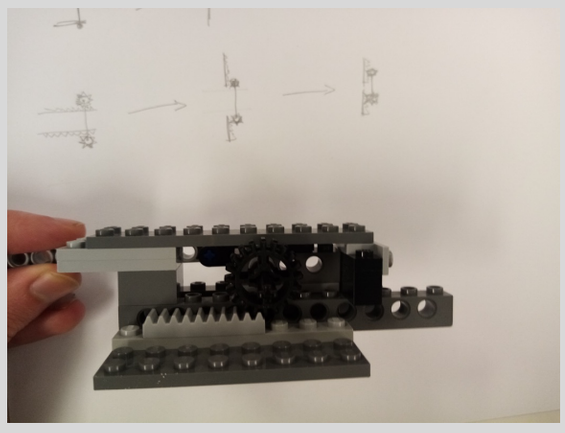
The first thing I thought of was the "Own Mechanism" from the "Intial Sketches" page which allowed me to change the distance between the turning base and the two boards on each side.

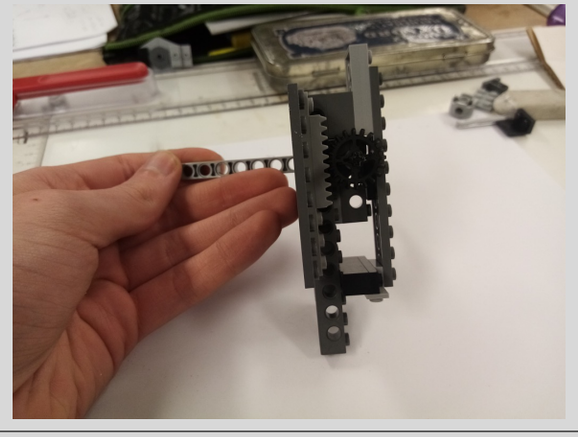
The other method I could use is to allow the drawing surface to slide forwards and backwards, thus changing its distance and so allowing it to fold normally. However, the only worry with this is that it will greatly increase its dimensions.

The first method I tried was the "Own Mechanism". I did a quick sketch of first, the three steps which it would need to go through, and then how it could work with gears [as seen underneath]. This design was supposed to move in 2 steps. First, the boards open so that they are in line with each other, then they slide along until they hit the turning board in the middle creating a flat drawing surface. To test this, I made a model which can be seen on the right:



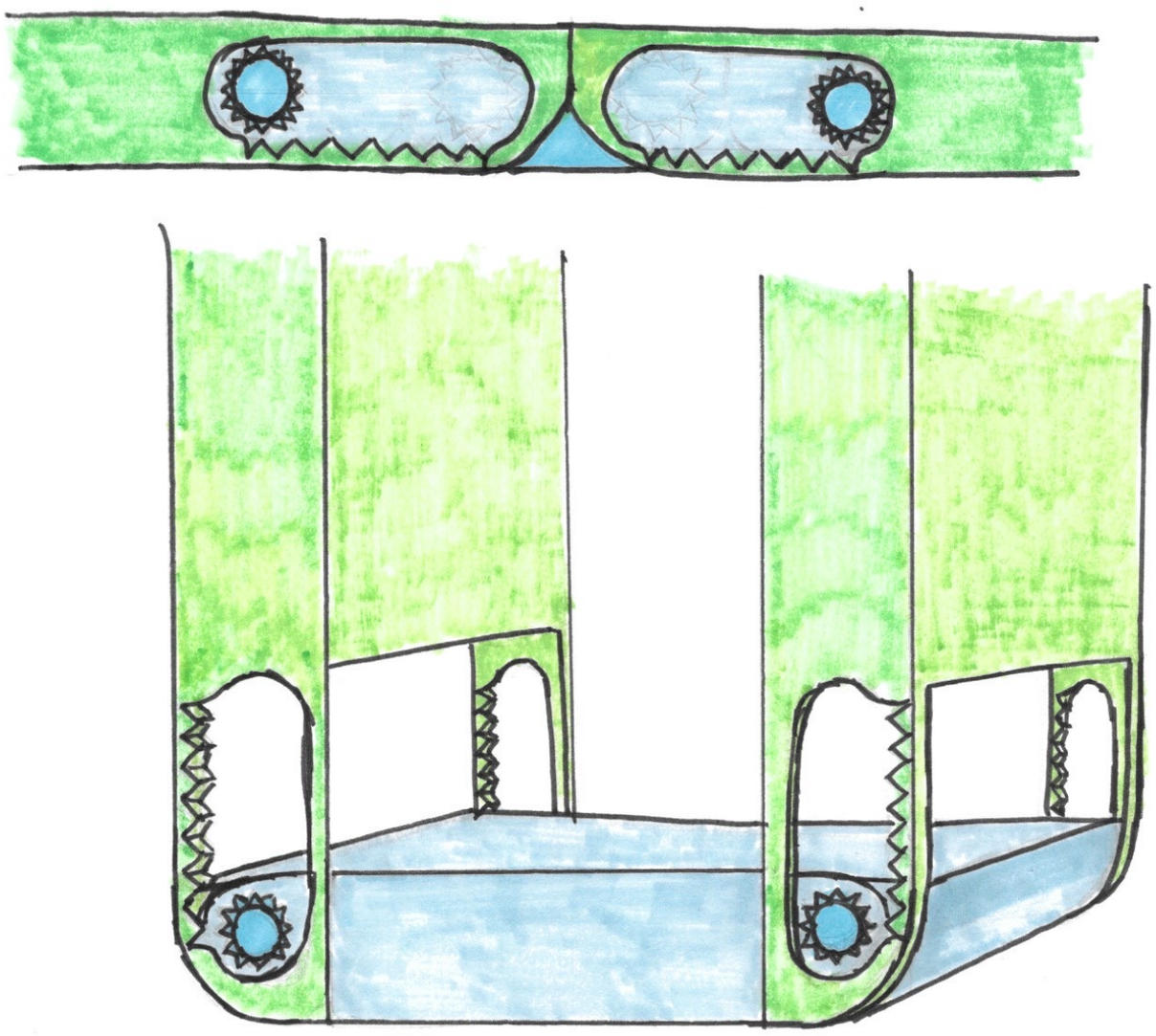




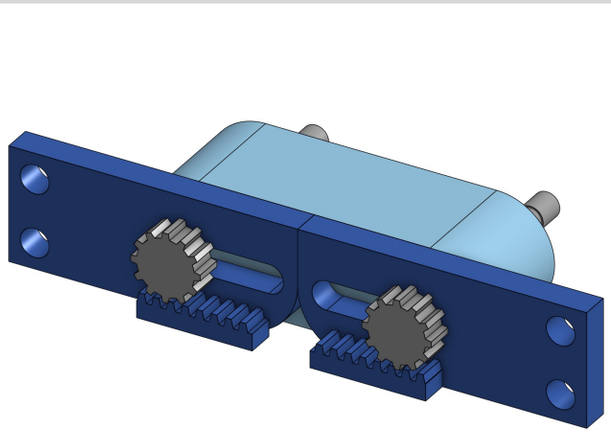


This model showed me how as the bearing moved the piece came closer together, so the distance between when it was perpendicular and horizontal was different which would allow my product to work. Furthermore, it made me realise that there would be no need to block and unblock the pivot meaning that you could skip the middle step and go straight from closed to open in a smoother motion.

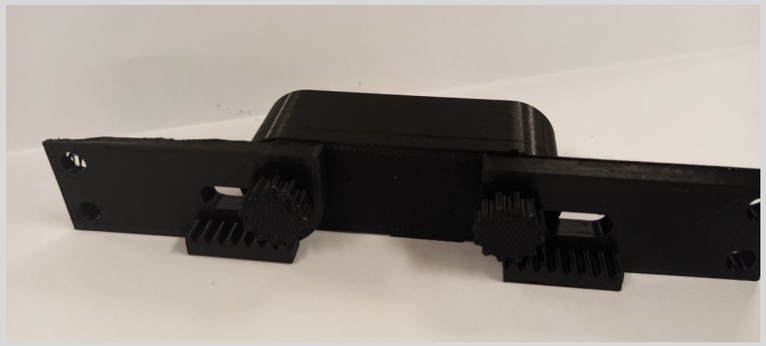
Designing And Modelling – Joint Mechanism:

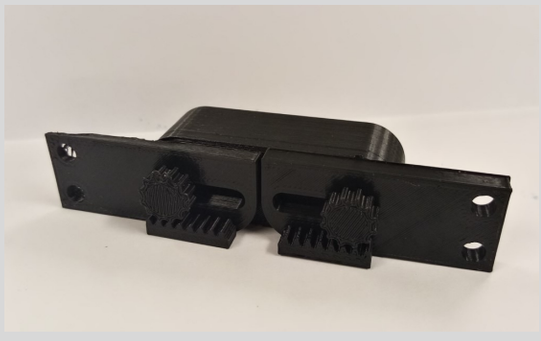


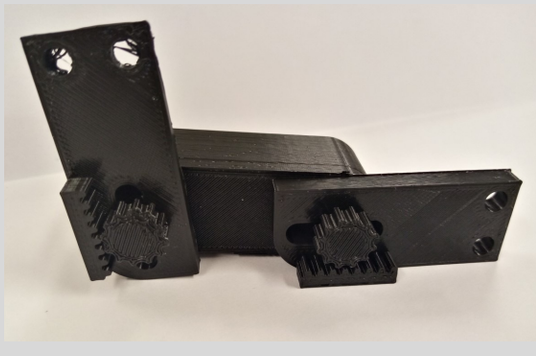
This design would use a gear rack attached to a gear inside the turning base.
This allows me to increase the distance between the material and the turning base in order to stop it from crumbling.



To test whether the mechanism would work or not, I modelled a simple CAD model of it on OnShape and exported it as an STL so that it could be 3D printed on the school's 3D printer.

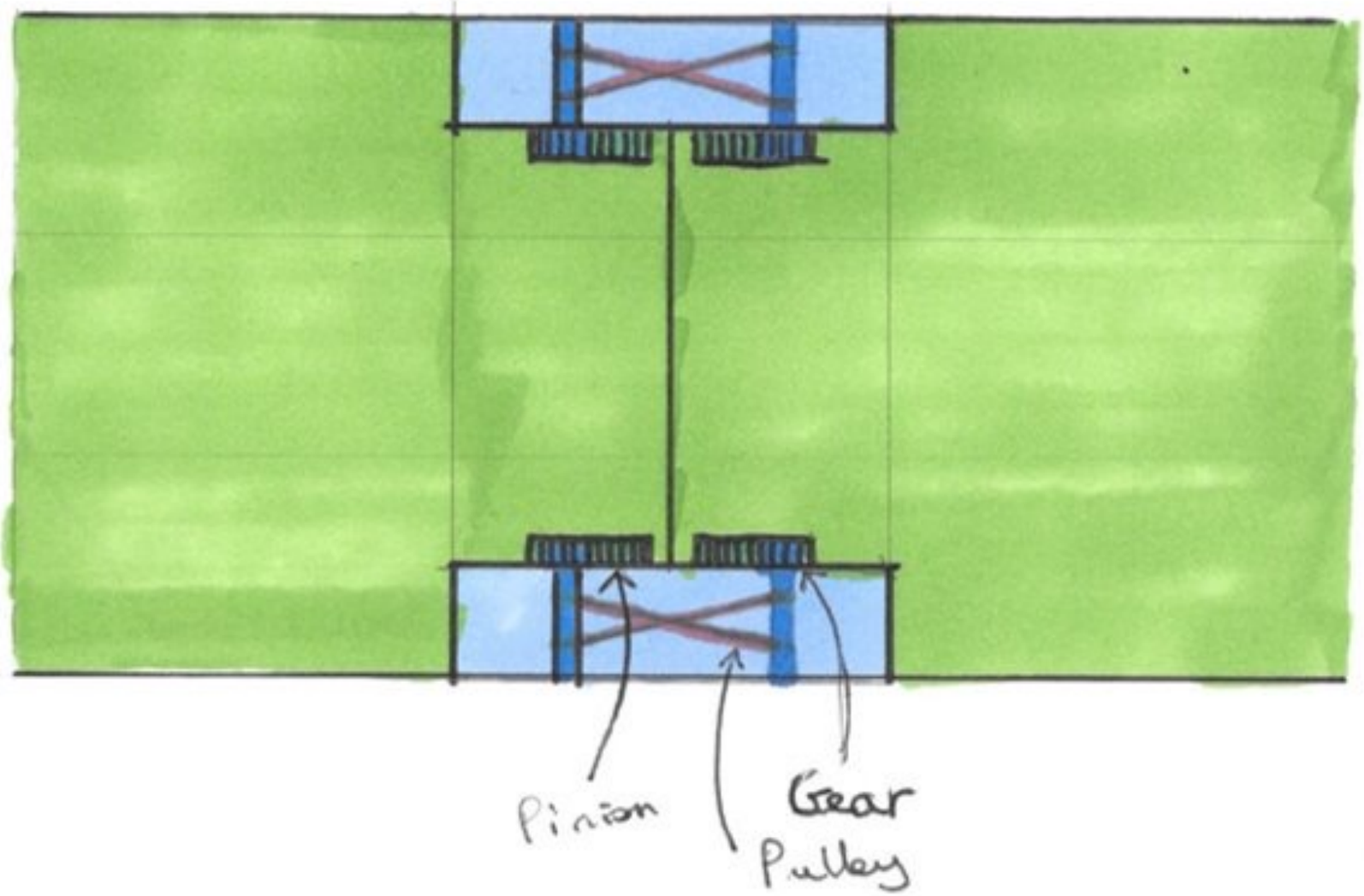






Looking at the model, 3 problems arose. Firstly, the 2 pieces could be pulled away from each other without anything restricting that movement or stopping that (as seen in the top image). Secondly, the two pieces would not always align correctly and flush together (as seen in the second image) but that was probably because of a misprint. Lastly, sometimes one of the two pieces would turn without moving away (as seen in the third image).

Designing – Joint Mechanism Variation:

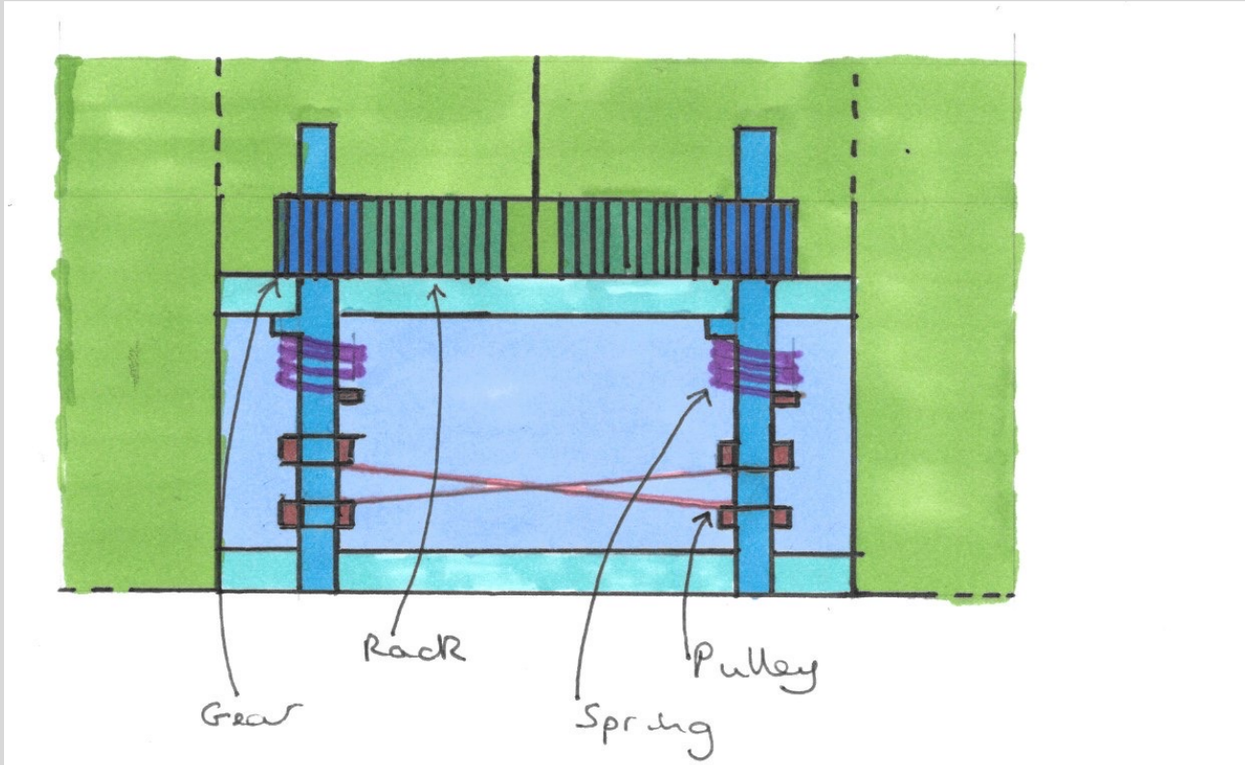


Exaggerated design with the Gear and pinion inside

A variation of the Joint mechanism from the previous page with the aim to hopefully simplify the design by having the two middle pieces slot together.

Designing – Snap Open Mechanism:

I wanted to make my board snap open by itself using springs so that the user could have an easier time of opening it. The design below shows a way in which I could do that using an improvement of the previous design:

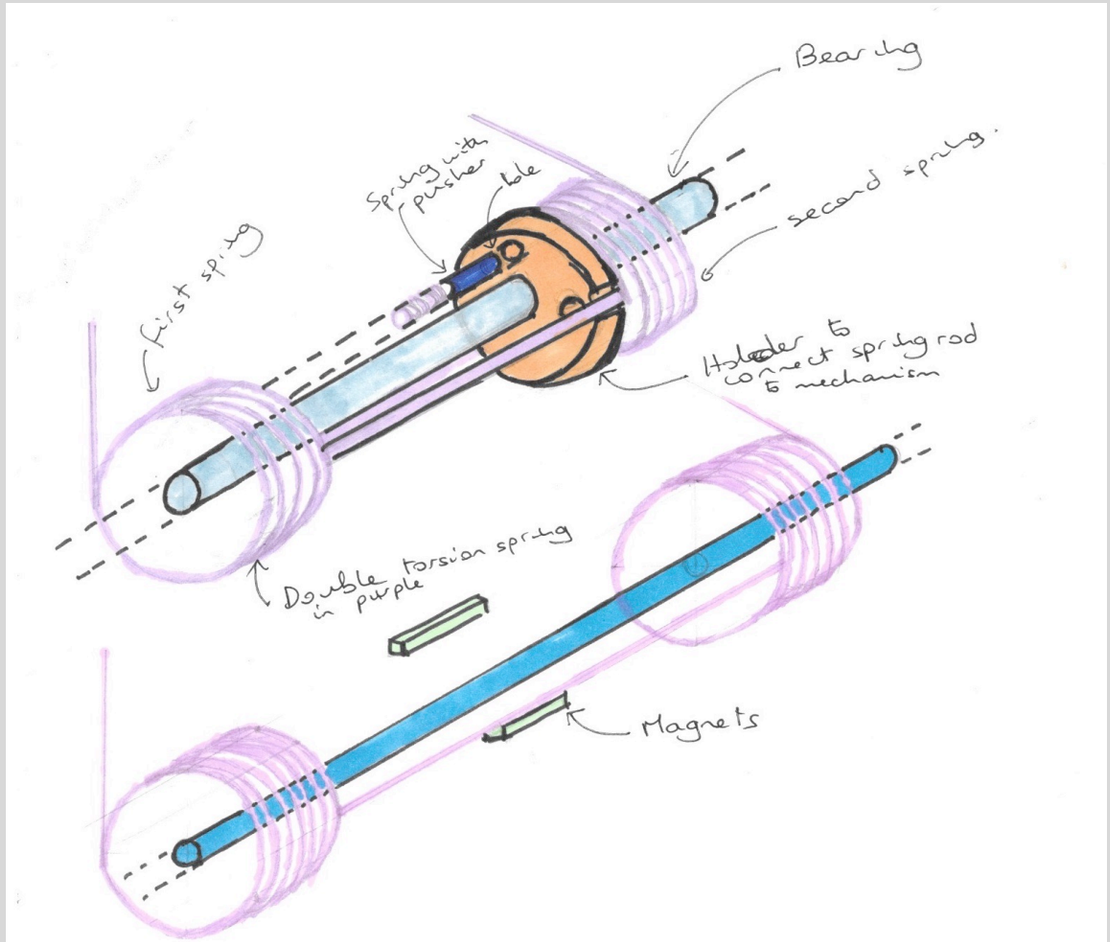


I wanted to include a way for my product to snap open using springs.
This would use a small but strong spring which turns the gear thus snapping open the board.

However, It was suggested to me that the board should contain a snap open and snap closed mechanism in order to make it even easier to use.

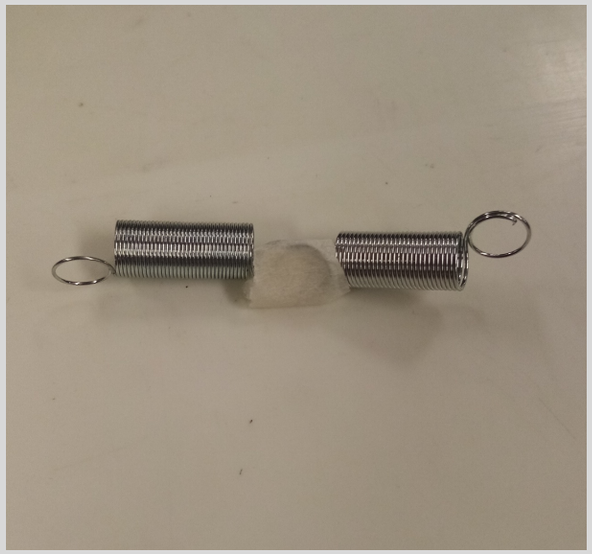
I started trying to find different mechanisms which could do this or different products which used this mechanism, but I was unable to find any. However, I did find out about double torsion springs which could be used in order to create this as they can create force in 2 directions by locking the middle point.

As I was thinking of using double torsion spring, the idea behind them was to lock the middle rod so that only one would push. And in doing so, it would turn one way, however, when it was unlocked, it would turn the other way.
One idea I had to do this was to use a mechanical lock, while the other was to use magnets in some way.



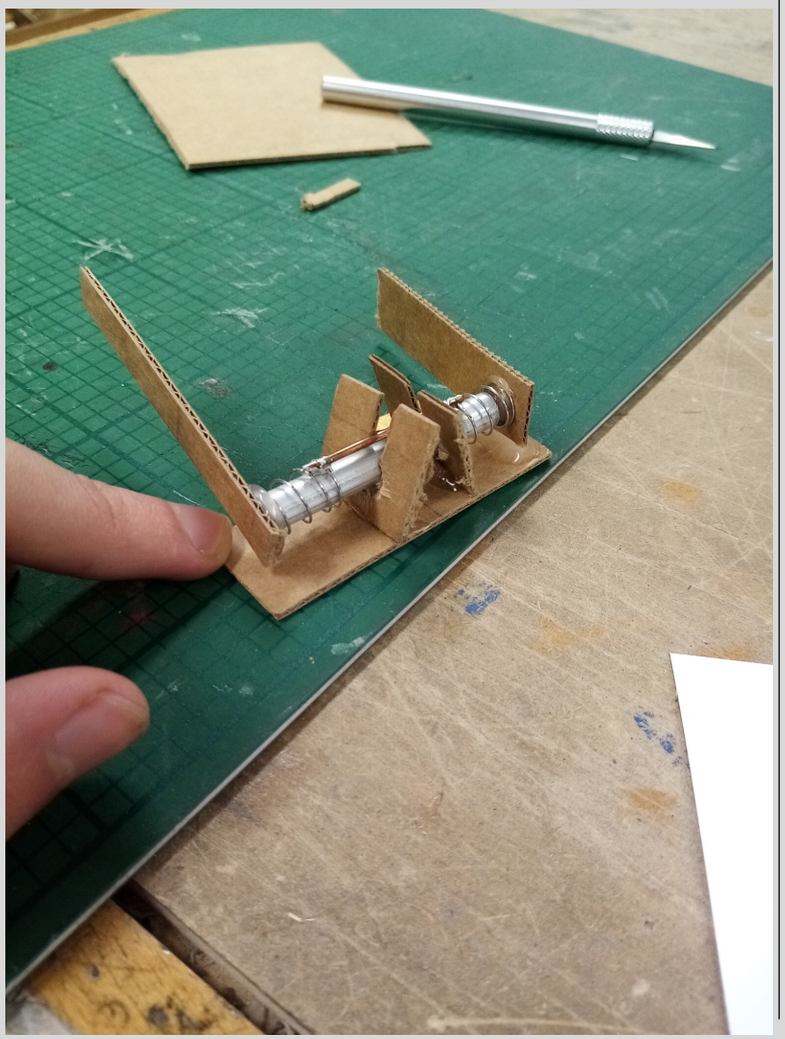
The first design uses something similar to the one that was derived from the Ballscissors.
It has a pusher which locks the first spring on one side allowing it the other weaker one to push instead.
The second design uses magnets instead.

Modelling – Snap Open Mechanism:

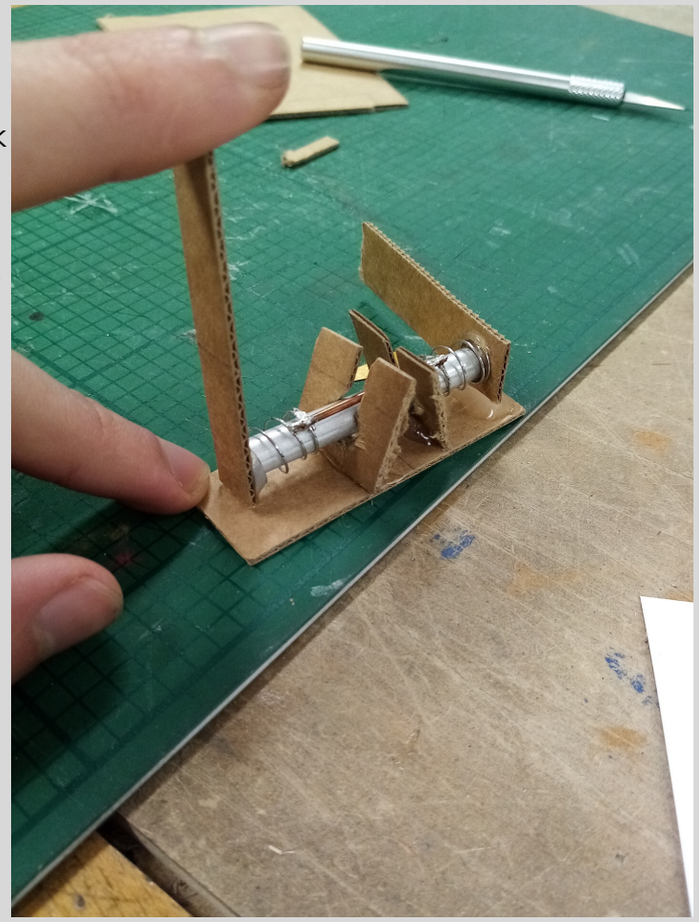


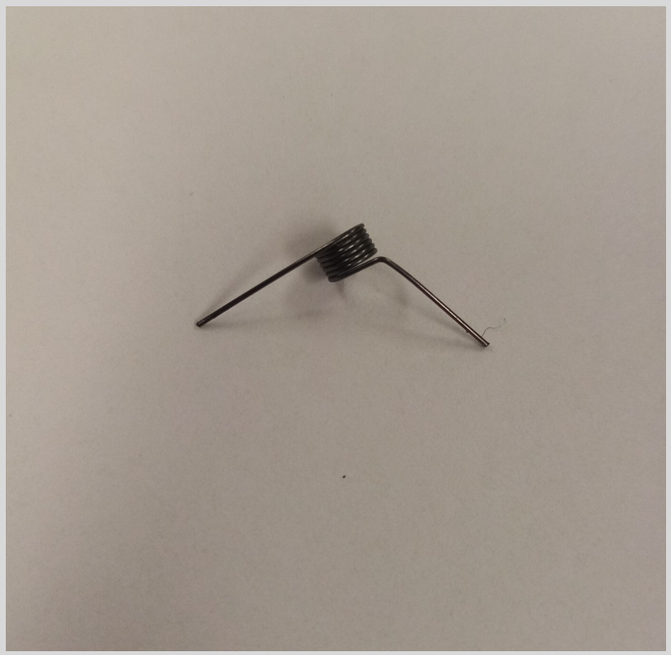
To begin modelling it, I got two simple springs and connected them together using some tape.
I did this to understand how the movement would work together and how the springs would interact if they were twisting against each other and if they would cancel out, which they did. This means that I would need to lock one of them like in my design on the previous page

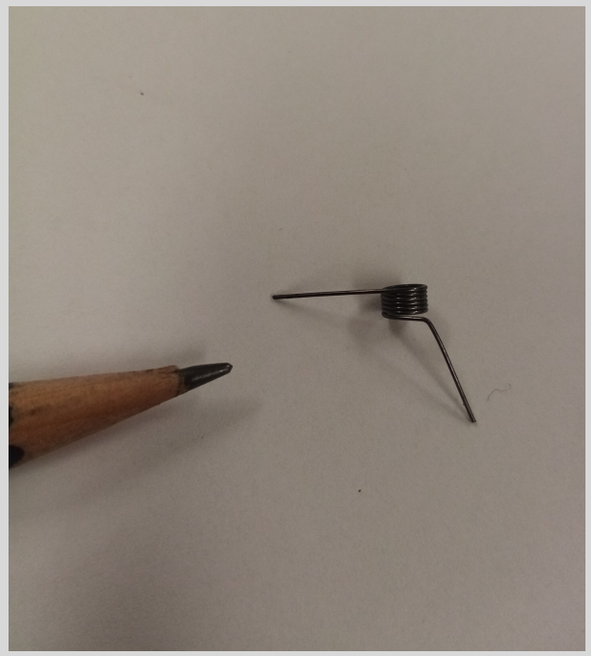
To begin the design of the mechanism, I started by soldering two springs together through a brass wire. This allowed me to connect my two springs together in a way which meant that they acted against each other. Then, I got some card and attached one to the pipe and spring and the other to only a spring, which then connected to the two clamps in the middle in order to stop it from twisting.



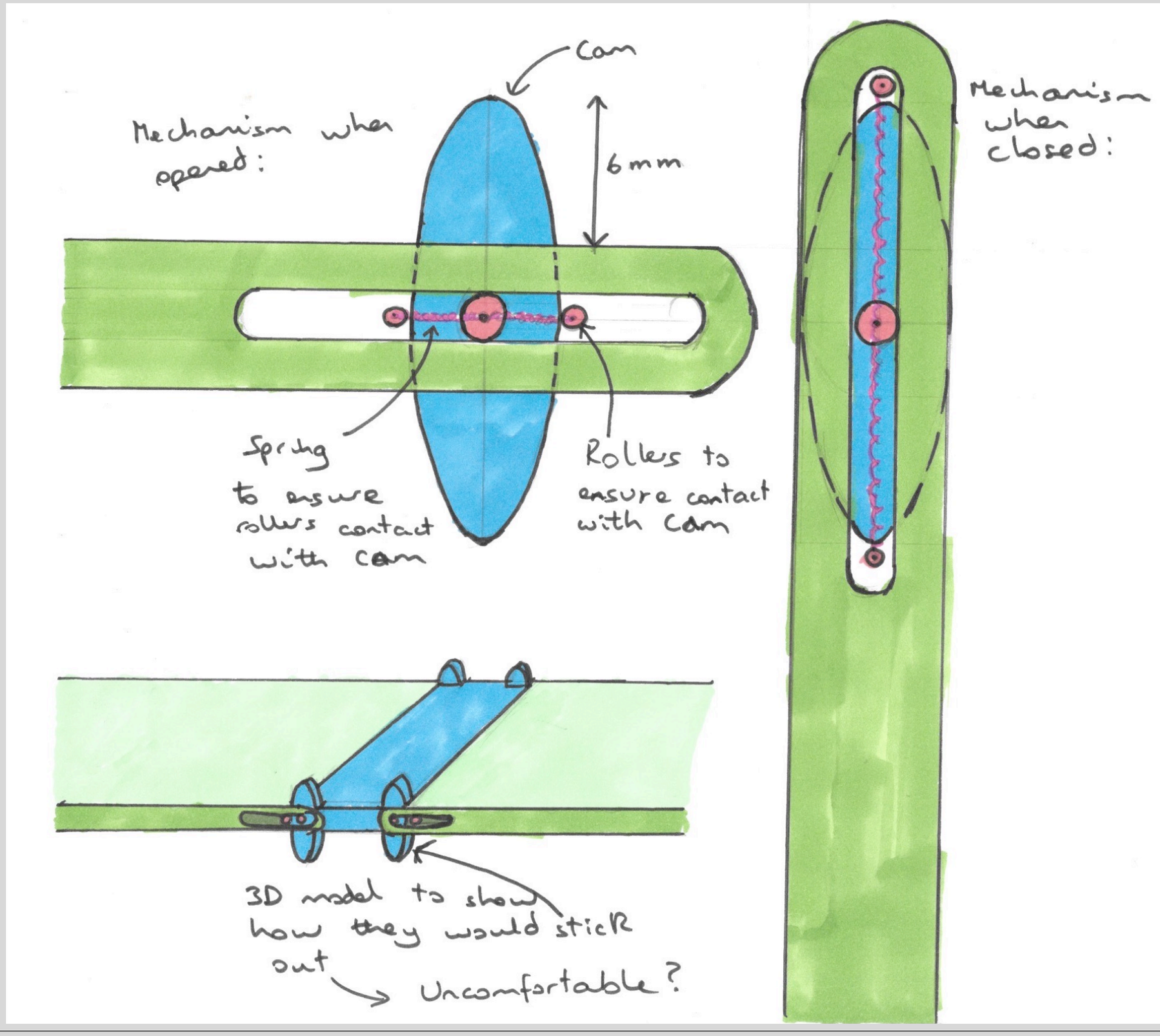
Making this model, I realised three things. Firstly, although the mechanism would work, it would require careful calibration of the springs to make it work so that they wouldn't counteract each other too much. Secondly, the locking mechanism would need to change, as it would not be able to interact properly and so lock properly in the current design. However, the biggest thing I realised was the complexity needed to build this:
To keep the thickness down, I would need this mechanism to be very small. Because of this, I was planning on using the springs which can be seen below. However, their diameter is about 4mm, and I do not have the tools to be able to make a mechanism that small with parts as accurate as I'd need there, so I would not be able to realistically create this. Furthermore, the springs would probably end up being too week given their size to reliably open and close the board.







Designing – Mechanism Using A Cam 1:

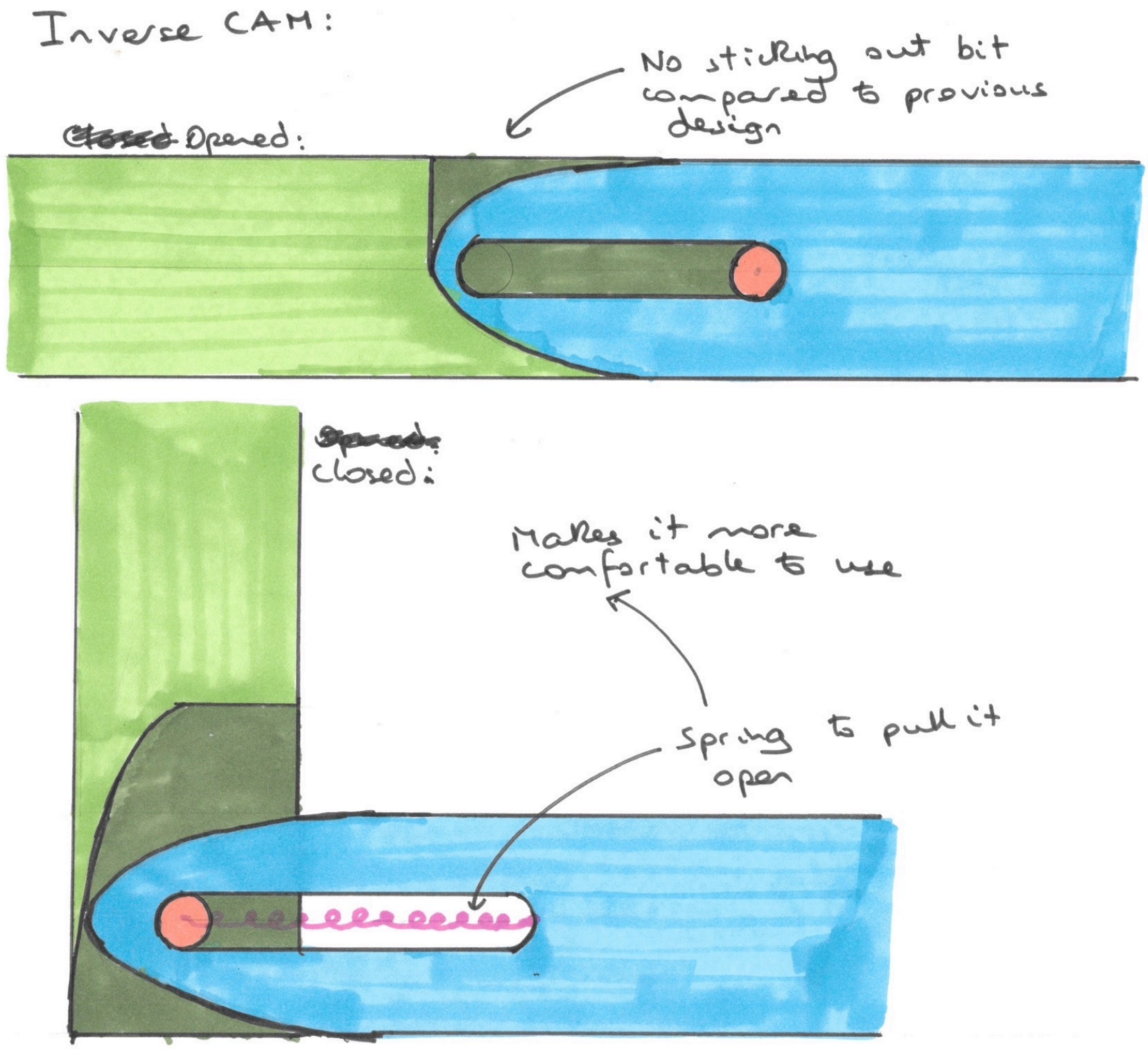


The idea of using a Cam came as a result of attempting to simplify the design of the mechanism.

At the place of using a rack and pinion, I thought about using a cam so that I would not have to create an extremely complicated mechanism like in the previous design which would be both extremely hard to create with the tools that I have access to at school and also be would be too easy to break for storage within a backpack as it needs to be stored within a backpack.

I thought to include springs in it to both keep the rollers attached to the Cam, and to facilitate their opening as they will try to move to the shortest position meaning that they will try and keep the board open.
Because of this, I will almost certainly not be using a snap open snap closed mechanism if this specific mechanism works as they cannot be used together, and I would face the same issues as with the gears where there is too high of a complexity. This is something I should avoid as greatly increasing the complexity is likely to result in a hinder to the user's comfortability.

Designing – Mechanism Using A Cam 2:



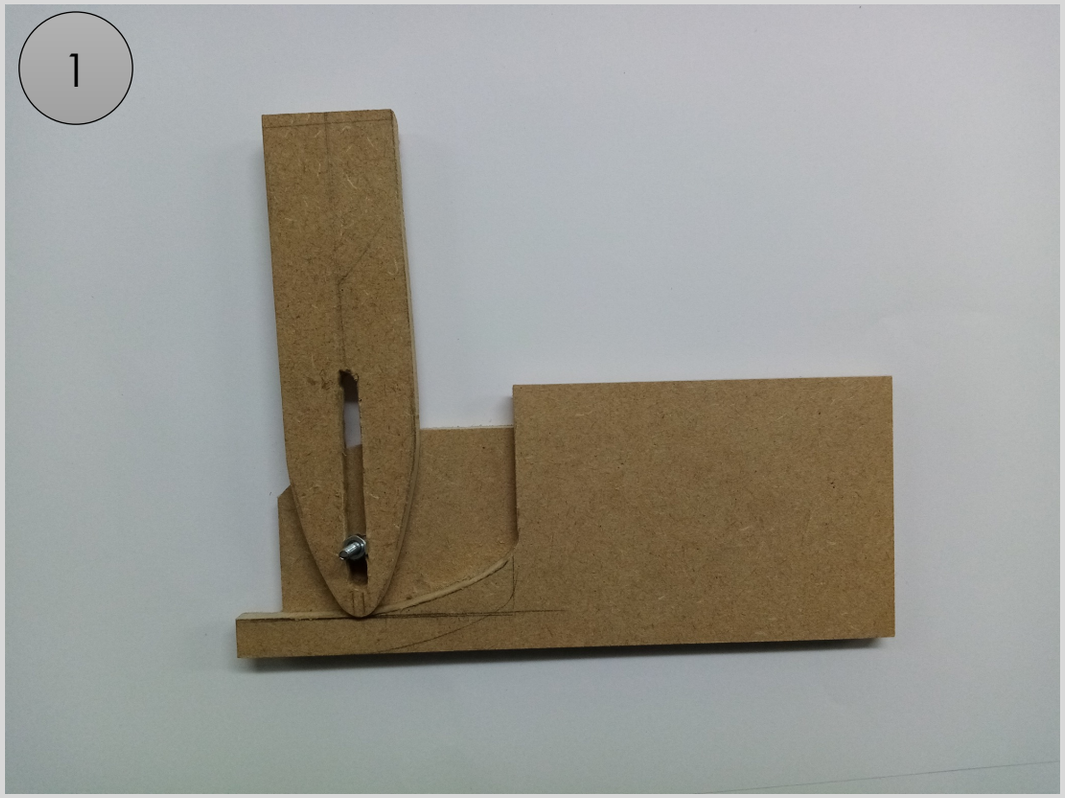
As the previous design had parts of the Cam sticking out by about 6mm both top and bottom which would hinder the design of my board.

Using an inversed cam, which has a negative of the profile, I could just use that as a guide for the main board so that it would lift up by 6mm, the same as the other design without its drawbacks.

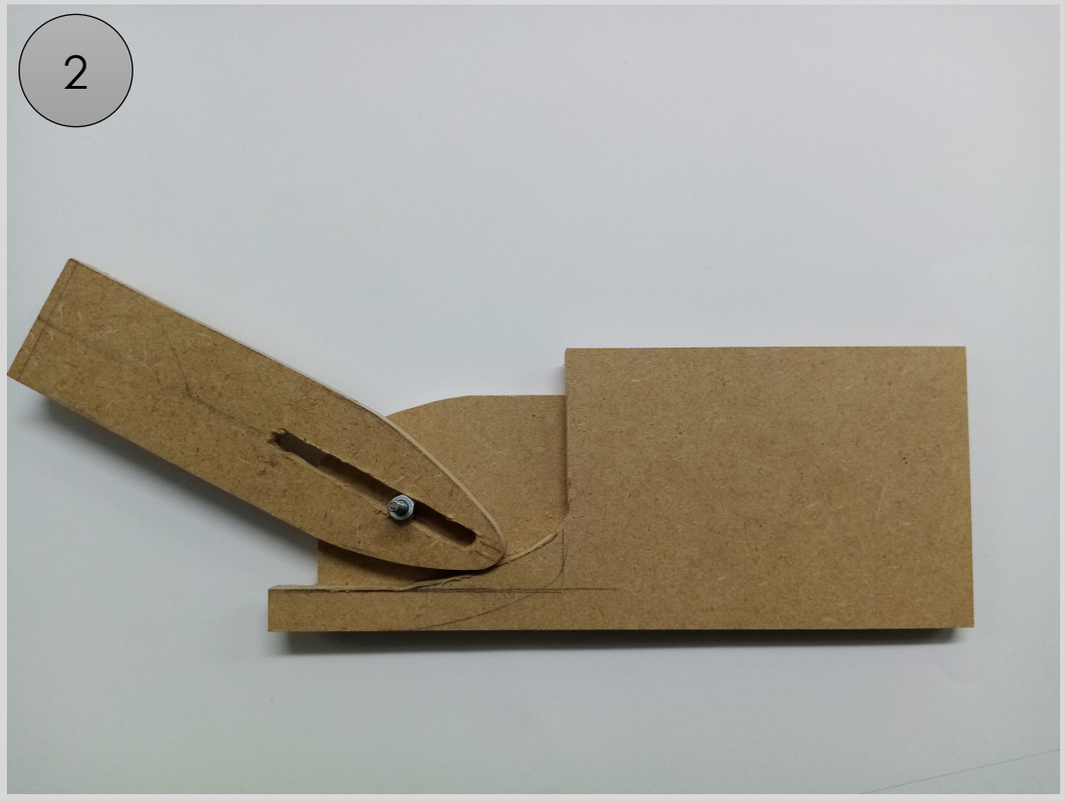
This would also have a spring which will pull attempt to pull the board open which will improve user experience.

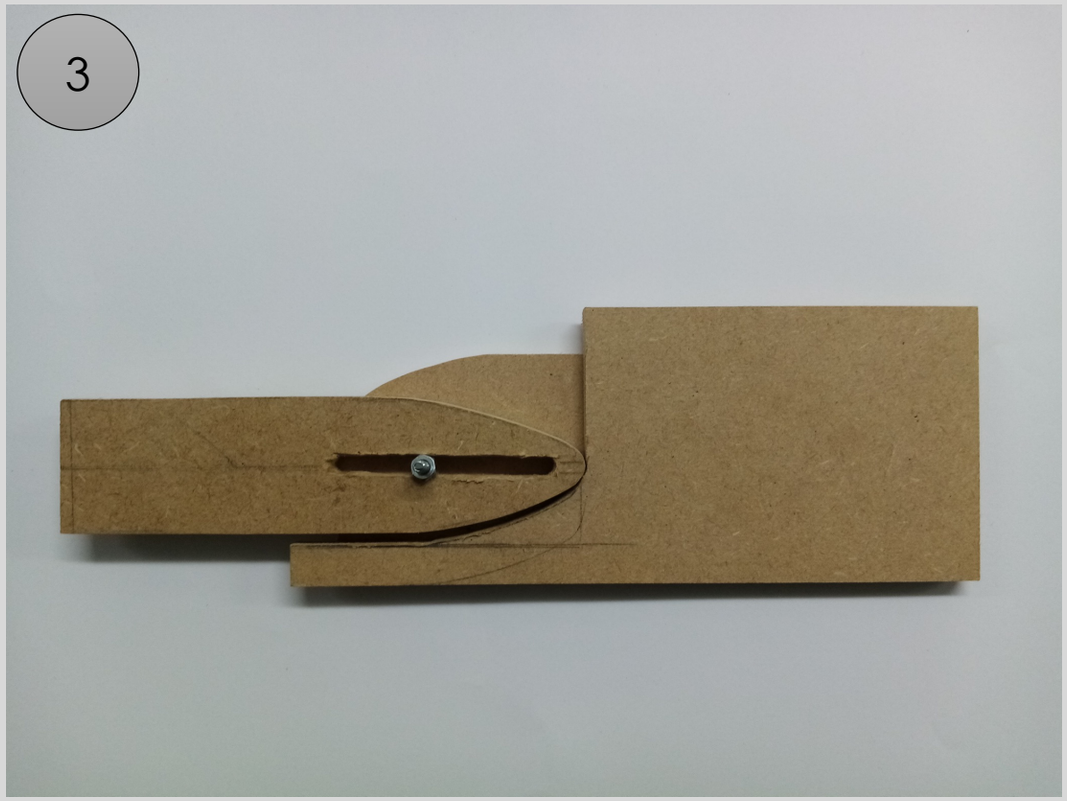
However, to check whether this would work or not, I will need to model it so that I can test its usability.

Modelling – Mechanism Using A Cam:

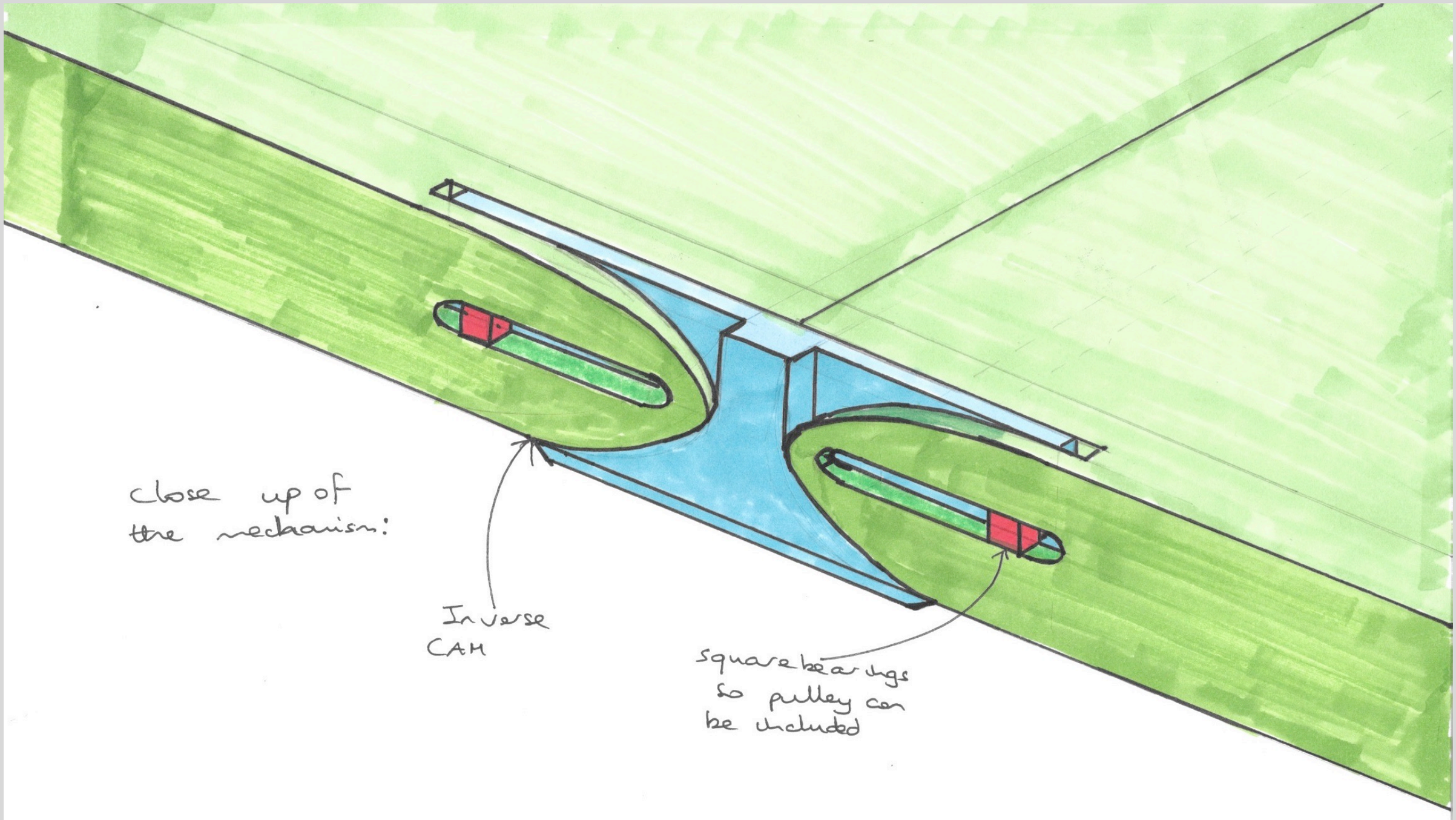


To make the model, I used MDF boards which I cut and sanded into the shape of the cam and inverse cams as the design describes.
As it shows from step 1 to 3, the method works really well and would not have any parts which stick out unlike the first cam design, however, some improvements need to be made:
Firstly, friction is a big problem with the current design, so I will need to either use materials which have a much lower coefficient of friction, or to use rollers in order to reduce friction. It is most likely that I will need to use the materials solution as fitting rollers in the design would be extremely hard given the very small dimensions of it.
The other issue with the design is something similar to what I faced in the rack and pinion: There is nothing stopping the piece from being pulled to the right in a way that would stretch and potentially break the bendable material. There are two ways in which this may be solved. I can either use a spring to hole the cam and push it towards the inverse cam, or I could add a track to my deign which would control where the board would go, however, this would need to be tested within my design to see whether it would be possible to be built given the small dimensions and whether it would break too easily, as with this small scale it is a big worry.
Lastly, to be able to put a pulley between the two bearings, I will need to change the top of the round bearing to a square shape so that it slides across the track but when it turns, it would also turn the bearing as well allowing me to put a pulley in-between them.

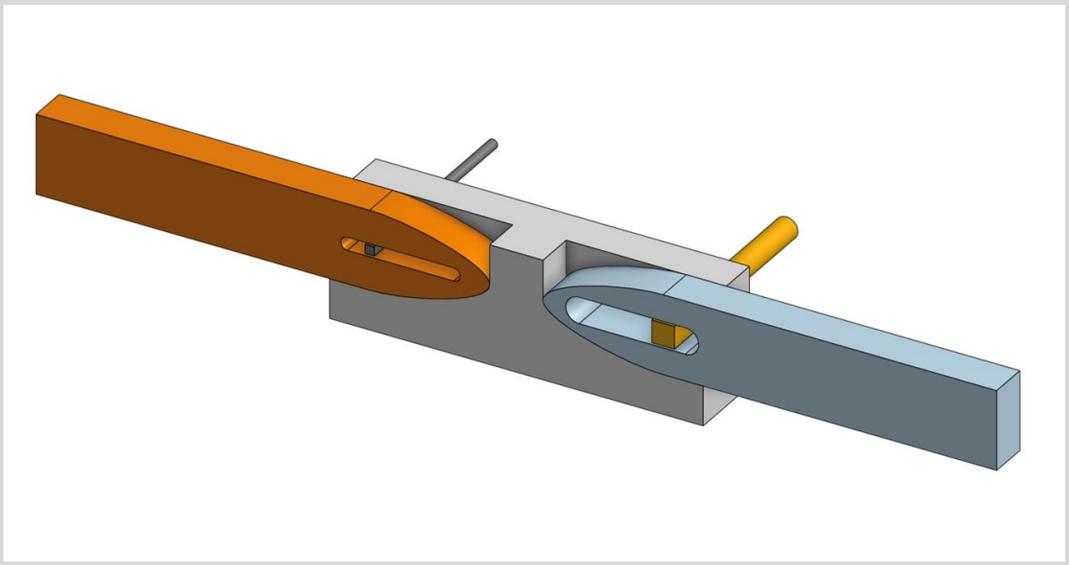




Designing – Finalised Design Of The Mechanism:



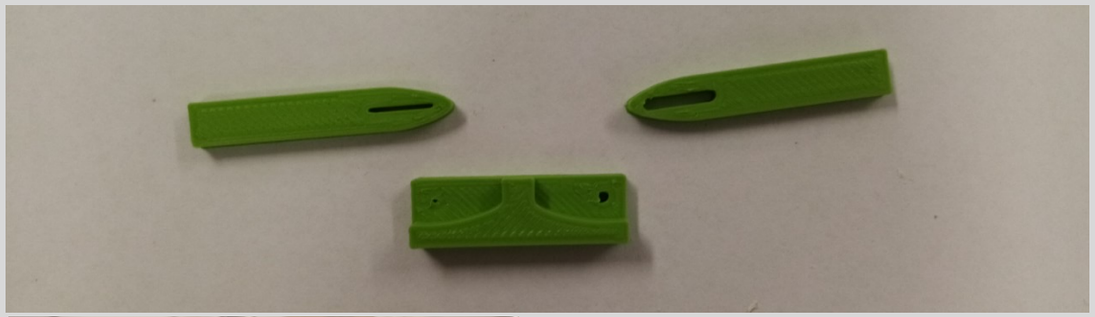
Modelling– Testing The Construction Of The Mechanism:

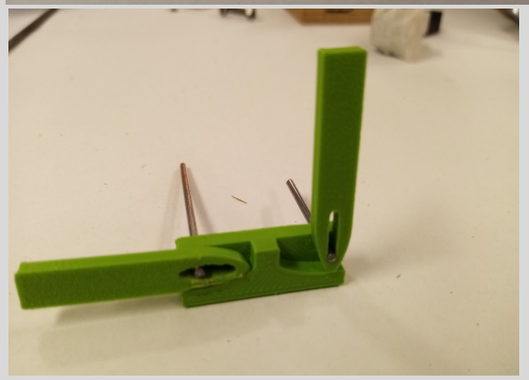


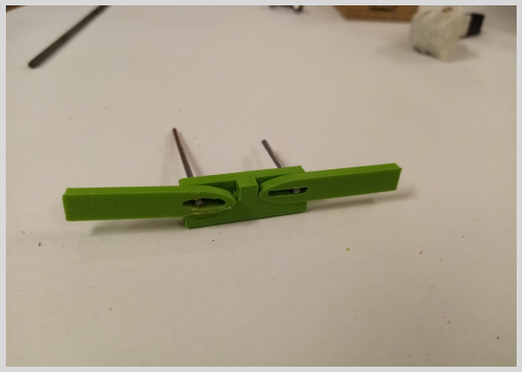
To model the new mechanism, I first made the model in OnShape so that it could be 3D printed.

I made the two sides of different widths for the track, one 1mm thick (on the left) and the other 2mm thick (on the right) in order to test which would be better to be used within my design.

I then sliced these parts in order to 3D print them so I could test them.





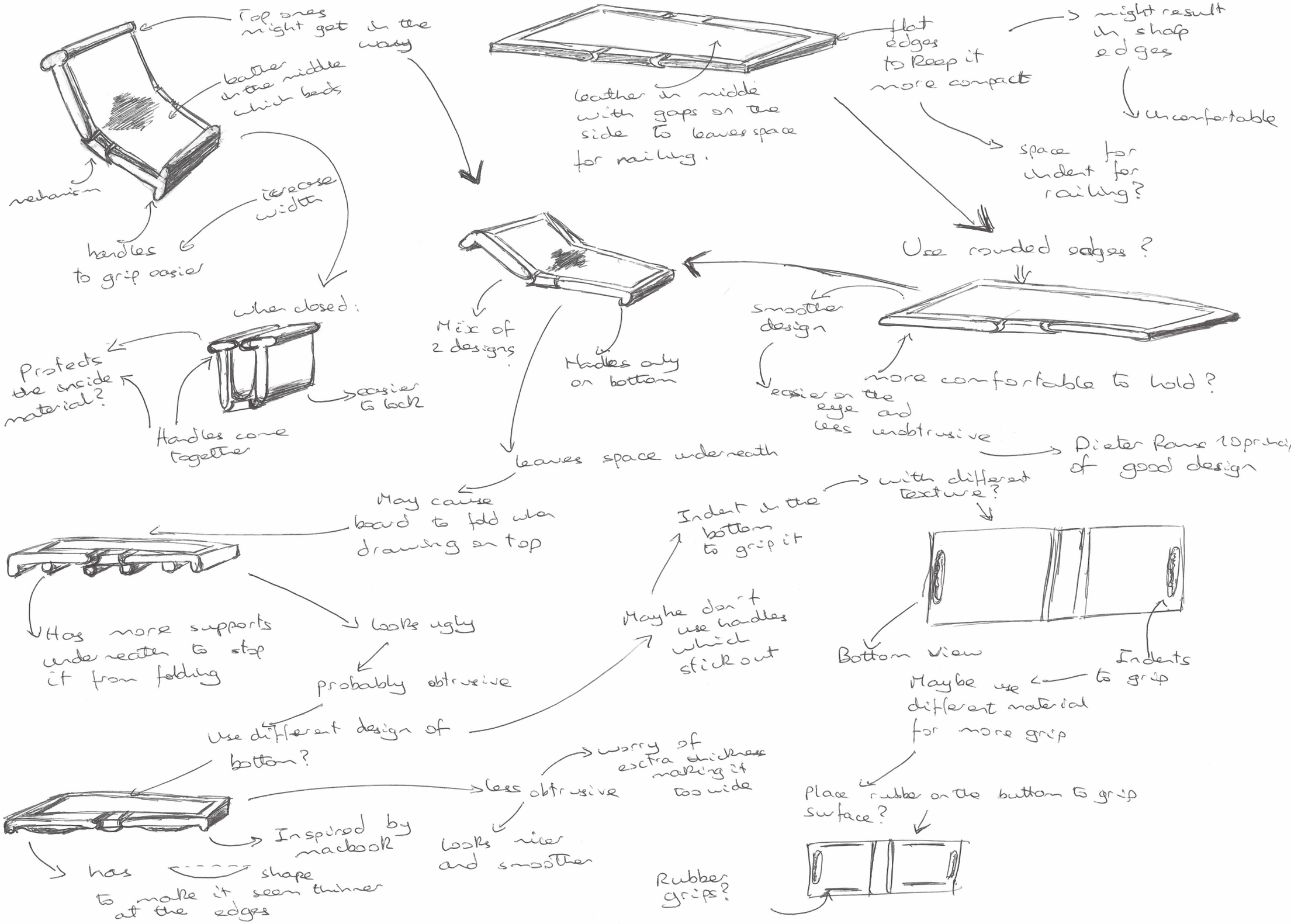


The 3D printed model worked much better then expected, excluding the small 3D printed rods which broke, the rest of it worked exceptionally well. The 3D printed parts did not have the same problems regarding friction that the fist MDF design had meaning that the parts moved very smoothly. Furthermore, as the mechanism worked this well at such a small scale, I do not think that it will be much of a problem at a larger scale. However, I will need to test whether the bottom plate which interconnects the two ends would clip in this design.

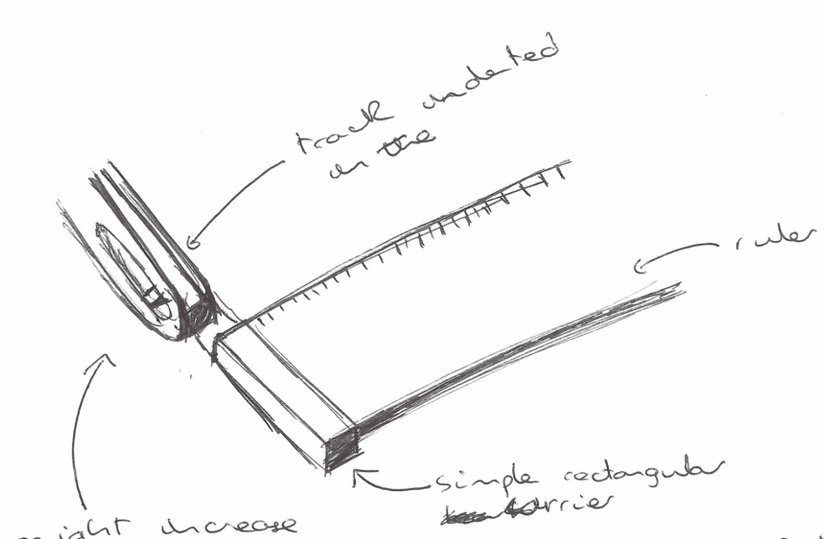
The next model I tried is using Puter and casting it using a wooden mold. The first image shows the first cast which didn't come out that well, while the second one was a bit better. However, some issues did arise with both. Firstly, I couldn't get the full curve to form as the Puter did not go into the gaps; this is a big issue as it means that the design can't fully work through casting.

The second cast did come out a bit better. However, it was still not good enough to ensure my design. Because of this, there are 3 options that I could use: First of, I could either increase its height so that there is more support underneath it for the Puter so that it creates the shape better. Secondly, I could 3D print the final design using a metal filament or something like wood PLA to ensure that it looks nice with the design. Or lastly, I could try and mill the shape out of some hardwood.
The main concern to decide what to use though is the durability, as over time, as it is used many times, it will wear down severely which means that it would have to be replaced. This means that I should ensure that the piece can be removed easily to be replaced in my final design.

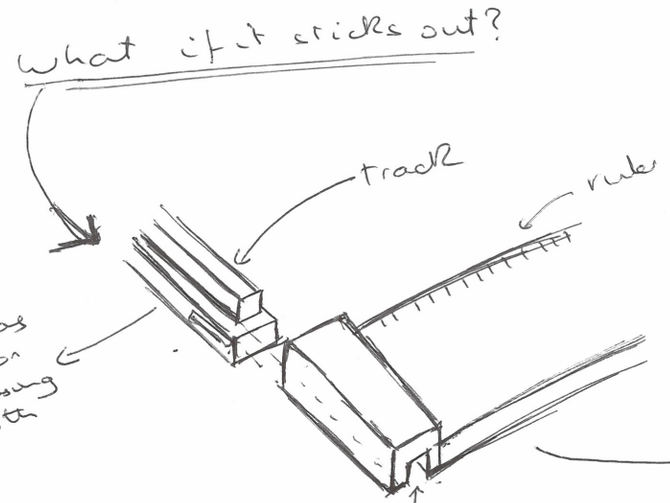
Designing – Starting The Overall Design:

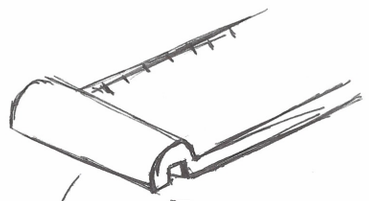


Designing – Rough Sketching Of The Holder For The Ruler:

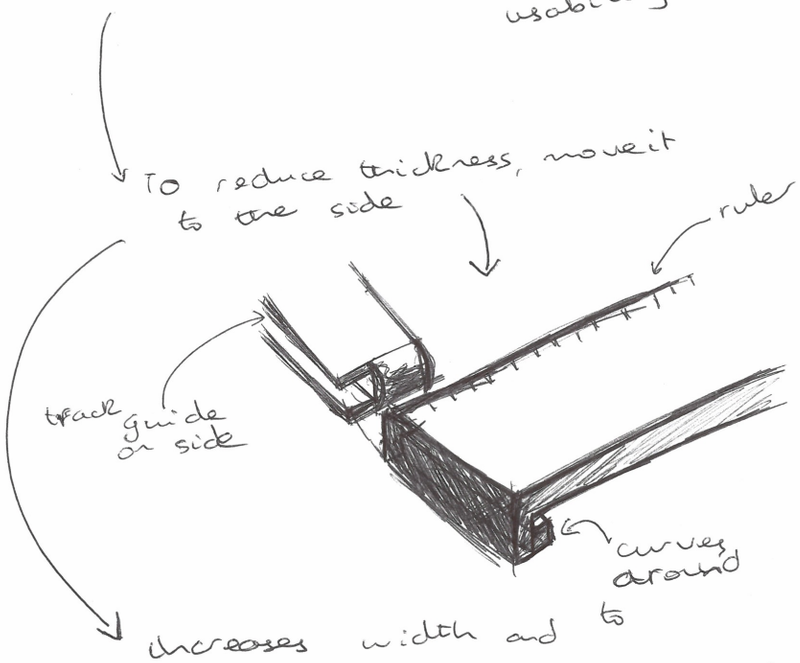


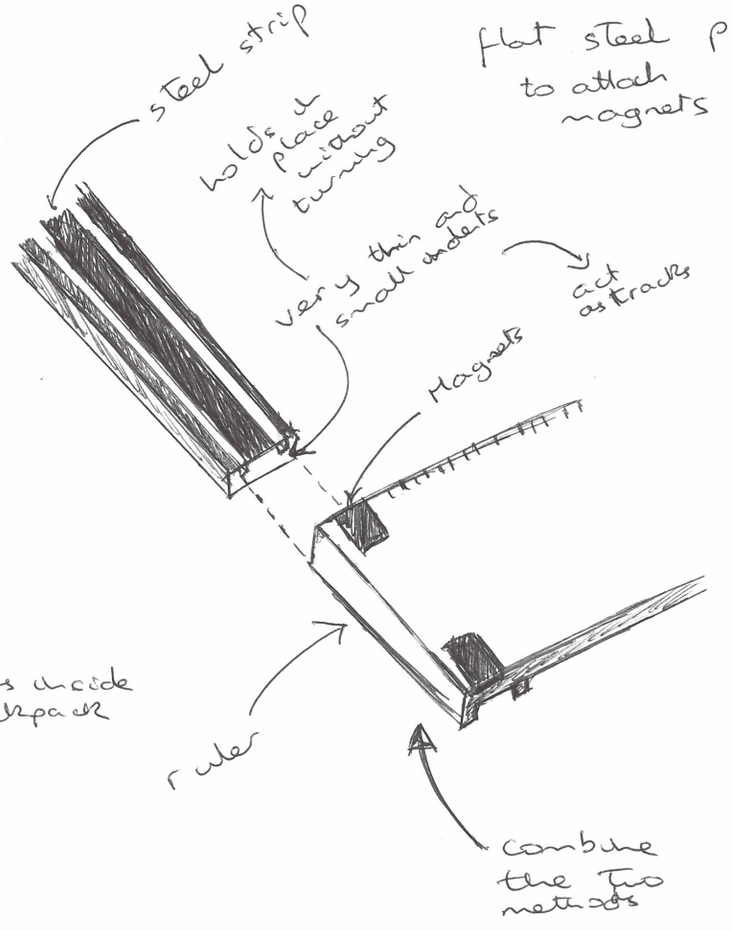
might increase thickness a lot as there needs to be space for mechanism → affects the comfort and usability? → Deter rooms unobtrusive desig

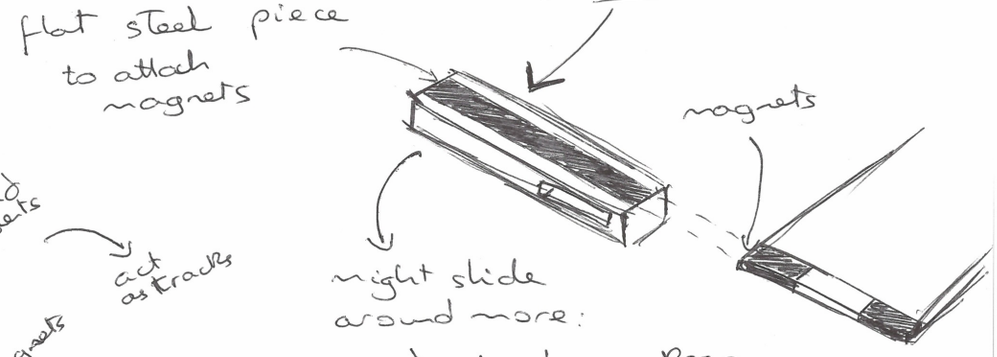




→ use the part that sticks out on ruler as handle

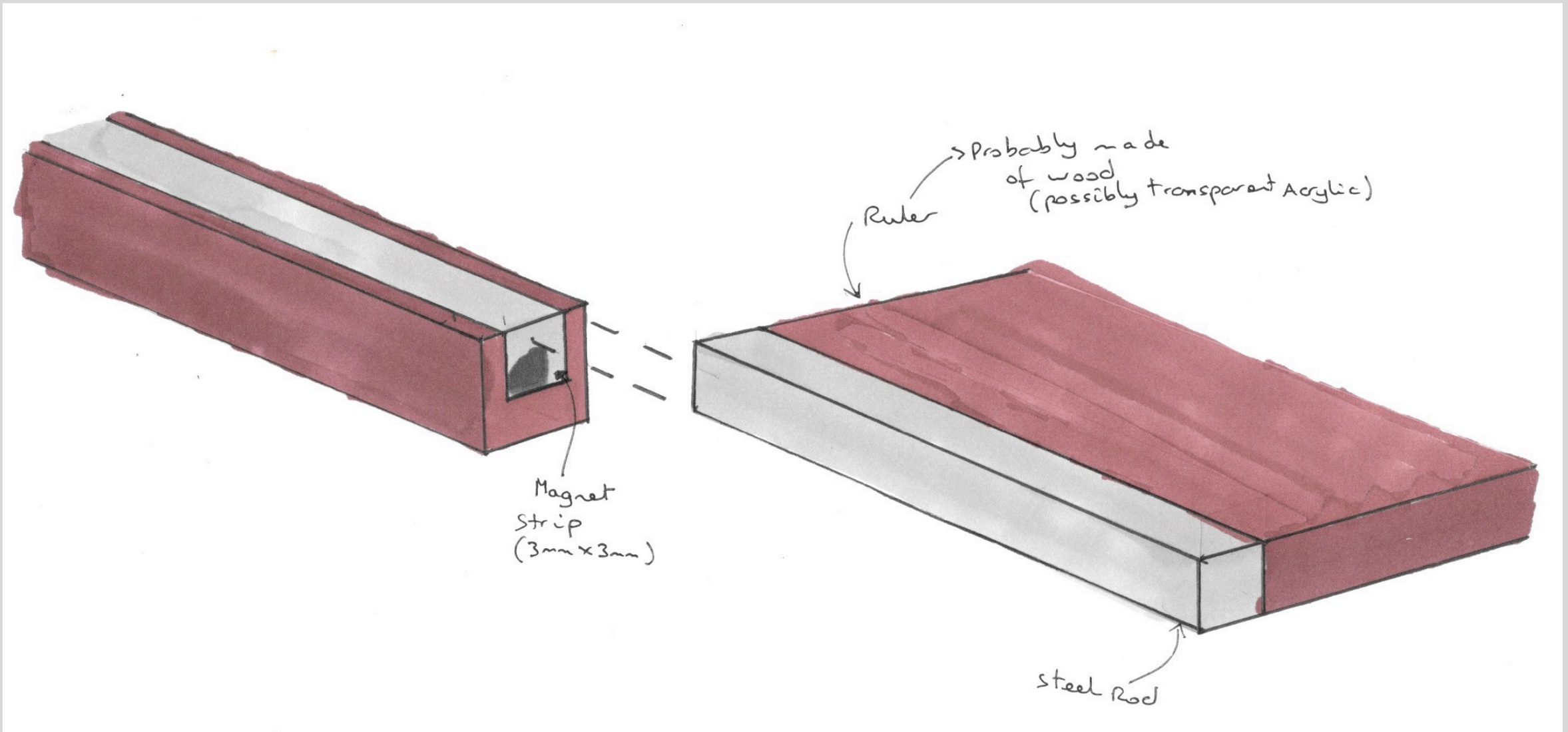




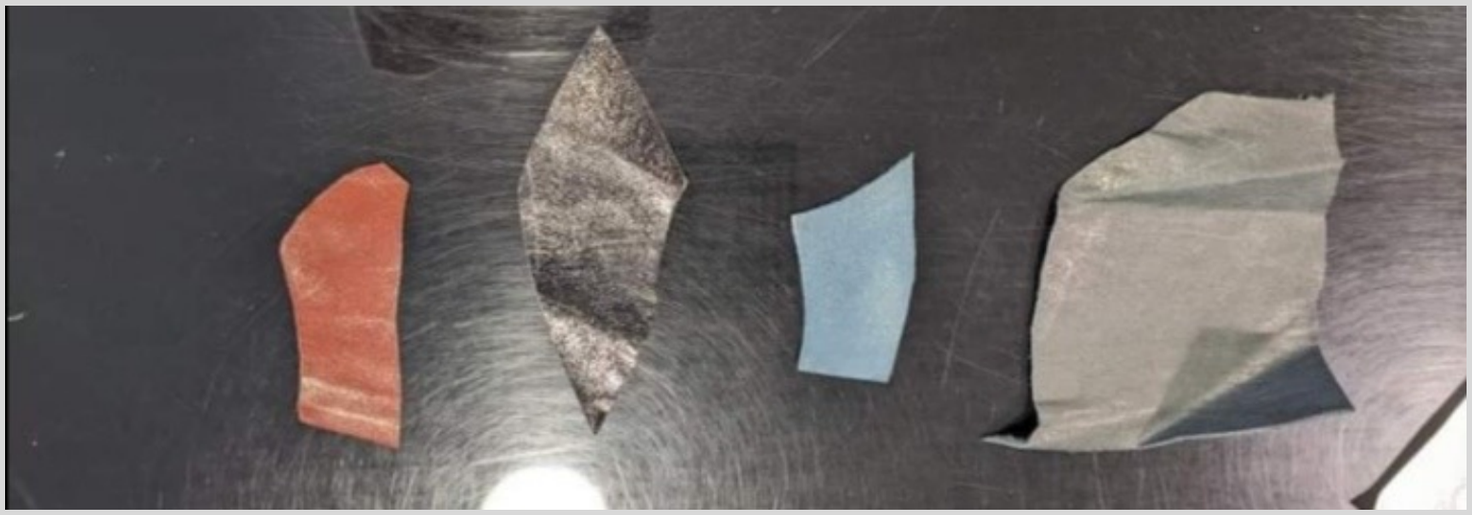


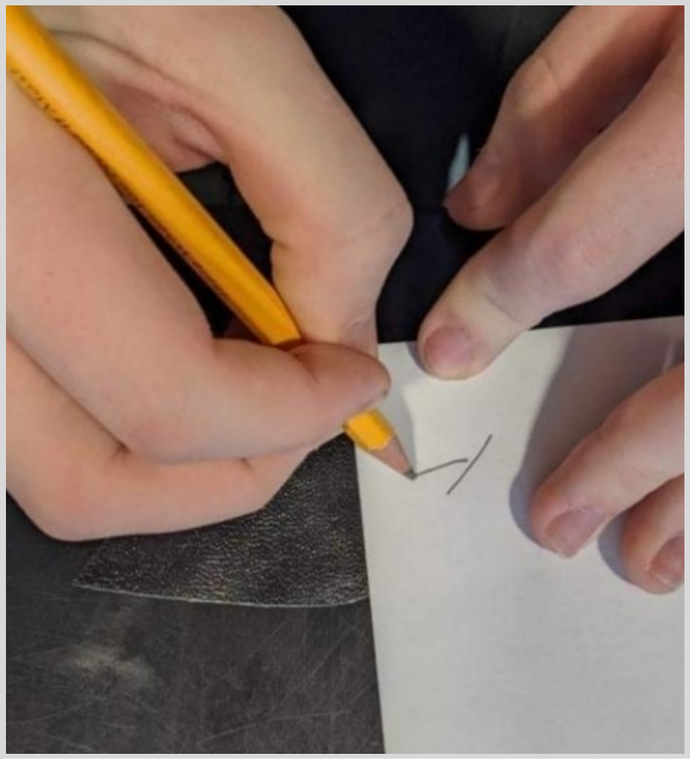
Developing The Design Of The Ruler:

To simplify the design of the bottom middle drawing on the previous page, I decided to remove the track indents and instead use an inset magnetic strip and so just have a steel strip on the ruler. This means that the strip would have to be about 3mm wide, but I could use a self-adhesive making it easier to assemble and disassemble.



Modelling – Looking At The Different Leathers:





Responses by Client:
The top image shows the client's order of the leathers from least favorite to favorite.
In regards to the 46 Vintage Calfskin, he commented that it's "too soft for my liking" and "the underside's weird", probably because he didn't like the felt like texture of the underneath.
The next one was the 121 petrol, for which he "hated it to bits" due to the color. Then, the 224 Black because "it's nice to draw", probably due to the plastic layer, but that also meant that it had "a plasticky feeling" which he didn't like. Because of this, he placed the 224 Cognac first, as it had the nicest feeling and "nice color". He didn't like it as much to draw on but found it nicer when there were multiple pieces of paper on top of it.

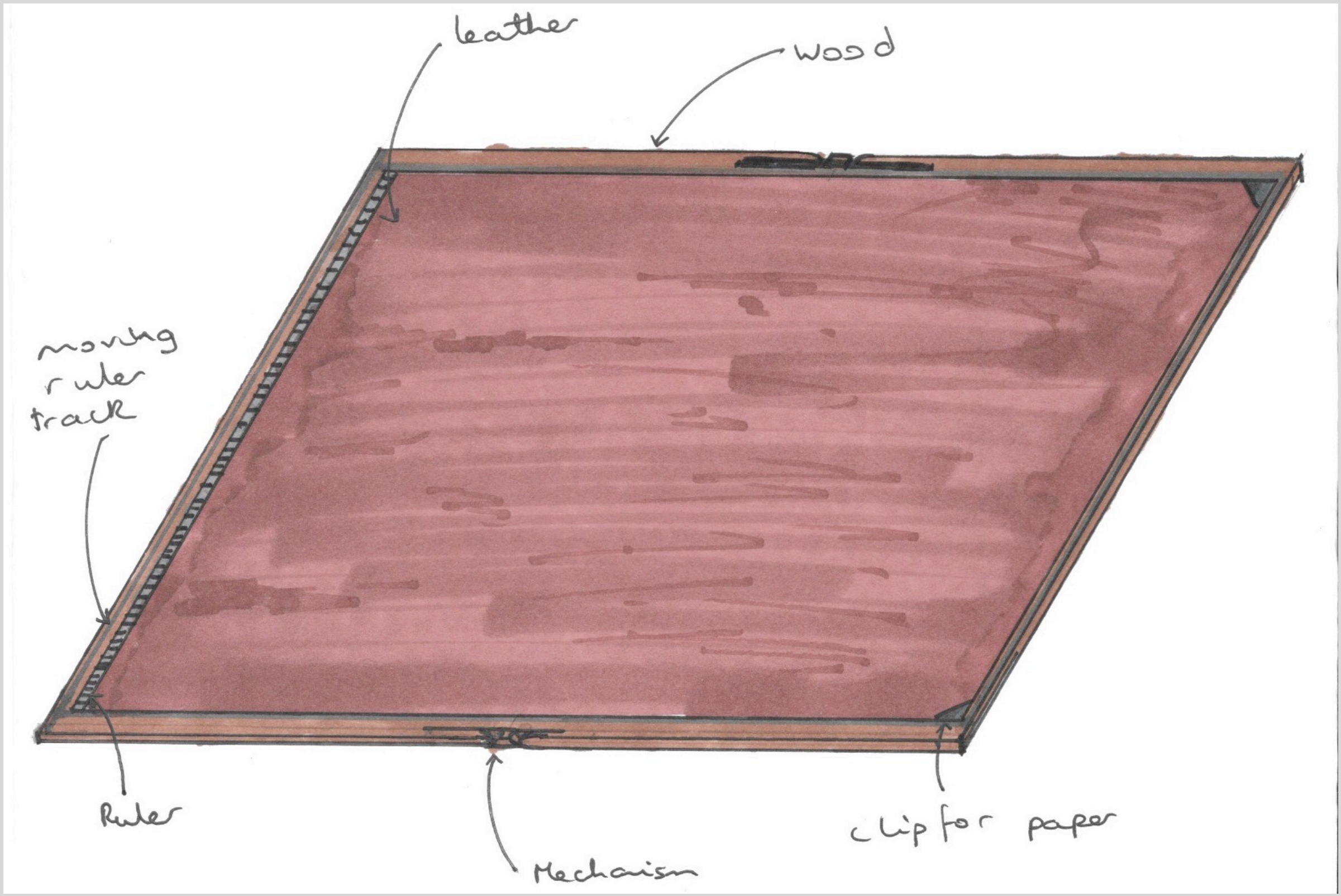
I got my client to test the 4 most different types of leather. These were, in the order of the image on the right, the 121 Medium Brown, 224 Black, The 121 Petrol, and the 46 Vintage Calfskin in black. The 121s are Lamb Skins and the 224 is a Goatskin which has a thin plastic layer on top.

The Vintage Calfskin (which can be seen being tested on the bottom left) is by far the stretchiest and softest one, meaning that it also bends the most and would be the easiest one to use in order to make the material fold. However, this also creates some problems: Firstly, it means that when you write on it, it feels very soft underneath the paper meaning that it becomes uncomfortable and often the pencil ends up cutting through the paper due to this. Furthermore, it might also end up affecting the final design as the leather could not be placed under enough tension.

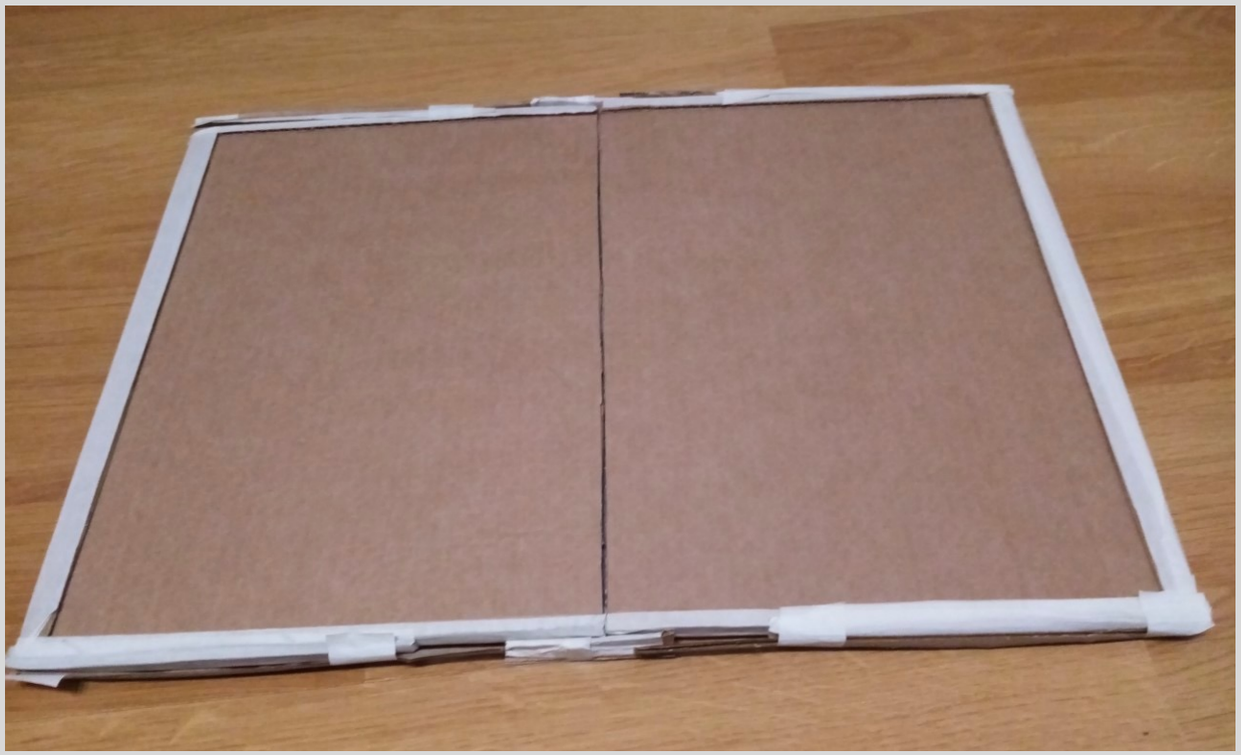
The 121 Medium Brown (which can be seen being tested on the bottom right) was a really nice leather to hold and touch meaning that it had a very nice tactile response to it. As the leather was also natural, it meant that over time it would end up with a much nicer finish then the 224 Black, as the plastic on the 224 Black would ruin over time, as by bending it multiple times it would end up snapping it and most likely ruining the top layer of the leather. Furthermore, the 224 Black could not be recycled as easily given the plastic layer and could not decompose meaning that it would go against the End of Life's point 1 and 3 in my specification. The 121 Petrol was just a different tone of leather that I wanted to include in there to show the consumer different tones which could be included.

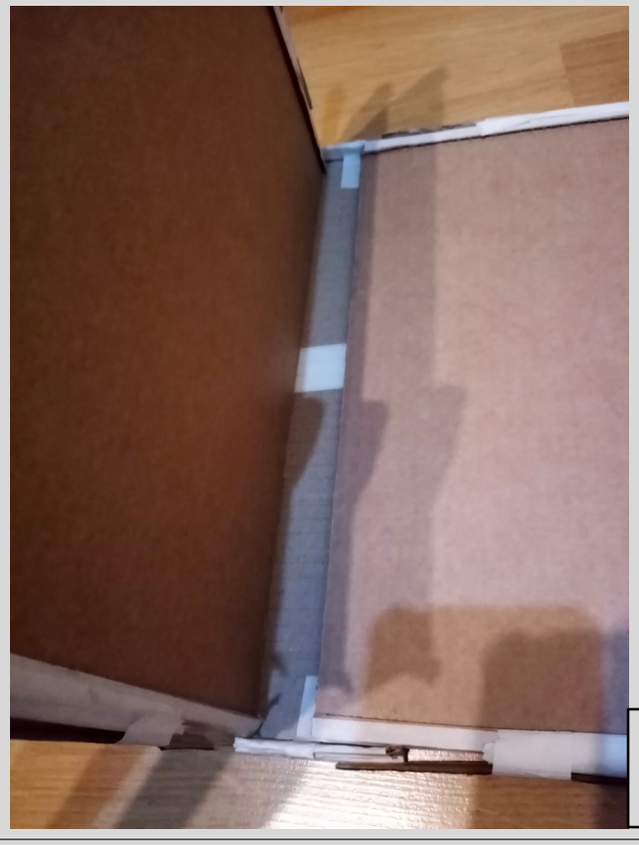
The only notable advantage of the 224 Black is that it's plastic layer results in it being slightly more comfortable to write on than the other ones, however, I do not think that it is a big enough advantage to make up for all its drawbacks, which go against the specification thus reducing the likeliness that I will use it. However, the choice in the end stands on the client, whose responses can be seen on the left.

Designing – First Coloured Sketches Of The Overall Design:



Modelling – First Cardboard Model Of The Overall Design:





To make this cardboard model, I used 3mm thick cardboard to create the base and attached it all together using glue and tape. Although I tried to build the mechanism, I was unable to using carboard as it always broke. However, that was not needed for this model as it was a just way to work out flaws and the overall size.

The overall size seemed fine, but I will need to check that different components can be built, such as the mechanism, the joint, the track for the ruler, and the way in which all the different pieces can be attached without using glues so that it follows the specification's end of life points.

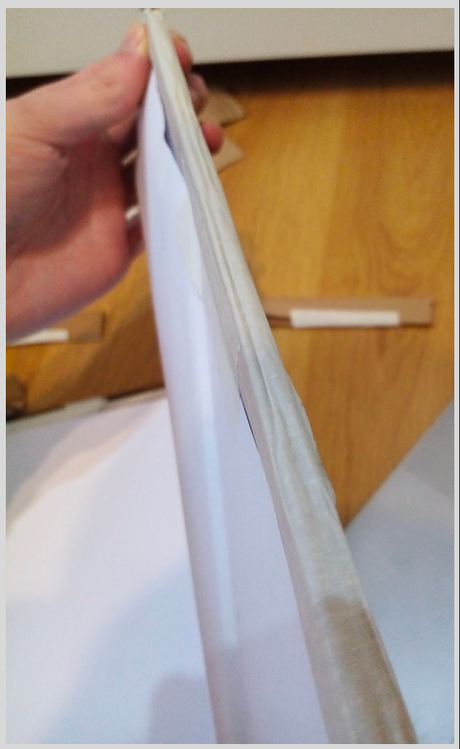
Another issue which I will need to check is the strength of the folding method and weather hardwoods can be used for it as it may break too easily meaning that a different material like aluminium should be used.

I will use this model later on to test different designs of the handle in a way which will allow me to test the client's reaction and comfort using them so that I can decide on which design I like the most to make the product more useful to the client market and its clienteles.

Client Feedback:

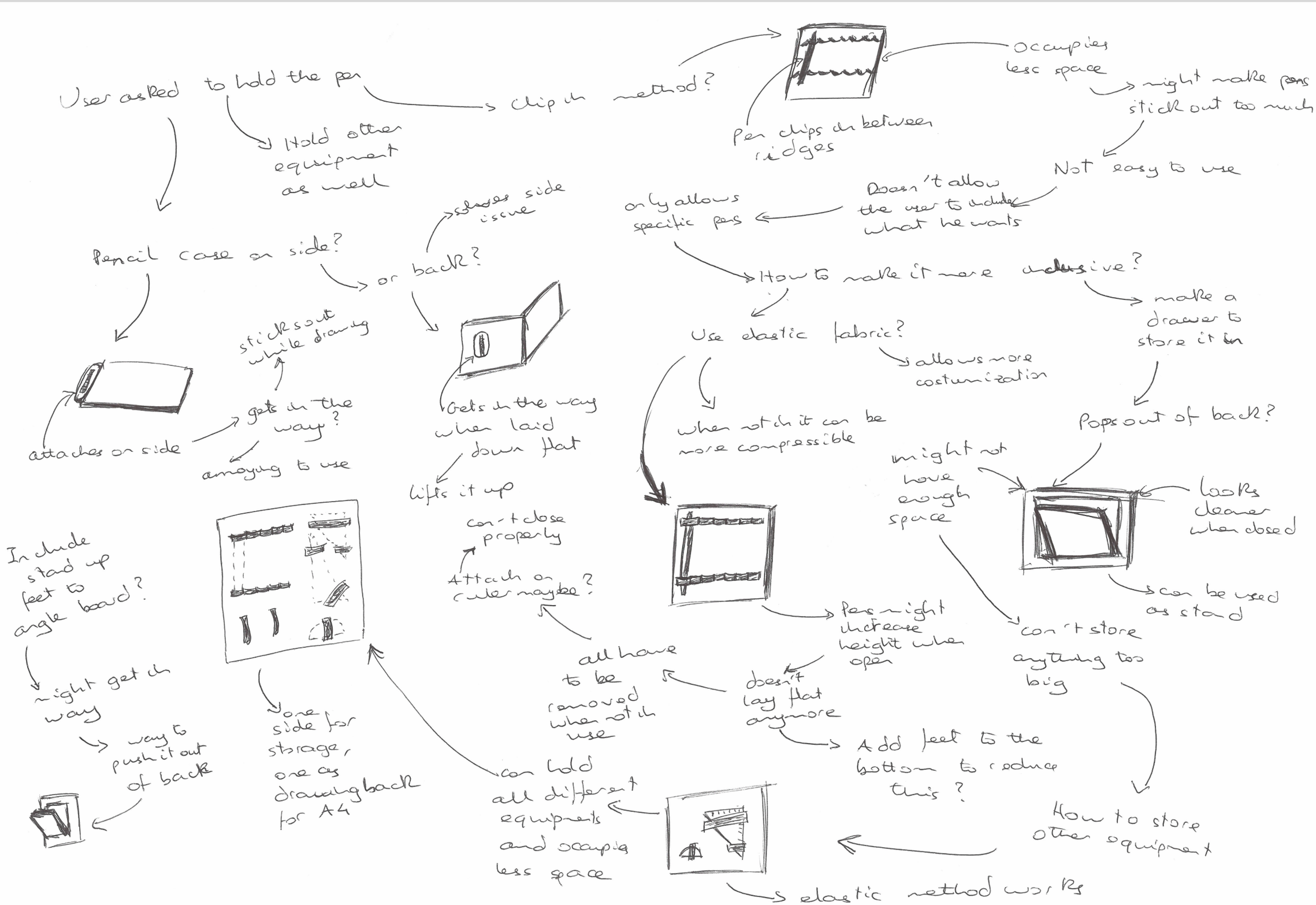
At this point, the model is looking good and it seems like it should work fine.

Modelling – Modelling And Testing The Handles:

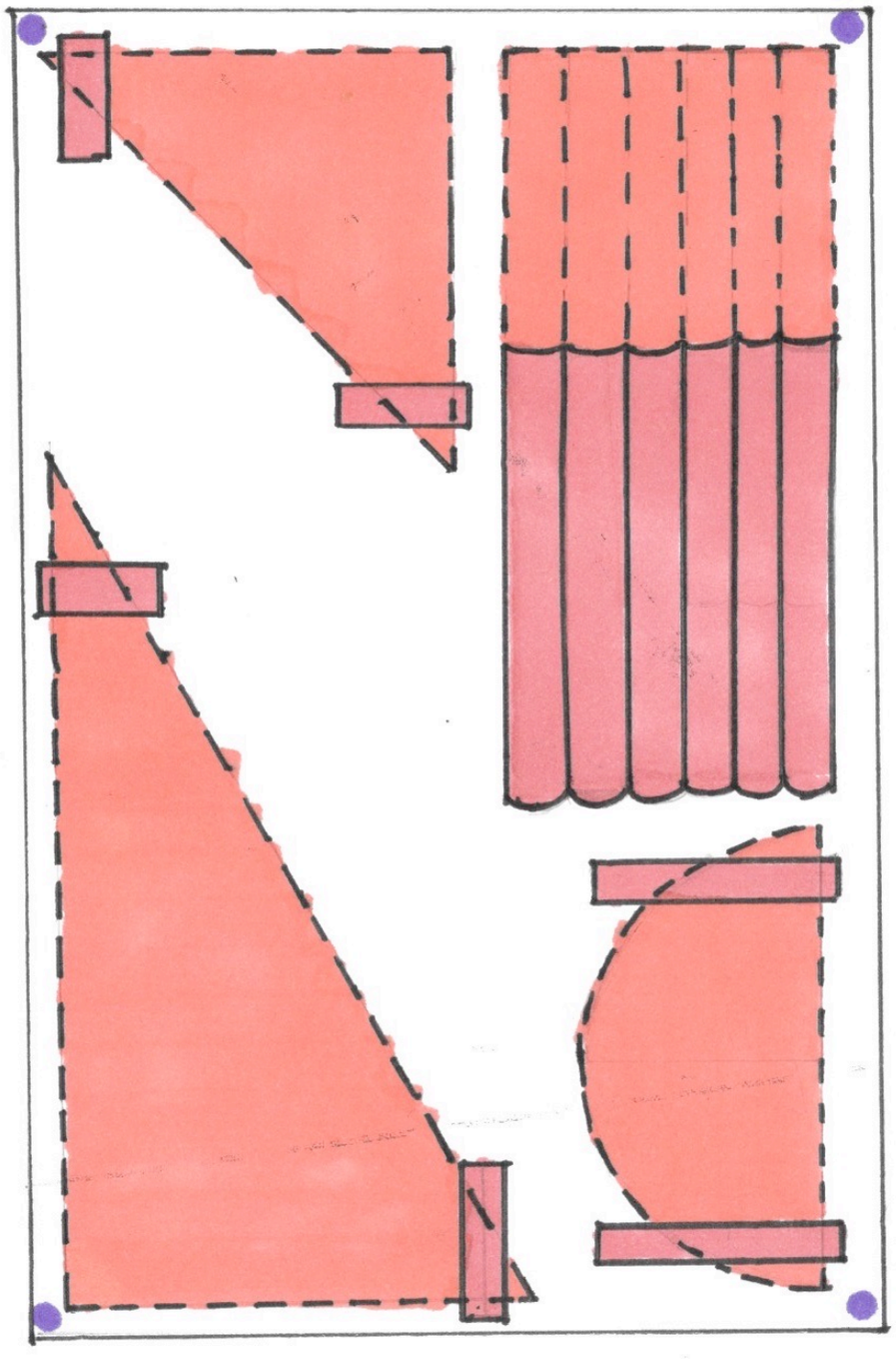
Model:	Circular All Around:	Below Only:	Extended Hole Cut:	No Handles:
Picture:				
Evaluation:	This design was the most comfortable to hold when opening and closing, however, many of the people I asked to test it said that the part that sticks out was annoying as it got in the way when they were drawing. Another problem that arose was that it stopped the board from being able to sit flat on the work surface meaning that it is harder to draw on. This problem also arose with the below only which is why both designs would require a solution to fix it such as placing additional support in other places which would further increase the height.	The Below Only design removed the top part of the first design so that it was more comfortable to draw on. However, this made it less comfortable to hold compared to the Circular All Around whilst also maintaining the same problem regarding it not making the drawing surface flat.	This design allowed the board to lay flat when opened thereby fixing the Circular All Around and the Below Only's problems. However, that meant that it occupied too much space when closed meaning that it was too big to properly fit in a smaller backpack. Another problem was that it was not as comfortable to use, but that may be due to the carboard being so thin.	This was the most basic design possible, however, it was one of the best ones as its only real drawback is comfortability, however, it is still comfortable enough to use and with some chamfering to smoothen the edges it will be comfortable enough. That is why I choose this as the handle which I would use in my finial design.

Inspiration And Research– Looking At Possible Attachments:

Designing– Attachments:



Designing– Attachments 2:



This is the probable layout in which I could place the different equipment for the drawing board.

There is storage for 6 pens/pencils, a protractor, a 60°/30° ~~tri~~ set square and a 45° set square.

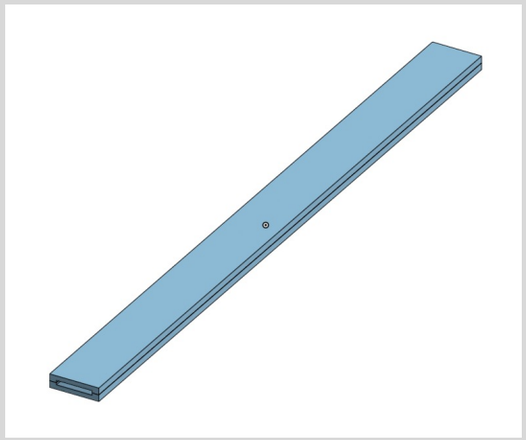
I am planning to use an elastic fabric to hold the pieces together as it will allow the user to customise it more.

However, there are 2 problems: Firstly, the pens would increase the height a lot (as can be seen in the drawing below) which means that the drawing board cannot lay flat unless everything is taken out. Secondly, as everything has to be taken out, it can be argued that it is useless.

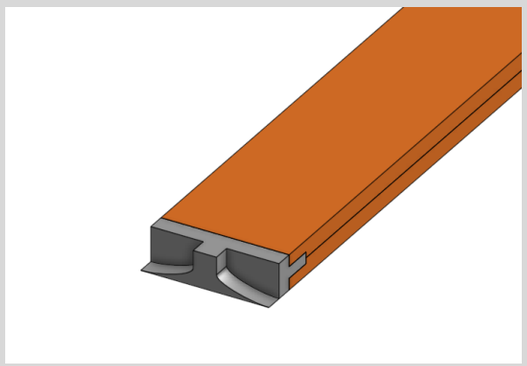
However, a solution would be to make it removable, such as by having magnets on the blue spots.

Although this design is nice, I do not think that I will include the holders as it is a bit gimmicky and not truly useful to the drawing board.

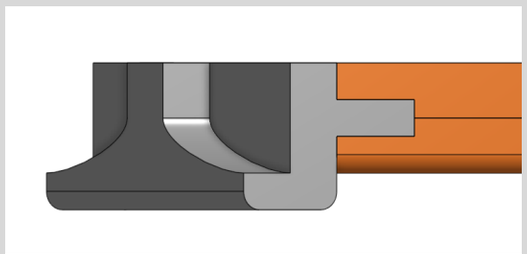
CAD Modelling The Wooden Design 1:



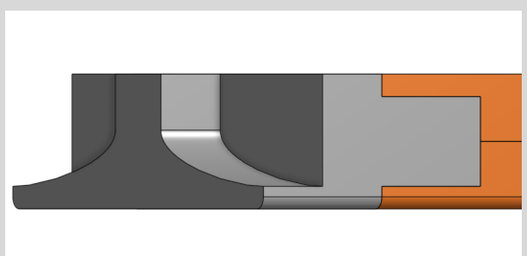
To start the CAD design, I began with the middle piece and modeled how it would be made out of two 3mm thick sheets of cherry wood. I placed a whole on each end in order to allow the Inversed CAM to slot in.



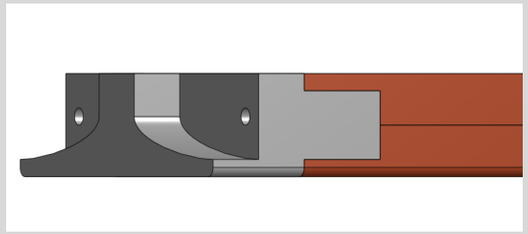
I then modelled and placed the Inverse CAM into the slot.



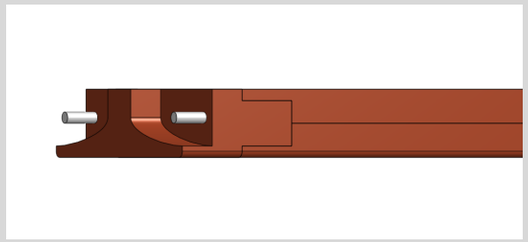
However, as there was little support, I needed to add material underneath it. The problem with this is that it stopped the board from being able to lay flat, and as I didn't want to increase its thickness, I needed to add material in another way.



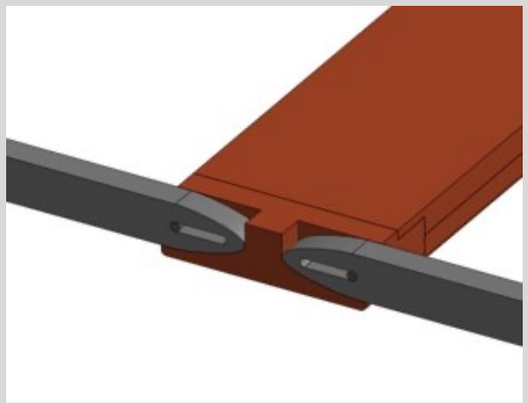
To fix the issue, I Increased the height of the male of the slot and raised the overall piece up by 2mm to add height. I also gave a continuous chamfer to the whole underside to make it more comfortable to hold.



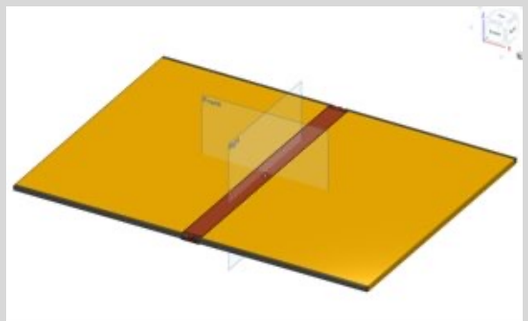
I then modelled the holes for the pivot and changed the color of the middle piece to resemble a more wooden surface.



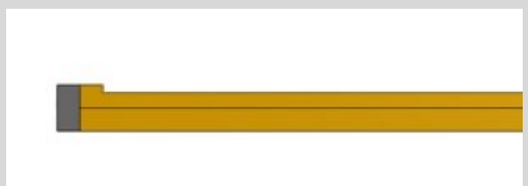
After this, I added the pivot bearings and changed the mechanism to the same wooden color to see how it would overall work.



I then modelled and added the inverse CAM to the design.

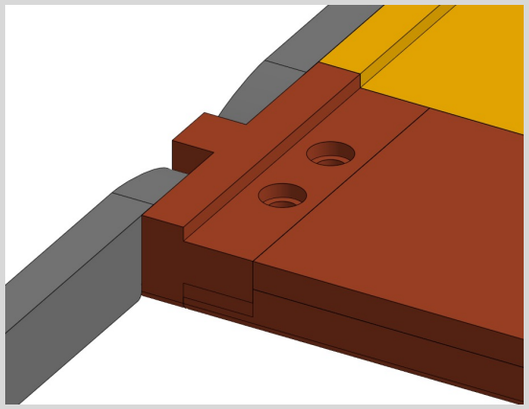


I then added the two large flat boards which act as support as two separate boards which are cojoined together.

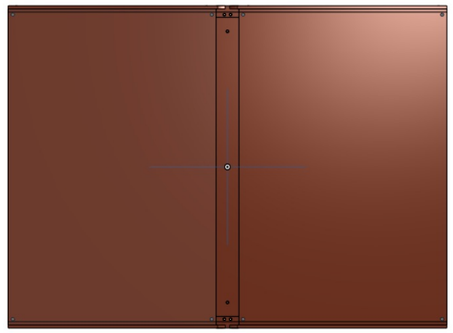


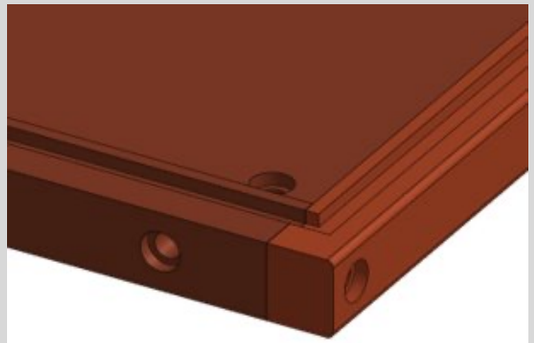
I then added an indent where the leather would be situated which was 1mm tall.

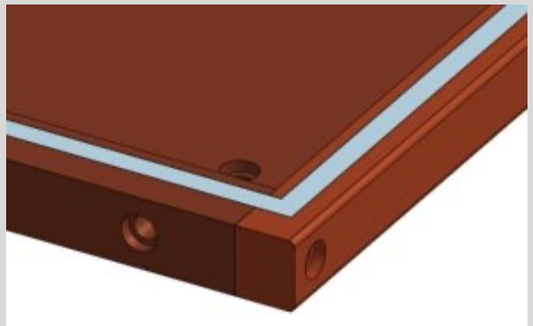
CAD Modelling The Wooden Design 2:



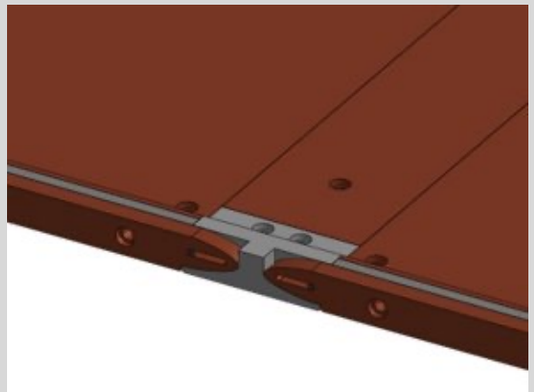
I then put screw holes into each side of the mechanism and into the board to show how it would be assembled together. I also changed the color of the end bit of the board and cut grooves for the steel strip to go into to hold the magnetic ruler..

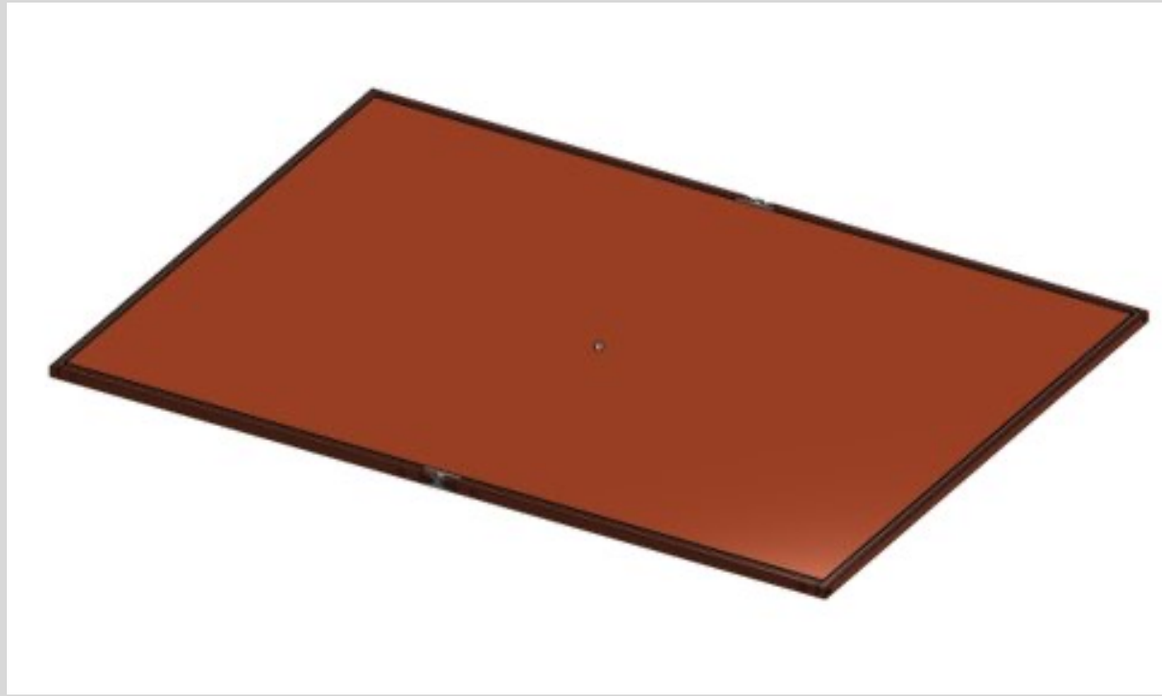




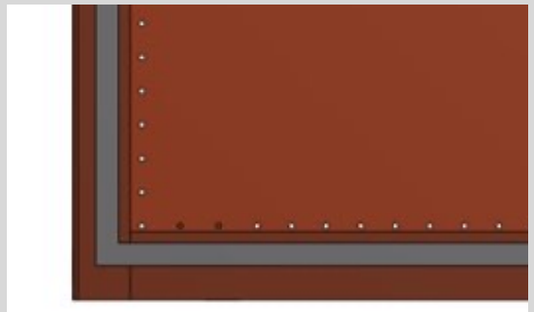


I then placed the steel strips into it which follow the edges of the board excluding where the inverse CAM is as I did not want to further weaken its structural integrity.

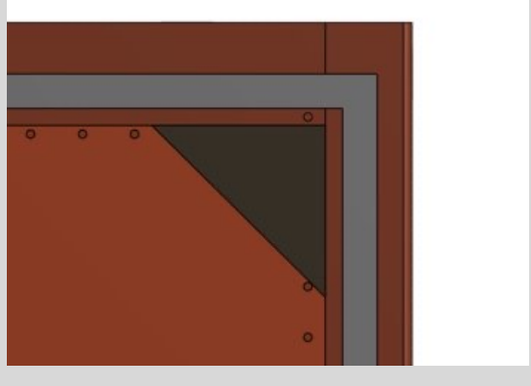




I then modelled the leather layer of the board to cover the whole A3 sized drawing surface.

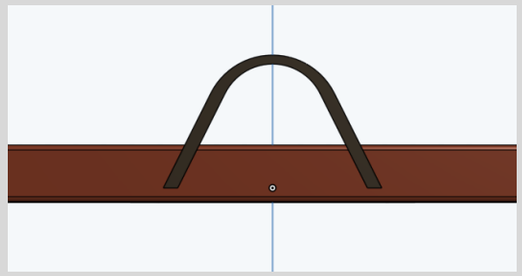


I then modelled the holes where the leather would be sawn into the wood to hold it still as I did not want to use adhesives, as stated in my specification.



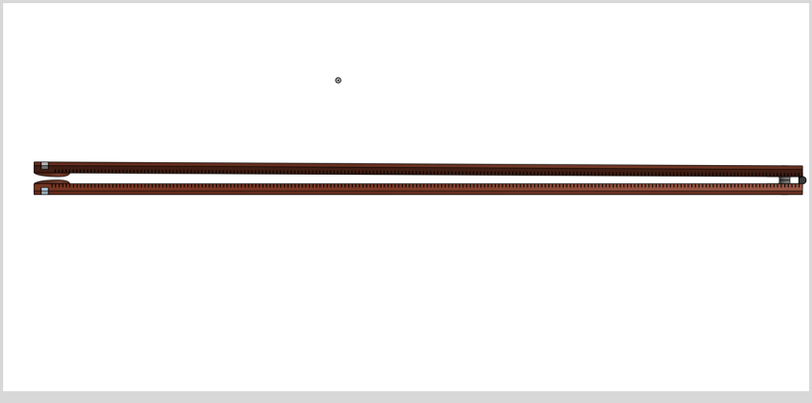
I then modelled the edges which would hold the paper. These would most likely be made out of a slightly elastic fabric which would just be sawn into the leather.

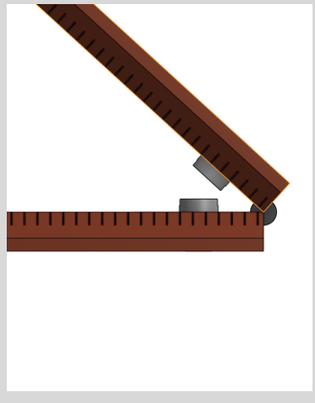
CAD Modelling The Wooden Design 3:



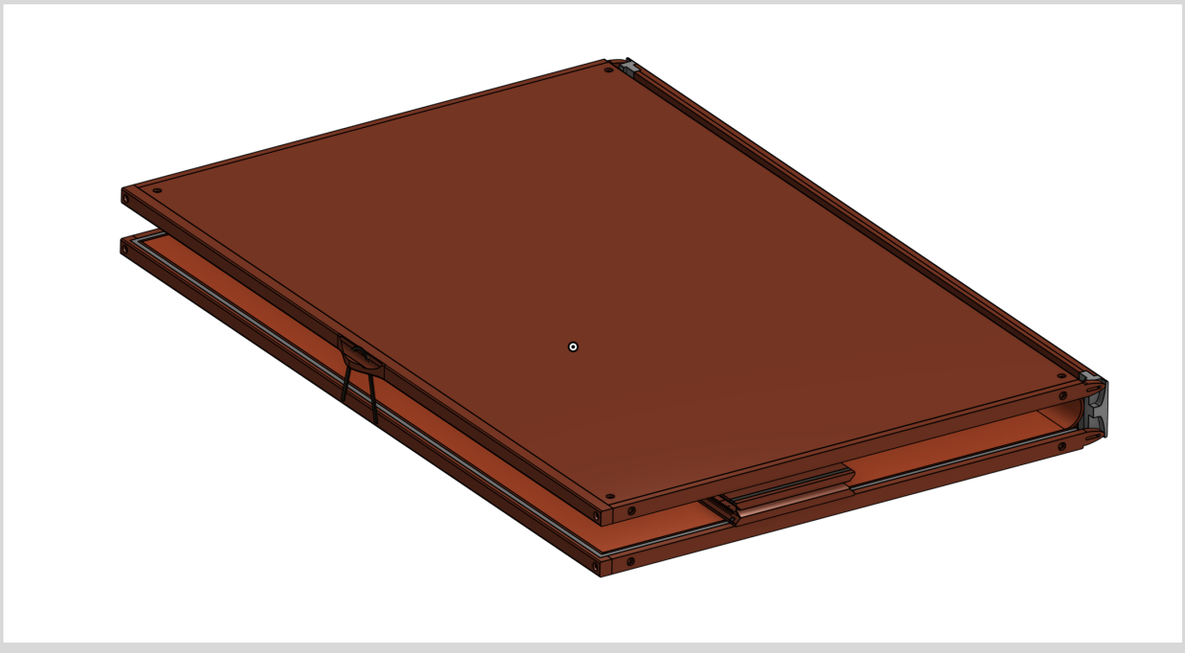
The next thing I modelled was a simple latch to hold it closed by using a simple lock which would have an elastic fabric band folded around it to tense it shut.

I then designed the ruler to be foldable and the same length as the long side of an A3 board and with a hinge in the middle which makes it wider when closed so that it sits flat within the board. The original hinge design (which can be seen on the bottom right) could not close properly which is why I had to increase the height of the ruler.





The picture above shows an assembly of the ruler and board being open, with the magnetic strip being able to slide perfectly along the tracks, and the ruler markings starting on the edge of the A3 paper.



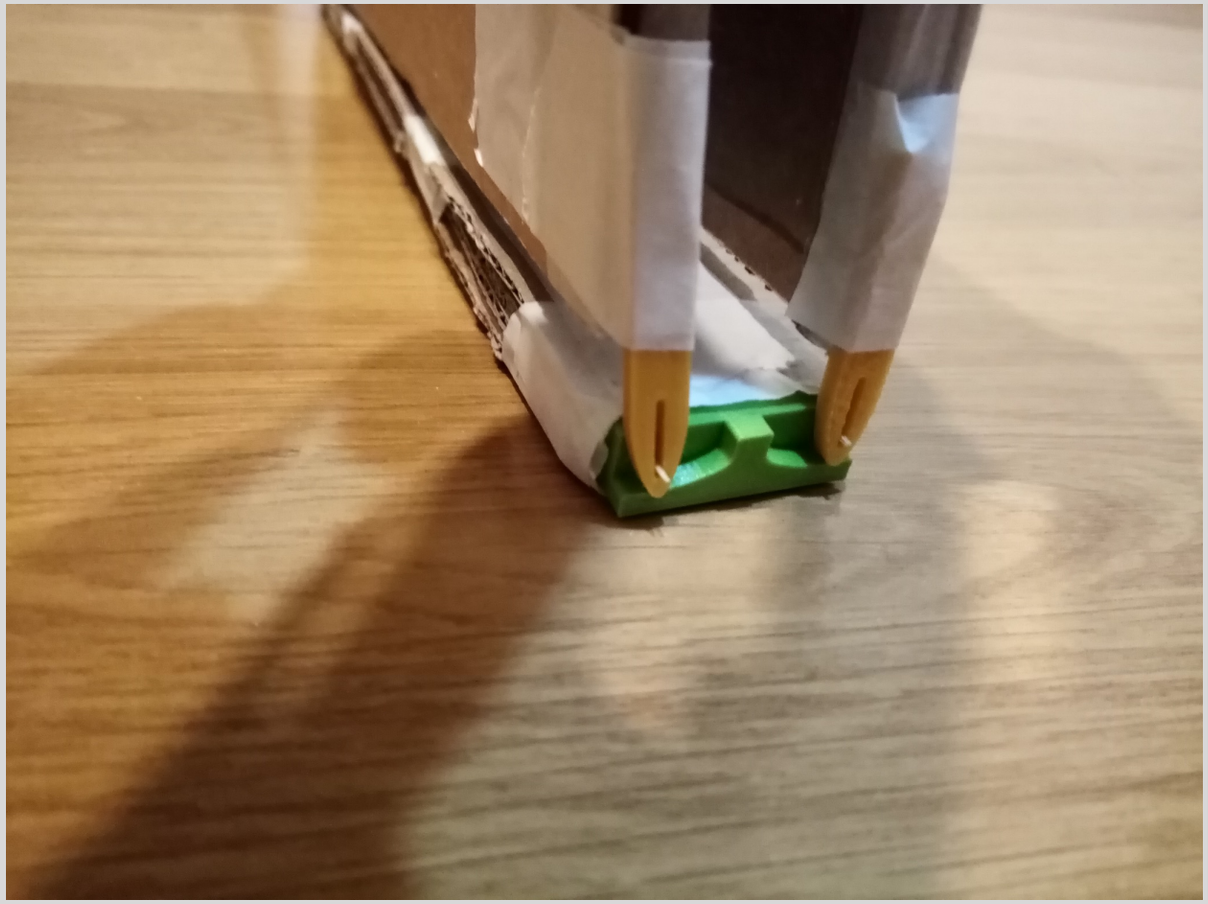
The picture above shows the board and ruler closed together. The overall Board is about 26mm thick when closed which means that it is not too large to be stored, but any possible future changes to the design should try to avoid increasing that.

Cardboard Model With 3D Printed Mechanism:

This model adapted the old cardboard model in order to have the 3D printed mechanism integrated into it to show how it would work. For this model, I just used chopsticks for the bearings, which didn't snap so the steel ones should definitely be able to handle the force exerted on them when opening and closing the board

It also created a substantial gap as expected which will allow the leather to go in it. The shape of it also helped to stay flat when opened as to start folding it closed requires a little bit of force meaning that it can support itself when opened thus improving user comfort.

This model has therefore shown that the theory behind this board works, but a worry with it is that the mechanism will result in the wood being sanded down over long periods of use so other materials will need to be researched to make sure that the product has a longer life and will not need to be replaced, thus lessening its environmental impact.



Looking At Materials For The Frame:

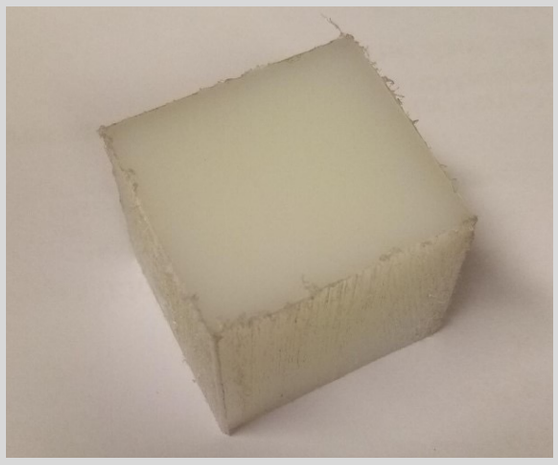
Looking at the design, I wanted to See different ways in which I could use different materials to change the style and color of my board. The first idea which comes to mind is to use a metal as it would be a lot stronger, but it would also have to be more lightweight, so the best material which comes to mind is aluminum.

The two products are both industrial designs which combine leather with aluminum. The one on the left uses a more greenish earthly color leather which has its attaching methods hidden below the frame to give it a more seamless look. The one on the left, however, uses a more cognac colored leather which is riveted onto the aluminum frame.

The leather's color could be a lot more customizable if I were to use aluminum as it is a lot more neutral color so the costumer could make it to their liking.

The product on the left shows how the aluminum could look with a white leather, however, the worries with using this color is that it would get dirty very easily as any small ink or marks would show, so darker colors may help to avoid that. The problem with darker colors though is that they would seep through the paper more making it harder to see fine lines.

Mass	Override	1.811 kg
Volume		670789.673 mm³
Surface area		423611.487 mm²

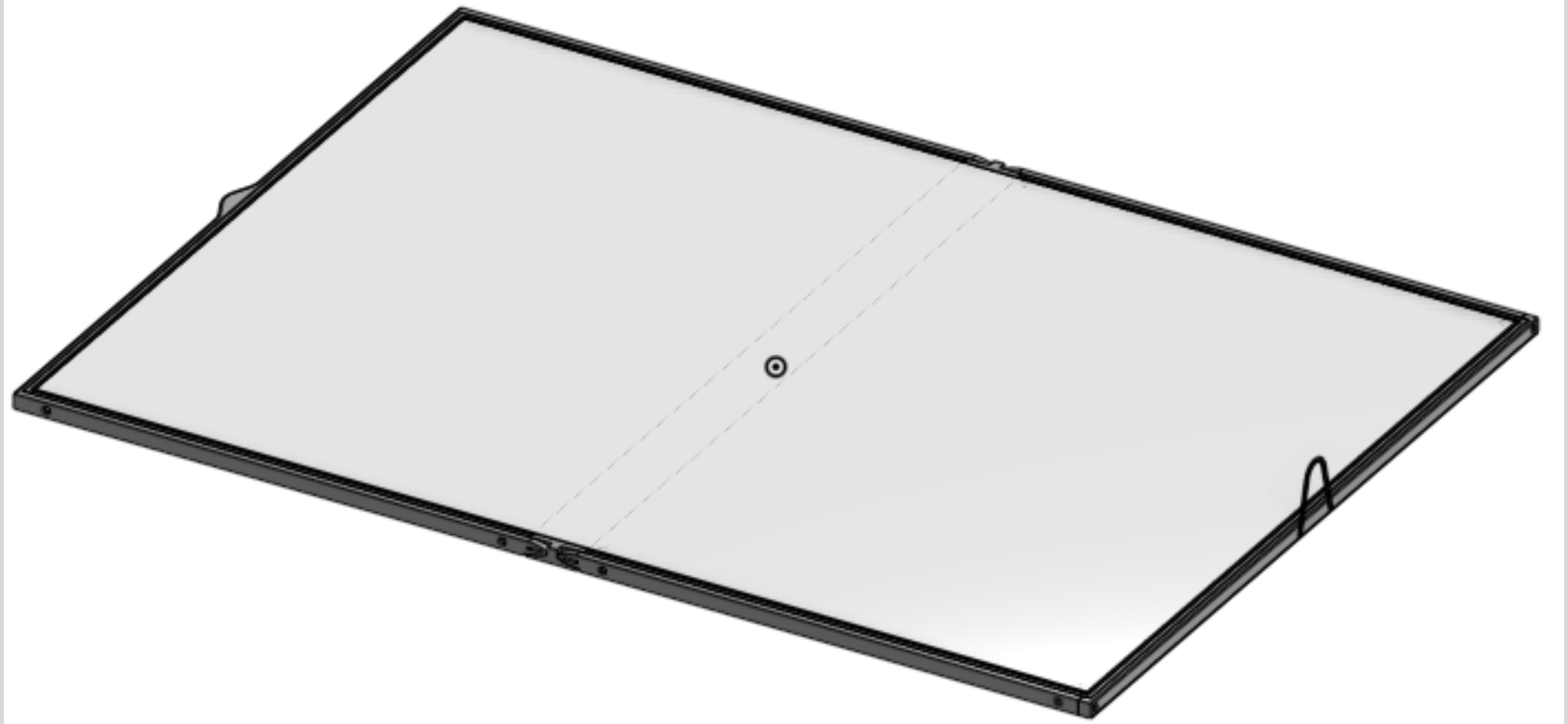


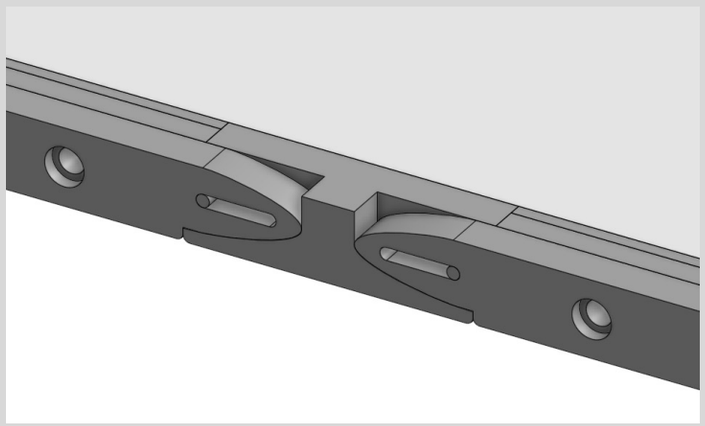
The aluminum frame could be milled out into its shape using a method similar to that of how smart phone or laptop cases are done.
This would allow me to add the indent required without having to use adhesives to attach multiple layers together and it would also produce a much stronger frame for the product. However, there are some concerns regarding the weight of the aluminum frame and weather that would result in too much force being exerted on the small bearings.

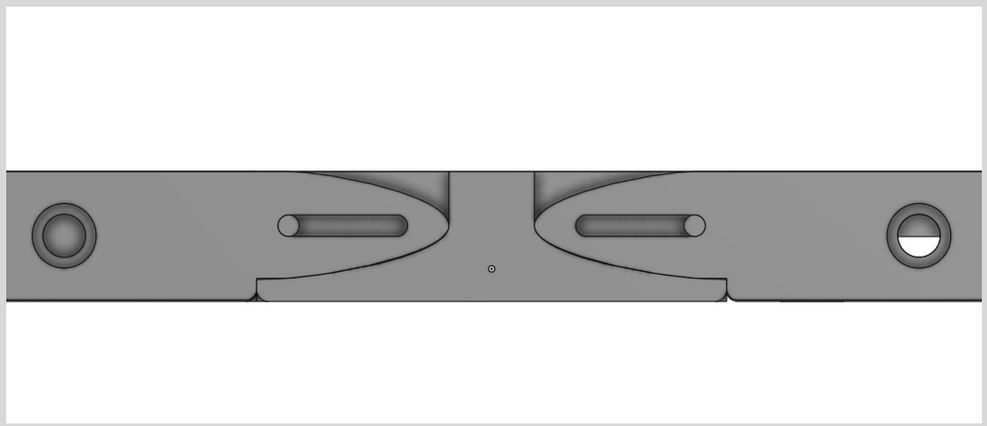
Having made a quick model of it (that can be seen in the next page), the
Board would weigh around 1.8kg if it was fully made out of aluminum, which is too heavy for the bearings, so to reduce the weight, I would have to make just the outmost frame out of aluminum and the inside bit out of a lighter material.
The first material I thought of was cork, however, looking at that material, I was found to be too heavy. However, two lighter materials could be used instead: those were Balsa and Cherry Wood as they were both really light but still durable and stiff enough that it would be comfortable to write on them.

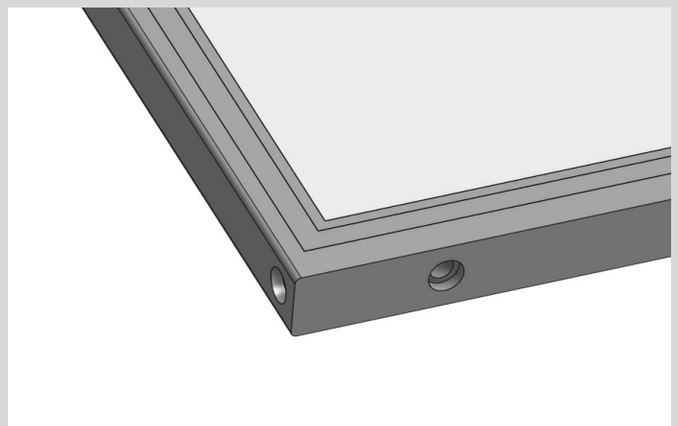
However, another possible solution is to use Nylon at the place of aluminum as it is a lightweight, strong and yet durable material which has good mechanical properties for this project as it is self-lubricating meaning that the mechanism would run smoother. Furthermore, it comes in a variety of colors making the overall board more customizable to the user, much like how the colors of iPad Airs or iPhones can be changed.

CAD Model Of The Aluminum Board:









Looking At Substitutes For Real Animal Leather:

Substitutes to real leather have been needed to be researched as it is not a sustainable resource and there are concerns regarding it being an animal product upsetting vegans.

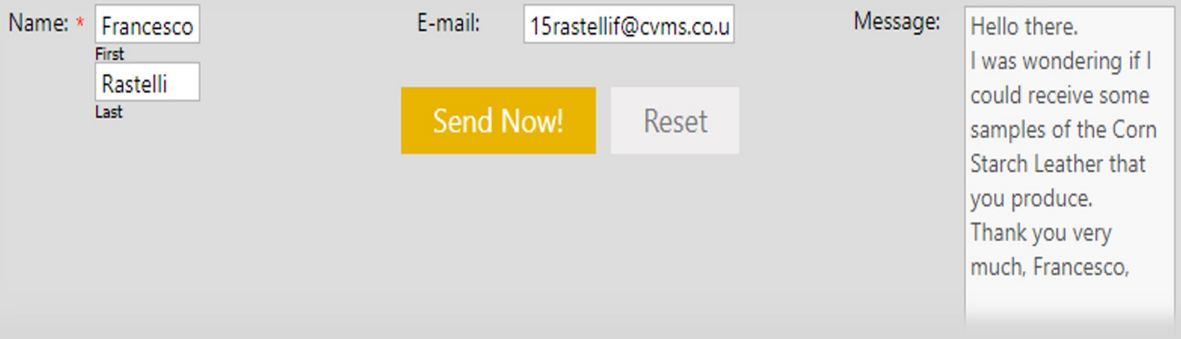
The first leather alternative I looked at is Faux Leather as it literally means "fake leather". It has a few advantages over real leather: Firstly, It is cheaper and has more color and texture options that real leather. It also requires less maintenance and is more stain resistant, and it is also a lot stronger and easier to work with. However, it does not last as long, it is more vulnerable to puncture and tearing.

Over time, it is likely that the faux leather will ruin much quicker and will have to be replaced. This is a worry as it is made of non-recyclable plastic which is mixed with multiple dyes, waxes and polyvinyl chloride, or polyurethane. This means that it will have a bigger effect on the environment when disposed of.

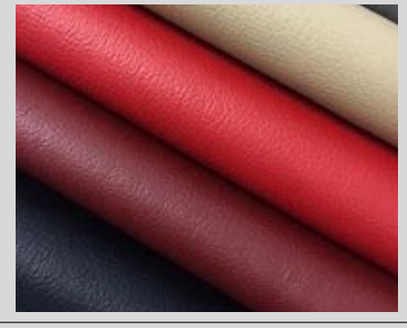
However, there are other substitutes of fake leather that can be used such as plant based or corn starch-based leather.

The next leather I looked at was plant-based leather from BZ Leather Company. They produce it through the use of corn starch out of a factory situated in Houjie Town in Dongguan City in the Guangdong Province, China. They use enzymes and micro-organisms to turn the corn starch into Propanediol.

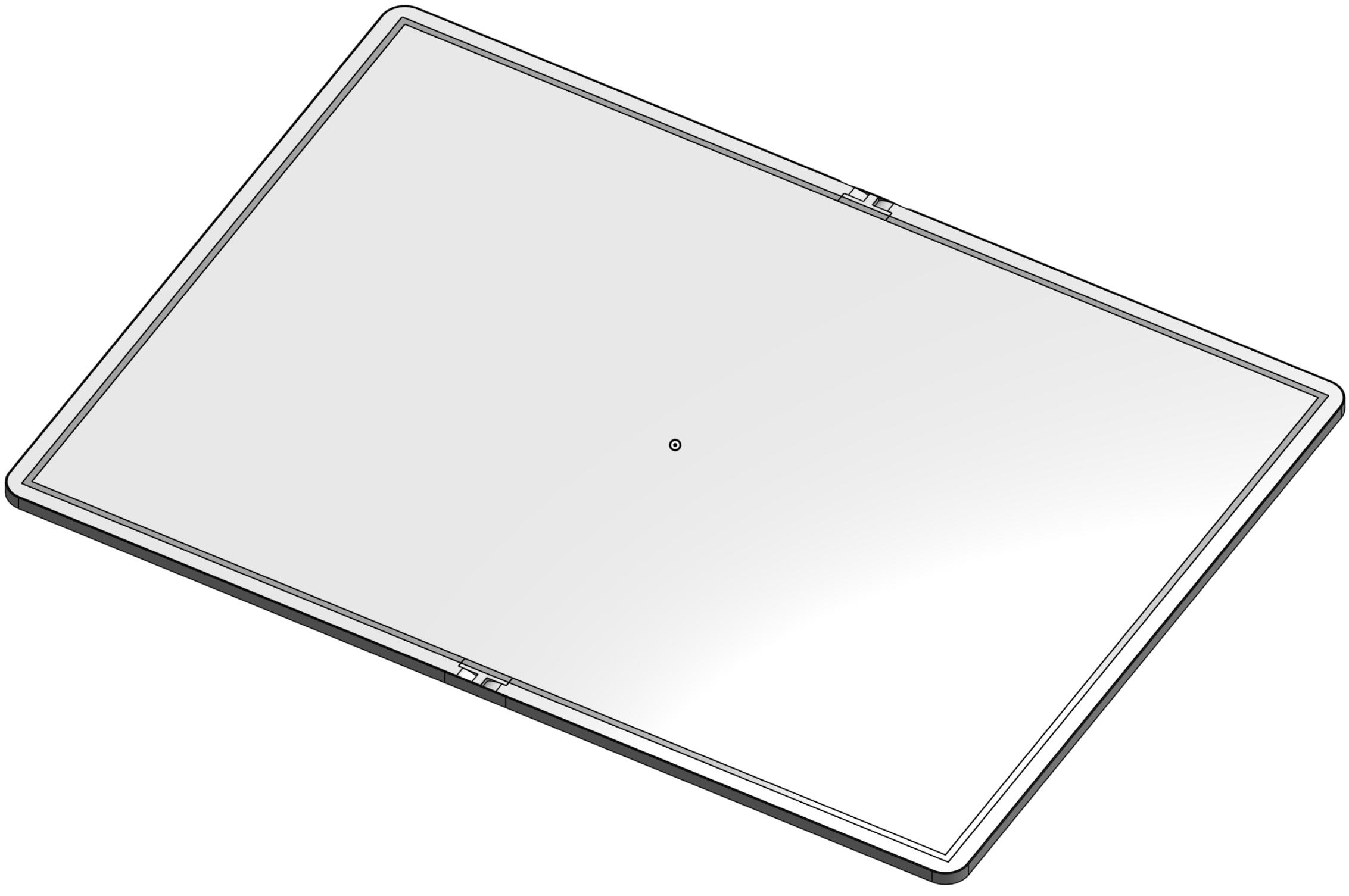
The same company also produces PETA approved leather which is either the plant-based leather seen above, silicone leather, recycled car windshields leather and cork leather. Out of these 4, the best option seems to be the corn starch leather as silicone is too soft of a material, the card windshield is made of PVB, and the cork leather looks ugly, Therefore, I emailed them asking for a sample of the leather but they unfortunately never emailed me back,



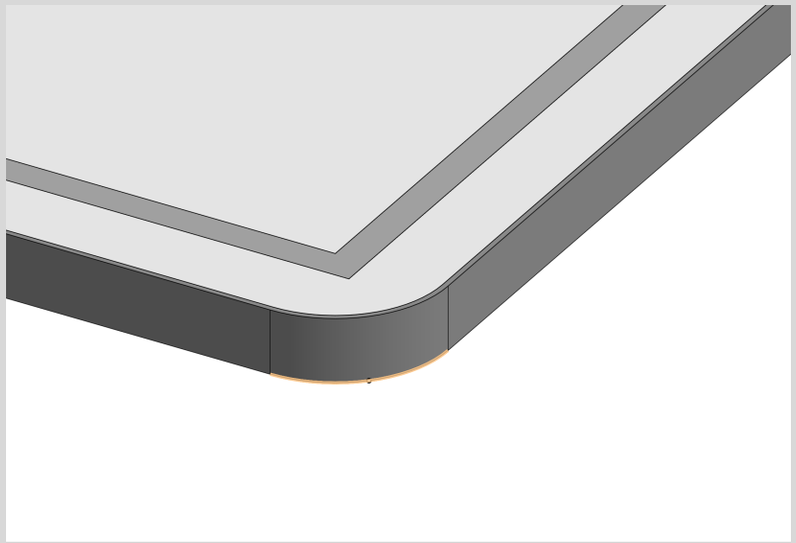
As they never replied, and I could not find any other easily available supplier, I decided to use Faux Leather instead which is much more easily available, and I could get a sample a lot easier. Such as on websites like Ebay where it would be deliver quickly comparted to having it sent over from places like China. Thereby reducing its carbon footprint as less distance needs to be increased,



CAD Model Of The Nylon Board – Overall Design:

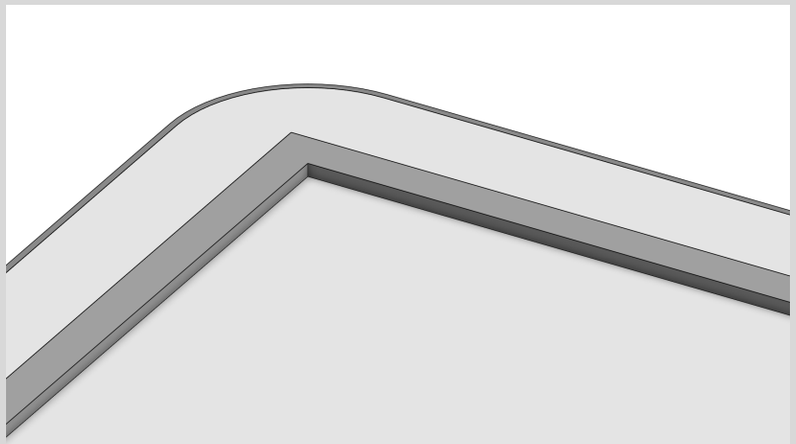


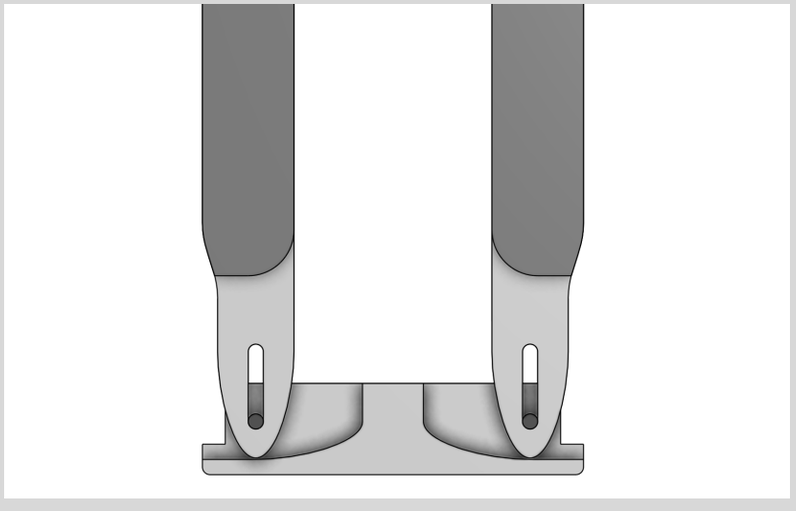
CAD Model Of The Nylon Board – Changes In The Design:



The first change I put was to change the corner edge to be more rounded and to increase the width of the bezel from 5mm to 10mm in order to give the user more space to hold it and also increased the overall structural integrity.

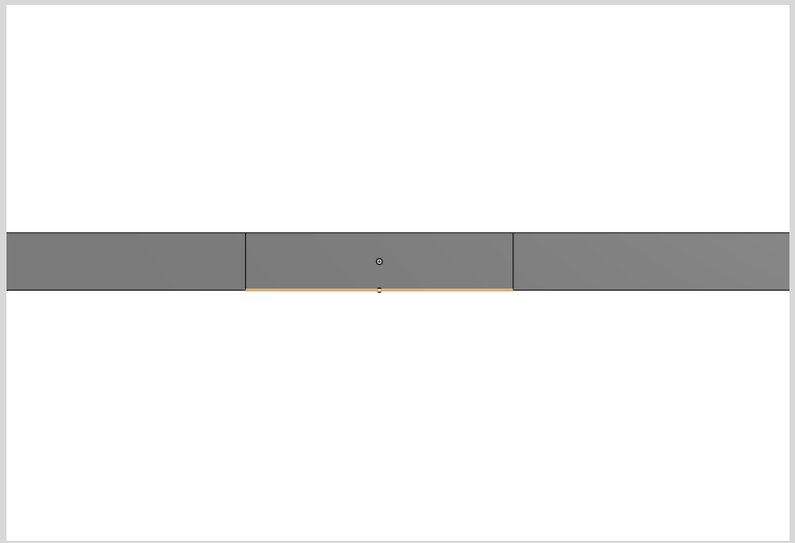
I also placed an aluminium band around the edges to give it a smoother and more modern look. This was greatly inspired by the design of Apple's iPad and the ability to use nylon to use multiple colours further reflects that like the iPad Air as my design could be customizable to have multiple colour choices.



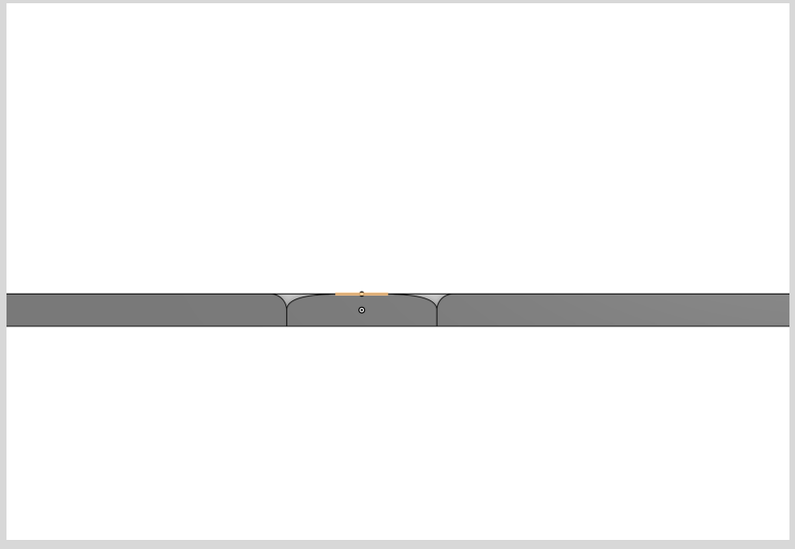


The next change to the design was moving the steel strip to the edge of the leather in order to simplify the design and to make it look cleaner. It also makes the design stronger as you do not have the thin 0.5mm bezel on the edges.

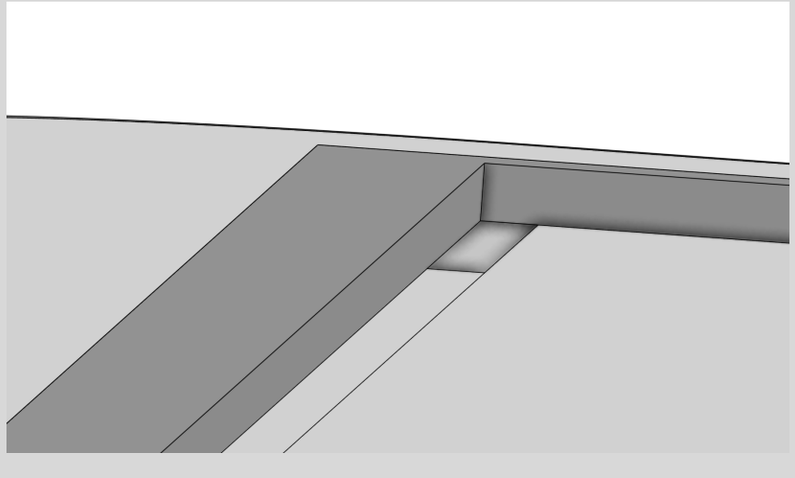
I then added a slight extrusion to the mid piece with an indent in order to create more support for it when opened.



The aluminium band around the edges originally had a square edge to make it flush. However, that would not work as it would stop it from being able to turn.

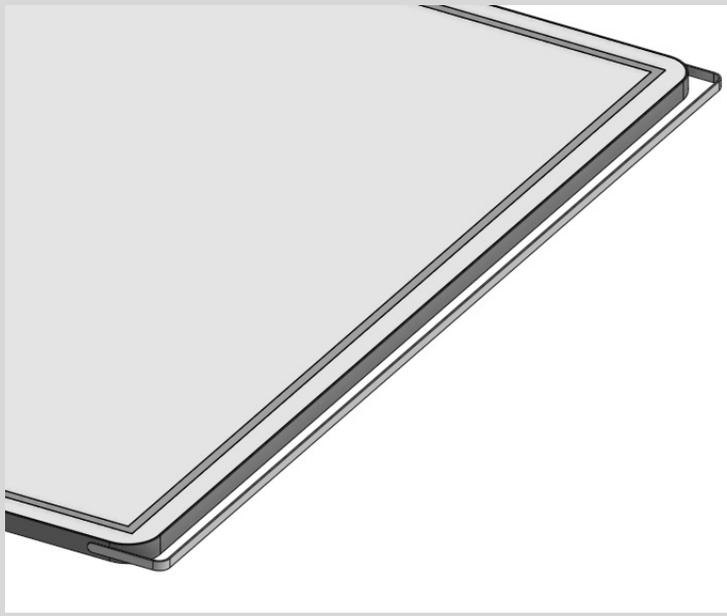


To fix that, I rounded the edges which also made it look a bit nicer when closed. This did not affect the board's mechanism and the cutaway meant that it would make it stronger as the aluminium is less likely to bend out of shape.



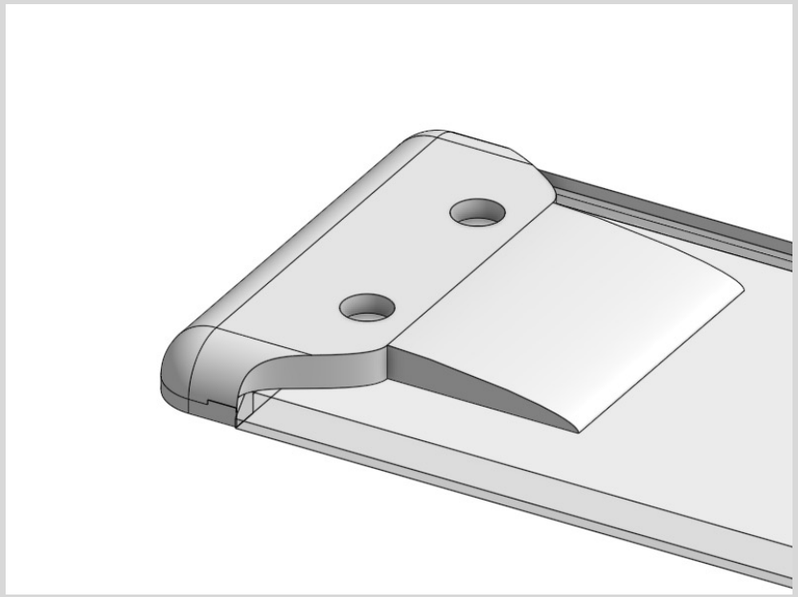
The last change I included was to add a cut-out underneath the steel pieces to use to clamp the leather to the board without having to use adhesives.

CAD Model Of The Nylon Board – Changes In The Design 2:



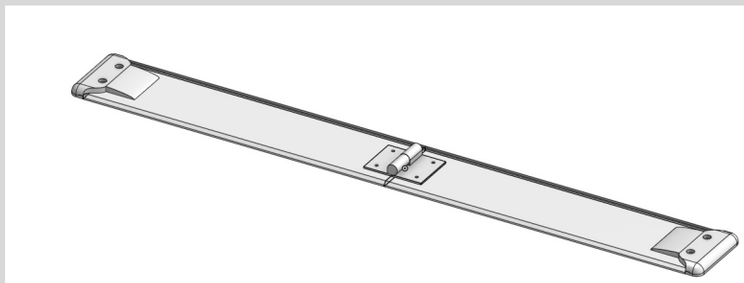
The latch was changed from having a sticking out bit to just being a large elastic band which covers the whole board.

This makes it easier to use and more stylish as it doesn't stick out at all compared to the wooden board one.



The end handle for the ruler was made wider to make it look smoother and more comfortable to hold.

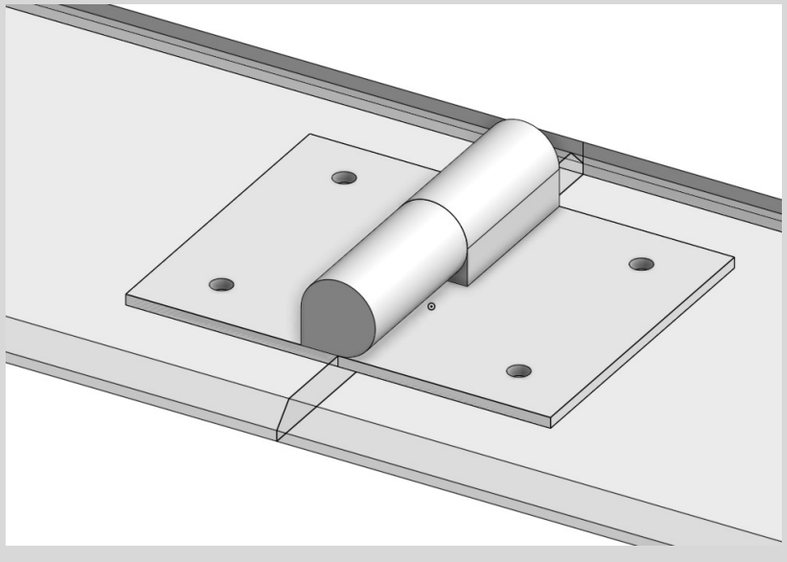
It also is just tall enough so that when it is closed, it will rest on each other meaning that it will support he board itself.



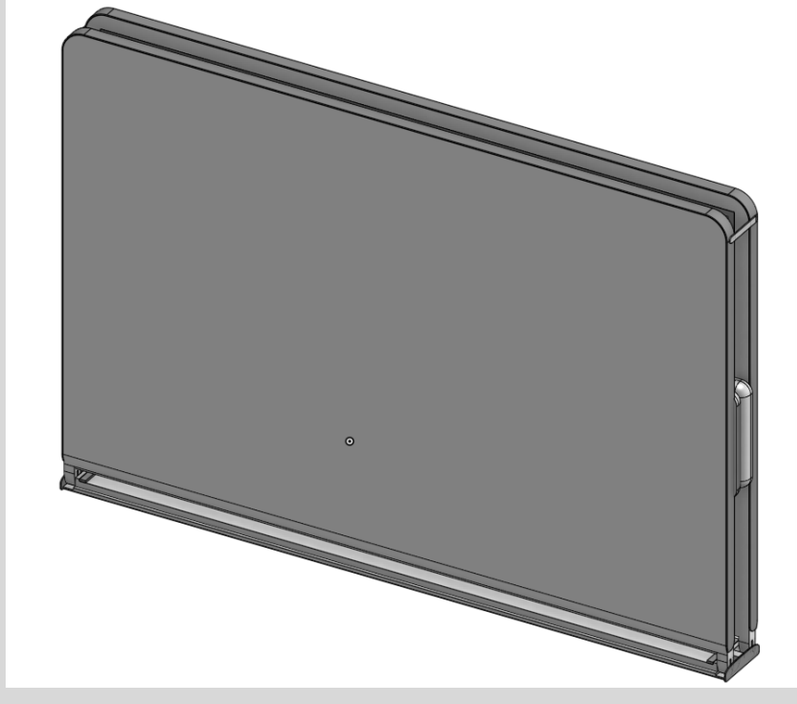
The ruler also changed as I changed the transparency of the material used so that you could see through it. I will probably need to use acrylic for this.

The picture on the left shows a side view of the board when it is closed and how the elastic latch would hold it and the ruler support it.

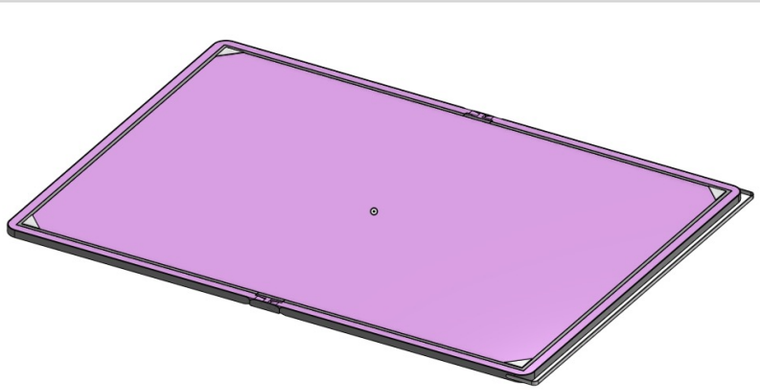
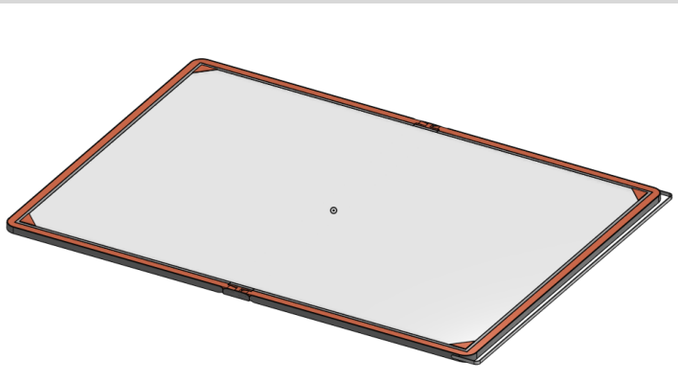
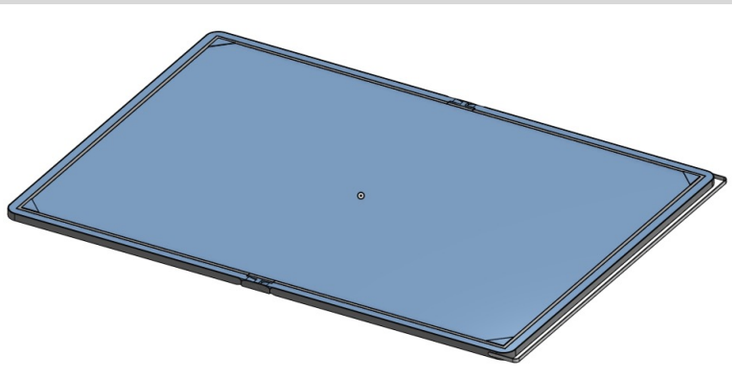
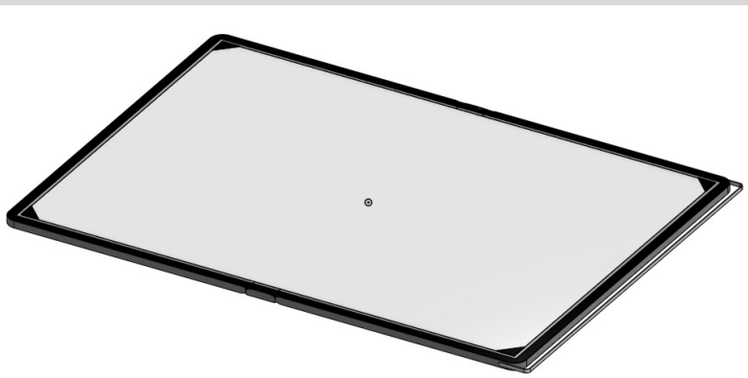
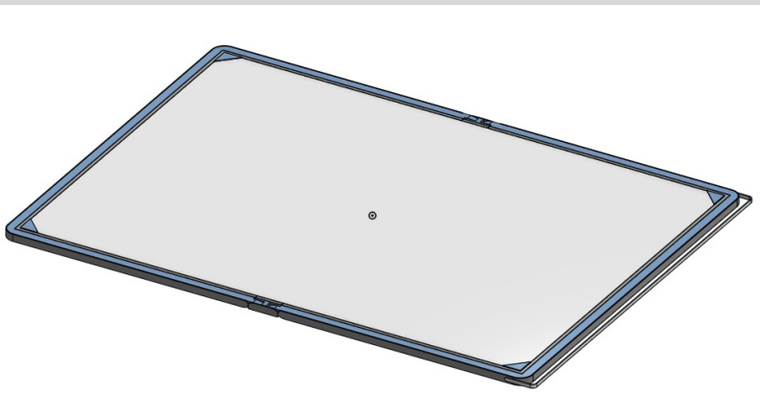
This picture instead shows a 3D side view of the board.

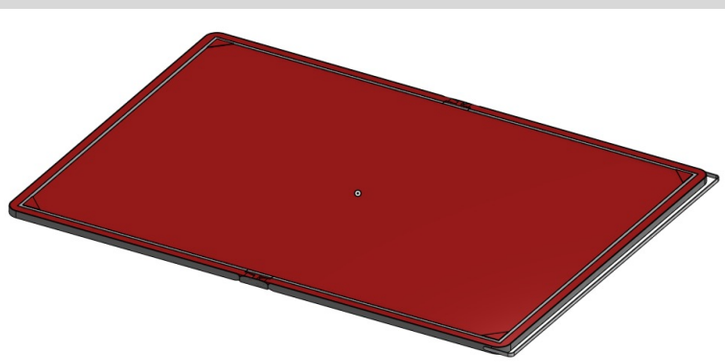
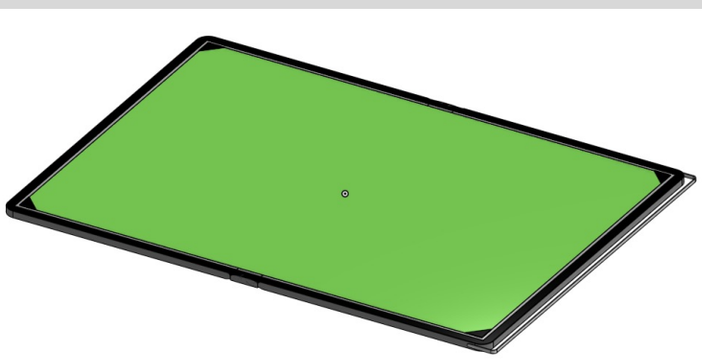


The hinge on the ruler was made bigger so that when the ruler would fold in half, it was just wide enough to fit in the frame so that it would support it when closed.



Possible Color Combinations And Client Reviews

Client Feedback:

First client:

Very cool. Very Swag. I like it.
I especially like the variation in color options. Very simple.
Could easily use it for drawing.

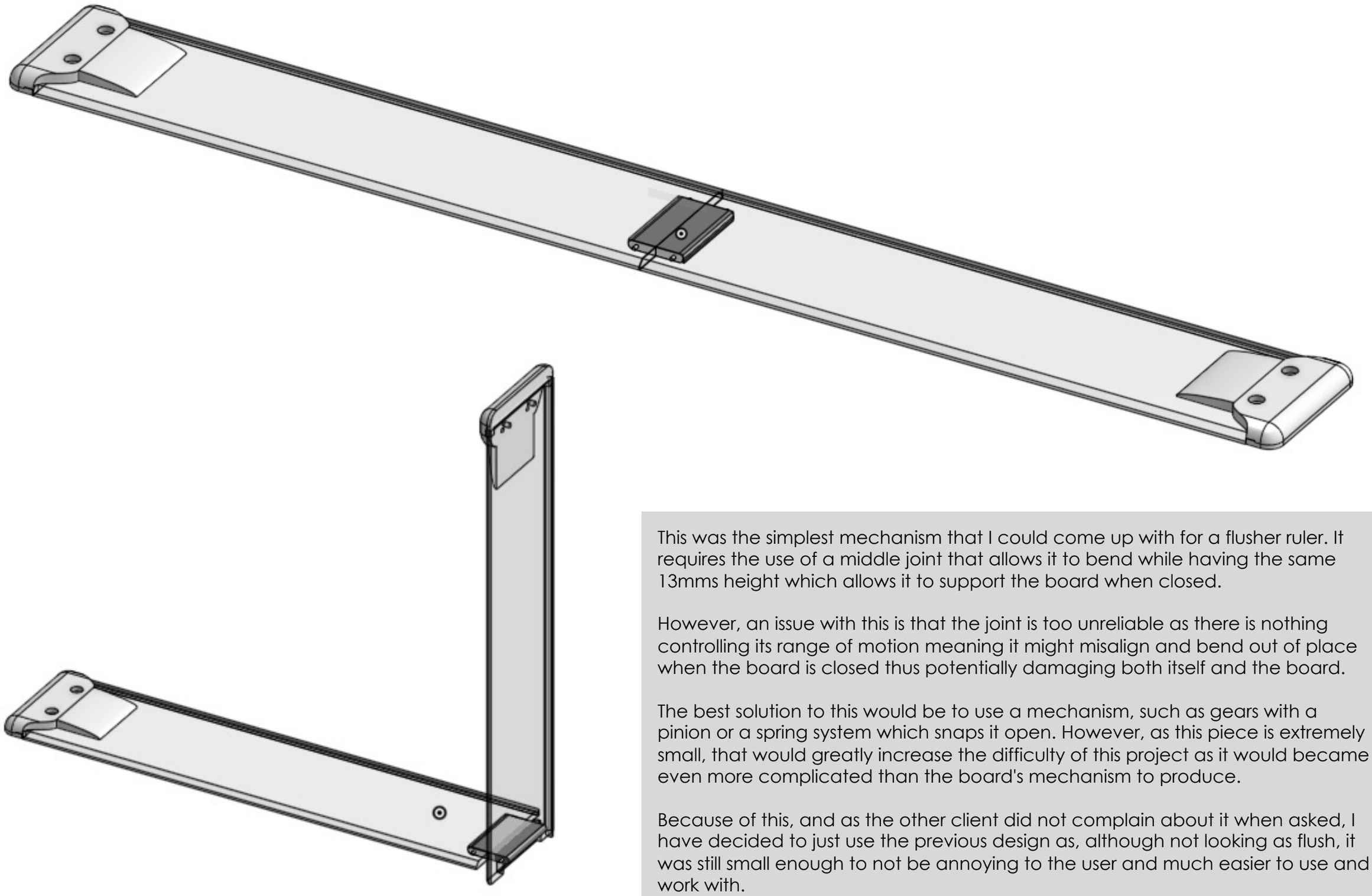
Second client.

I like it. Sleek and not bulky.
Looks like it could easily fold and be carried with no effort.
I like the grey with black border(no4)
Overall I think it meets with the initial criteria my only query is the hinge on the ruler looks a bit like an afterthought. Would be nicer incorporated better of flush. And will it engage with the edges of the board somehow?
13:48

Thank you sir. It will use magnets to engage the ruler with the edges of the board.
13:51

Excellent. I think it's developed very well. Hope the prototypes performed as expected.
13:54

Possible Changes Based On Client Feedback:



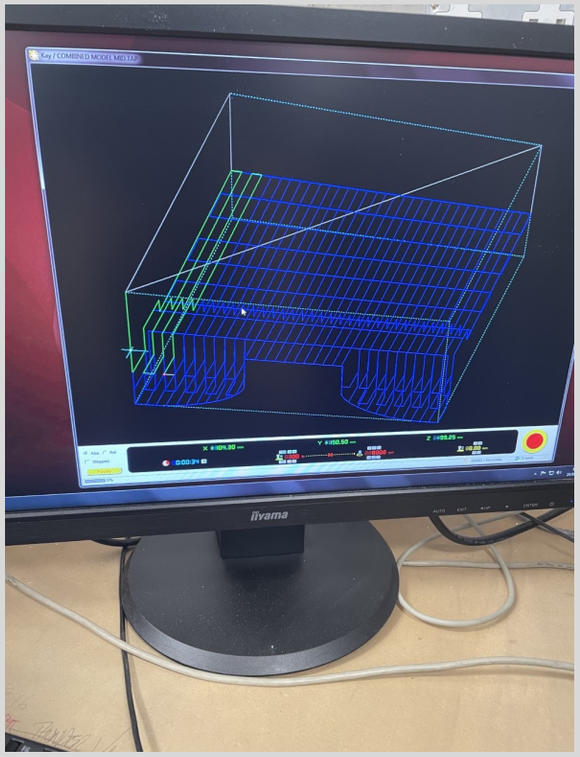
This was the simplest mechanism that I could come up with for a flusher ruler. It requires the use of a middle joint that allows it to bend while having the same 13mms height which allows it to support the board when closed.

However, an issue with this is that the joint is too unreliable as there is nothing controlling its range of motion meaning it might misalign and bend out of place when the board is closed thus potentially damaging both itself and the board.

The best solution to this would be to use a mechanism, such as gears with a pinion or a spring system which snaps it open. However, as this piece is extremely small, that would greatly increase the difficulty of this project as it would became even more complicated than the board's mechanism to produce.

Because of this, and as the other client did not complain about it when asked, I have decided to just use the previous design as, although not looking as flush, it was still small enough to not be annoying to the user and much easier to use and work with.

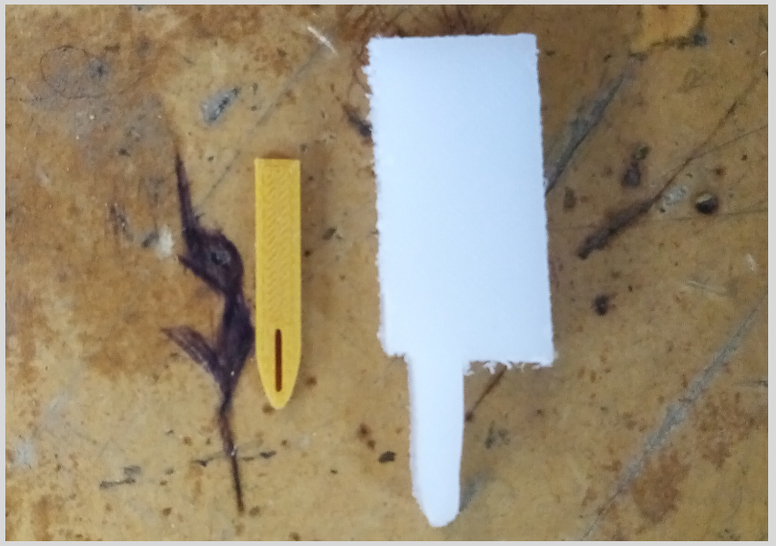
Testing The Fabrication Of Nylon:



Because of the restrictions of the CNC machine that I could use, The only part that I would be able to mill was the Inverse CAM of the mechanism. This means that I would have to cut the other part by hand, thus greatly raising the complexity of the design.

Unfortunately, the milling of the rounded shape did not turn out that well (as can be seen in the image below), probably due to the nylon deforming as it was heated because of friction, so it will have to be made in another way.

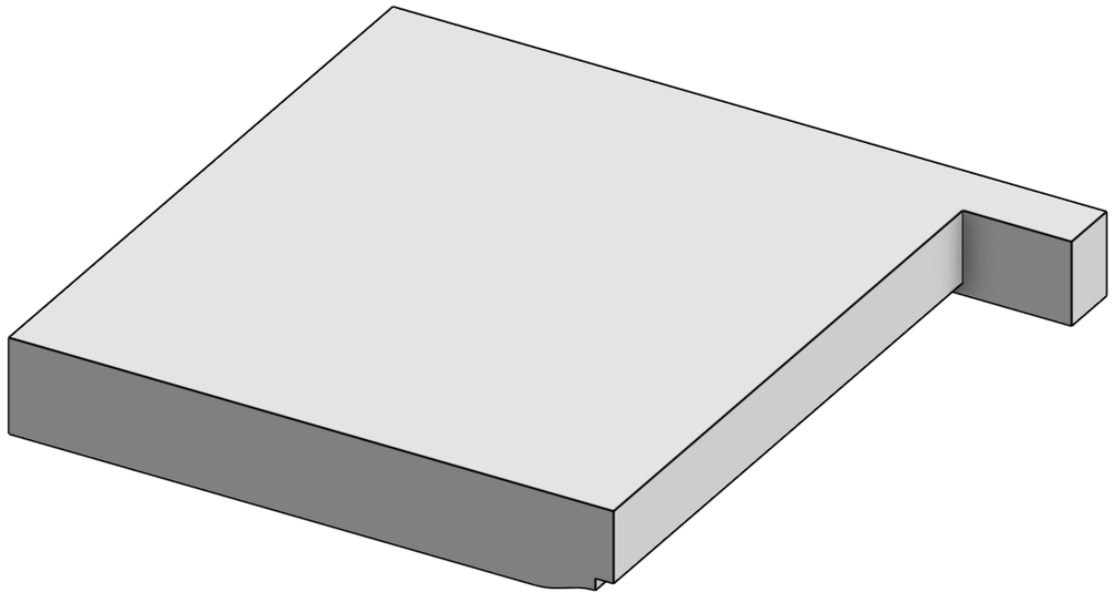
There are two solutions to these problems. I either cut and sand them being extremely careful to make sure that all the pieces are exactly the same, or I create a route for a multitool with a cylindrical filer to follow which will create the exact same shape for every piece (thus using a routing method). Both techniques were tested, as can be seen on the other side of the page.



The piece that I tried to cut out of nylon and sand did not come out. It worked a little bit however it would be very hard to make multiple pieces like this that all have the same shape to be used within the mechanism so I will have to test the routing method to see if it would work better.

Before making the piece, I had to 3D print the router for it using PLA which had a 3mm slot in it for the Dremel tool to follow. This router was made so that the 6mm circular sanding bit would follow the curve of the actual piece. The nylon piece was then firstly milled (as can be seen in top left image) in order to make it fit in the 3d printed piece.
This method ended up coming out with the piece being slightly rounded but still much better than the other two, however, the 3d printed piece melted so it did not come out perfect.

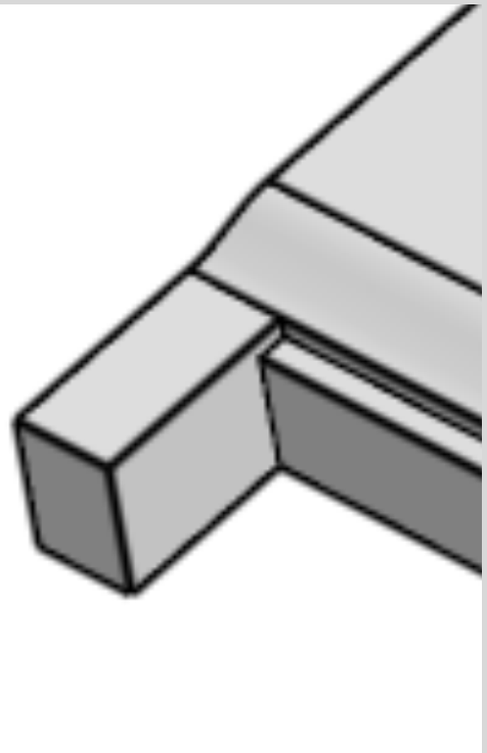
Testing The Fabrication Of Nylon 2:

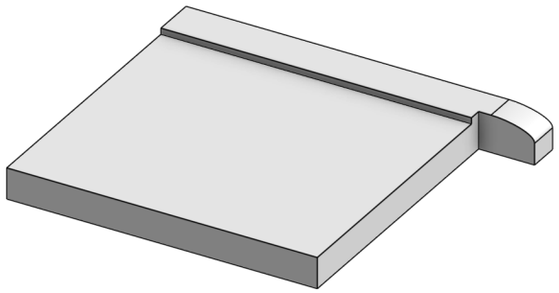


The four images below show the models which will be used to mill the nylon.

The ones on the left are the bottom pieces which will be milled first while the ones on the right are the top pieces.

The inverse cam has been removed to simplify the design as it is not actually needed for the mechanism. At the place of the actual cams, the mechanism just has a rectangle which will then be filed and sanded by hand to make the mechanism and the slot inside it will also need to be drilled by hand.

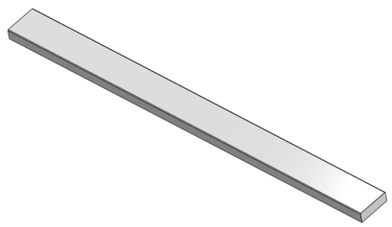


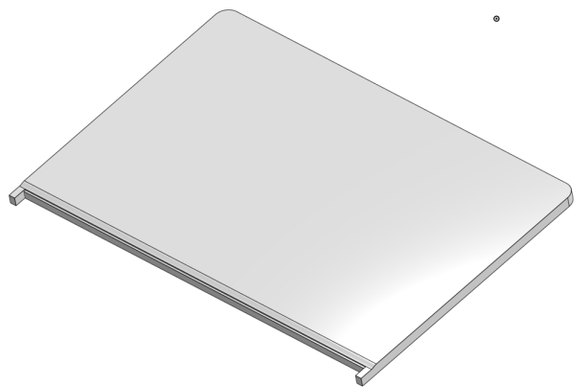


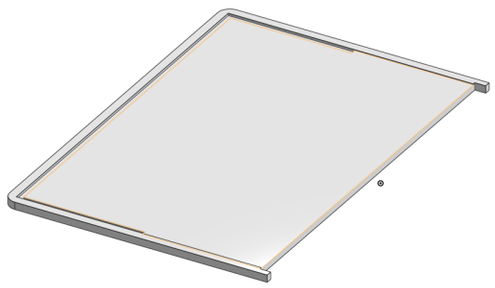
A solution I found to the problem regarding the milling of the piece was to do the underside first and the mill the top part so that I would not have to deal with overhangs.

Although this would have worked in theory, in practice, the piece did not come out as well as expected as can be seen on the bottom left image,

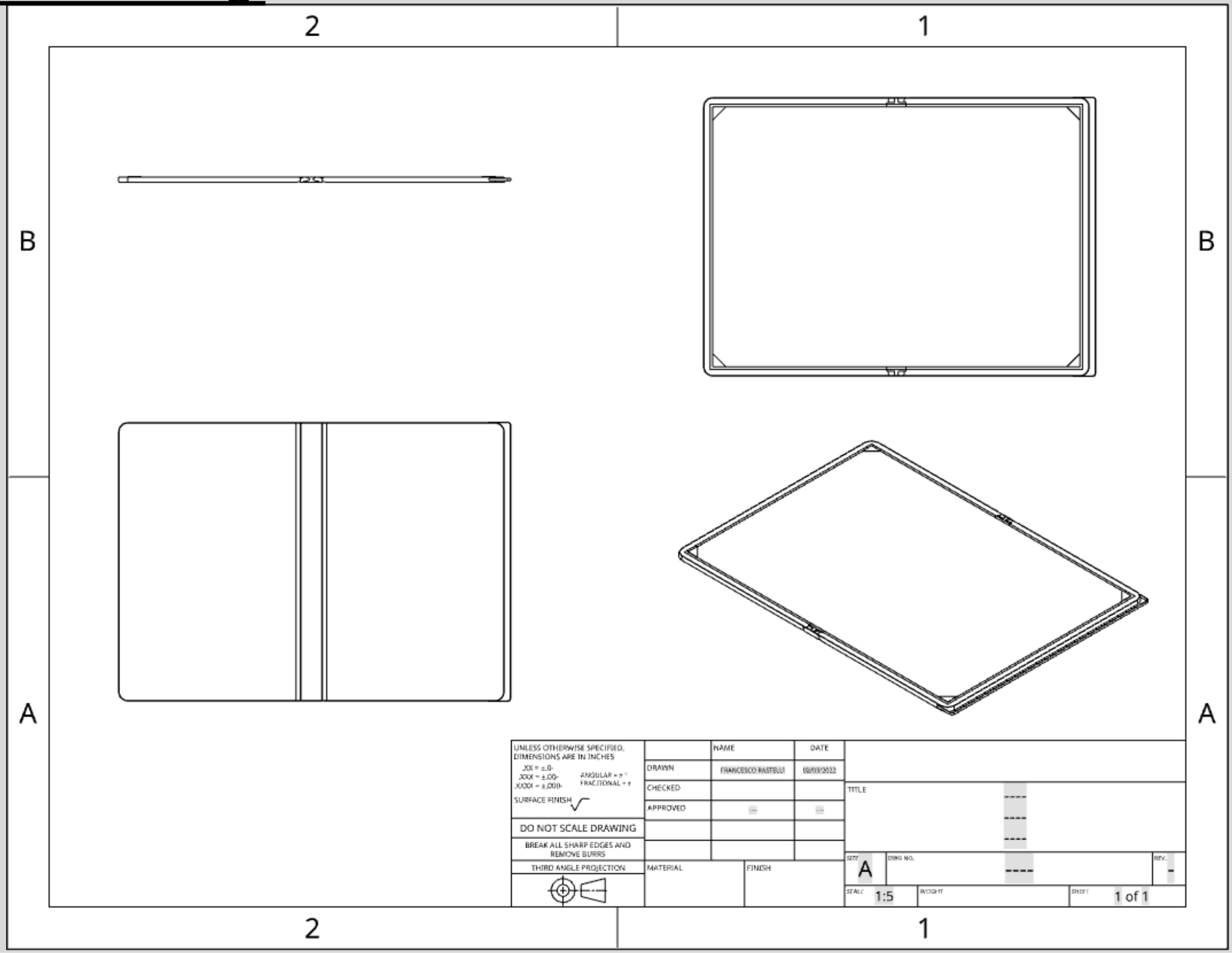
This means that milling would not work for the rounded edge of the mechanism meaning that it will need to be made by hand.



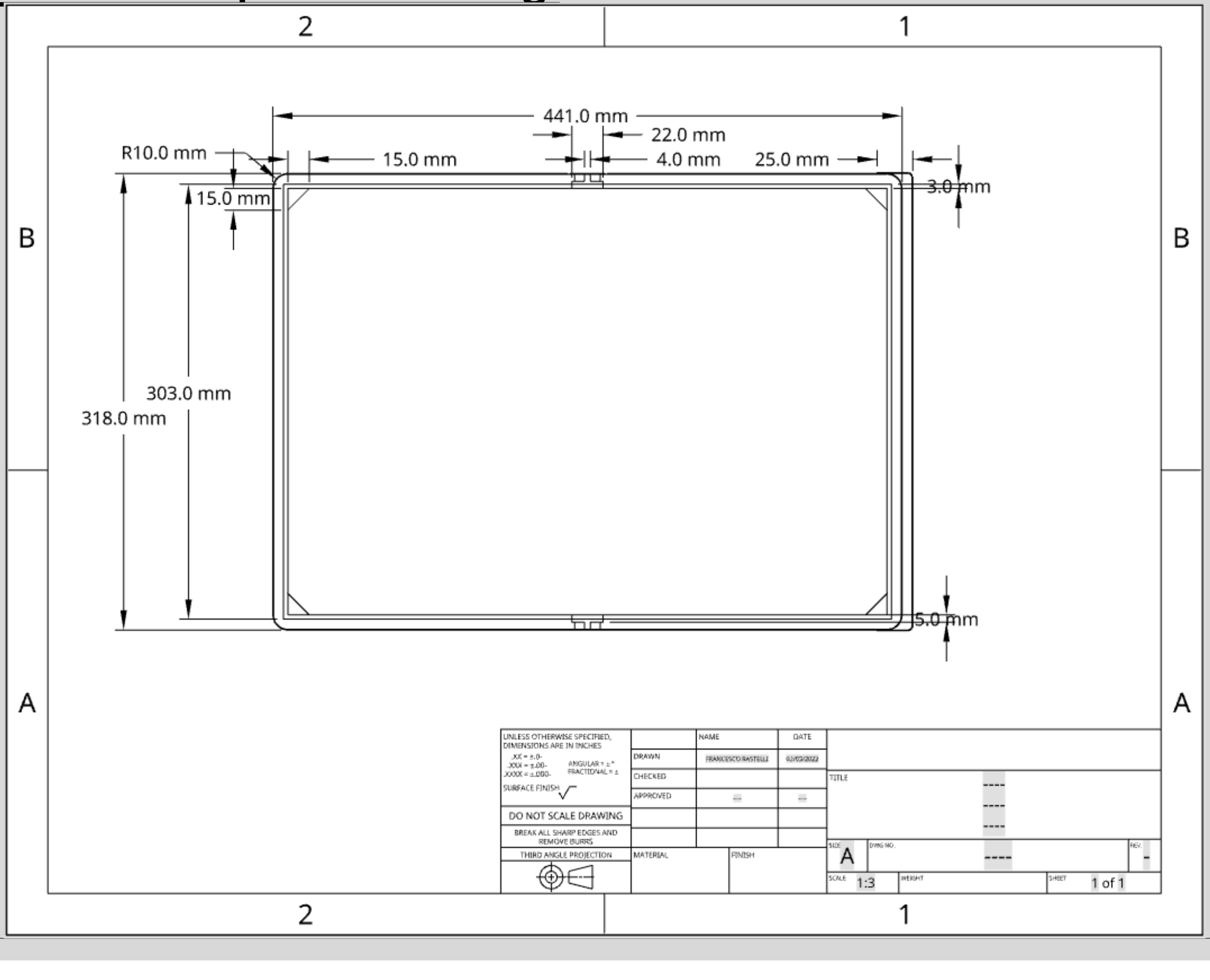




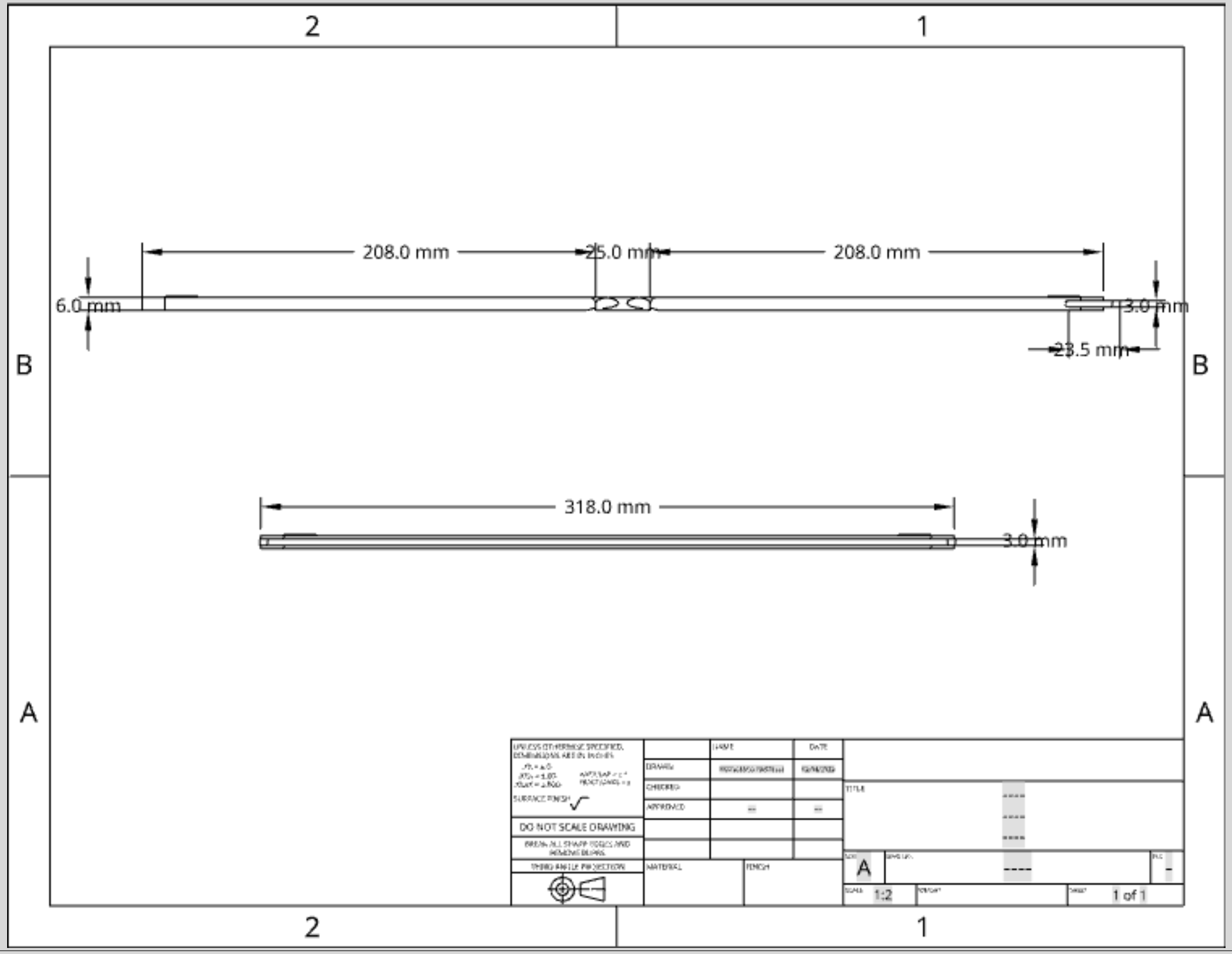
4 Views Drawing:



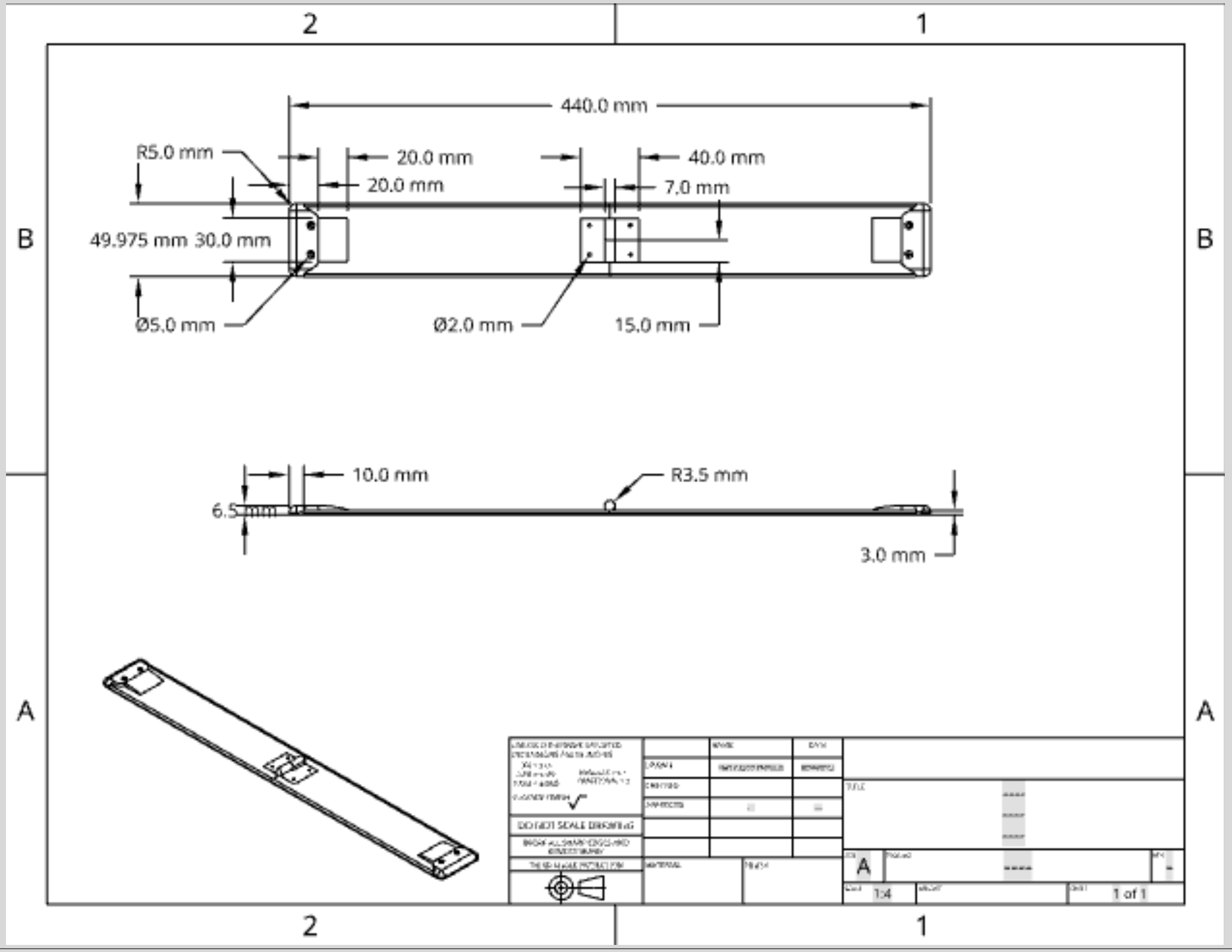
Dimensioned Top View Drawing:



Dimensioned Side And Front View Drawing:



Dimensioned Ruler Drawing:



Cutting List And Production Plan:

Cutting List:

Part	Quantity	Material	Dimensions (mm)
Middle Board	1	Solid Nylon in White	22x318x5
Sideboard	2	Solid Nylon in White	208x318x5
Aluminum Strips Long	2	Aluminum	750x5x0.5
Aluminum Strips Short	2	Aluminum	22x5x0.5
Steel Strips	6	Stainless Steel	303x3x1
Leather	1	Faux Leather in White	440x320x1
Bearing Rods	1	Tungsten Steel Rod	100x0.5R
Fabric Strip	1	Elastic Fabric	15x50 (as thin as possible)
Ruler End	2	Solid Nylon in White	40x50x6.5
Ruler Body	1	Transparent Acrylic	440x50x3

Production Plan:

Step:	Description:	Materials:	Tools	Expected Time:	Health And Safety
1.	Lower and smooth the surface of the Nylon board from 10mm to 6mm using the CNC machine and the 4mm bit.	Solid Nylon in White.	CNC Machine.	4 hours.	Make sure the cover is properly closed before starting the machine.
2.	Mill the side board's indentations using the 2mm bit and the rounding of the edges and of the indent.	Solid Nylon in White.	CNC Machine.	6 hours.	Make sure the cover is properly closed before starting the machine.
3.	Repeat Step 1 and 2 for the other sideboard.	N/A.	N/A.	N/A.	N/A.
4.	Lower and smooth the surface of the Nylon board from 10mm to 6mm using the CNC machine and the 4mm bit.	Solid Nylon in White.	CNC Machine.	3 hours.	Make sure the cover is properly closed before starting the machine.
5.	Mill the Mid board's indentations using the 2mm bit.	Solid Nylon in White.	CNC Machine.	2 hours.	Make sure the cover is properly closed before starting the machine.
6.	Smooth the edges of the board using lower grit sandpaper first (around 150 to 300) and then higher grit sandpaper after(600 to 1200).	N/A.	100 Grit Sandpaper. 600 Grit Sandpaper.	30mins.	N/A.

Production Plan 2:

Step:	Description:	Materials:	Tools	Expected Time:	Health And Safety
7.	Use a belt sander to sand out the rough shape of the inverse cam and then use a file and sandpaper to get it exact.	N/A.	Belt Sander. File. Sandpaper.	2 hours.	Wear goggles and tuck in your tie when using the belt sander. Be careful of kickback from the sander and of sanding yourself.
8.	Cut out the 1mm slot in the cam using a 1mm drill bit (and moving the Sideboard forwards and backwards.) and drill the holes in the Mid board.	N/A.	Pillar Drill.	45mins.	Wear goggles and tuck your tie in. Be careful of kickback from the drill.
9.	Add a chamfer to the edges of the boards to make it more comfortable to hold using a file and sandpaper.	N/A.	File. Sandpaper.	2 hours.	Be careful not to hurt yourself if the file skids off the board accidentally and results in you hitting something,
10.	Cut the Aluminum Strips to the correct lengths.	Aluminum.	Tin Snips.	10 minutes.	Be careful not to cut yourself with the tin snips or the sharp edges that occurs because of cutting the aluminum. Also be careful of the tinsnips clamping your fingers and of aluminum strips flinging off after being cut.
11.	Attach the Side Boards' aluminum strips to the edges of the Side Boards by sanding the nylon and aluminum edges with high grit sandpaper first and then low grit.	N/A.	Nylon Bonder. 100 Grit Sandpaper. 600 Grit Sandpaper.	50mins.	Wear gloves as to not damage your skin with the Nylon Bonder.
12.	Insert the Bearing Rods into the Mid Board through the Side Boards.	N/A.	Mallet.	10 minutes.	Be careful not to hit your fingers with the mallet.
13.	Attach the Mid Board's aluminum strips to the edges of the Mid Board by sanding the nylon and aluminum edges with high grit sandpaper first and then low grit.	N/A.	Nylon Bonder. 100 Grit Sandpaper. 600 Grit Sandpaper.	30mins.	Wear gloves as to not damage your skin with the Nylon Bonder.
14.	Cut the leather to the correct dimensions.	Leather.	Cutter.	45mins.	Be careful not to cut yourself with the Cutter.
15.	Saw the fabric into the leather and in the side.	Elastic Fabric.	Needle and Thread.	30mins.	Be careful not to injure yourself with the needle by accidentally stabbing yourself.
16.	Staple the leather onto the nylon.	N/A.	Stapler,	15mins.	N/A.
17.	Attach the Board's Steel strips to the indents of the Board using a similar method to the aluminum strips .	N/A.	Nylon Bonder. 100 Grit Sandpaper.	50mins.	Wear gloves as to not damage your skin with the Nylon Bonder.

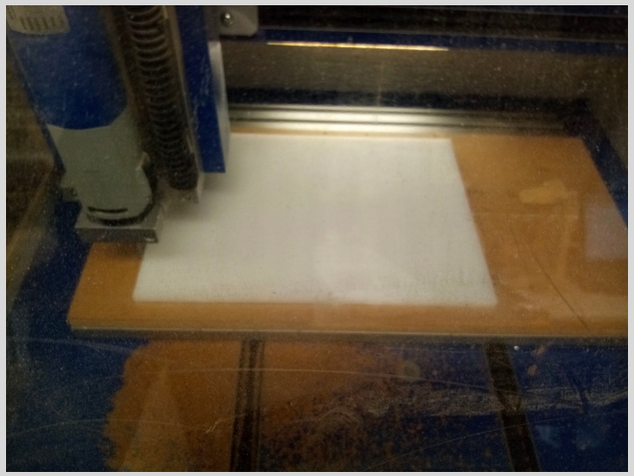
Production Plan 2:

Step:	Description:	Materials:	Tools	Expected Time:	Health And Safety
18.	Laser Cut the ruler's shape and mm markings.	Transparent Acrylic.	Laser Cutter.	15mins.	Make sure that the lid is closed and that the fumes disperse before removing the laser cut pieces so that you do not inhale any fumes.
19.	Add the chamfer to the ruler's edge using a 45° block and passing the acrylic on a belt sander. Then smooth the surface using high grit sandpaper.	N/A.	Belt sander. 1200 Grit Sandpaper.	30mins.	Wear goggles and tuck in your tie when using the belt sander. Be careful of kickback from the sander and of sanding yourself.
20.	Cut the indent for the hinge to go in using a handheld router.	N/A.	Handheld router.	15mins.	Wear goggles and tuck in your tie when using the handheld router. Do not touch the spinning cutter when it is running. Make sure that the acrylic is clamped down properly. Be careful of shards of acrylic being thrown away.
21.	Cut and sand the Nylon blocks to shape for the Handles.	Nylon.	Hacksaw. 100 Grit Sandpaper. 1200 Grit Sandpaper.	2 hours.	Make sure to tuck in your tie and be careful not to cut yourself with the Hacksaw's blade.
22.	Use a tap to create a screw inside the holes for the hinge and nylon block.	N/A.	Tap.	30mins.	N/A.
23.	Screw the handles and hinge to the ruler.	Screws.	Screwdriver.	20mins.	Be careful not to stab yourself with the screwdriver.
24.	Attach the magnetic strip in the gap between the handles and the acrylic.	Magnetic Strip.	N/A.	10mins.	N/A.

Quality Control:

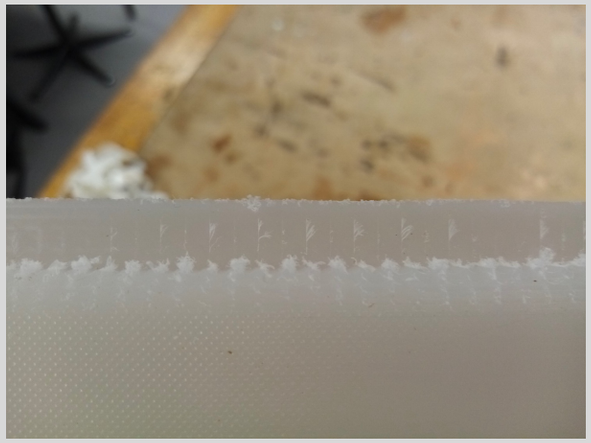
Control:	Tools:	What it is used for:
Height, width and length of pieces.	Ruler and Vernier Calipers.	Used to check the thickness of the board, and its dimensions to make sure that the pieces would fit together. Check the height and length of the metal strips so that they fit neatly around the side of the board without being too large or too small.
Check for the smoothness of the sanding.	Orbital Sander.	When sanding the surface, check for any bumps that need to be smoothed down more so that you can smooth them with the orbital sander.

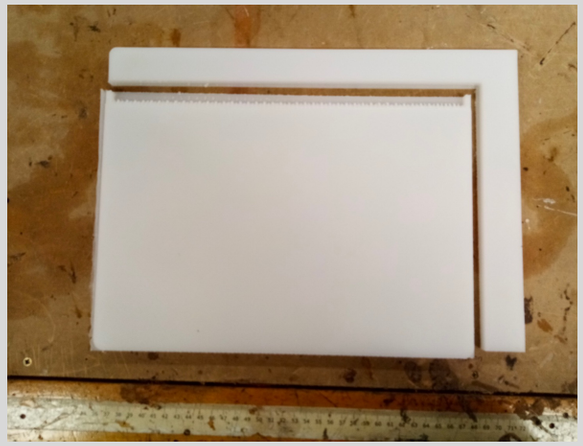
Diary Of Production 1:

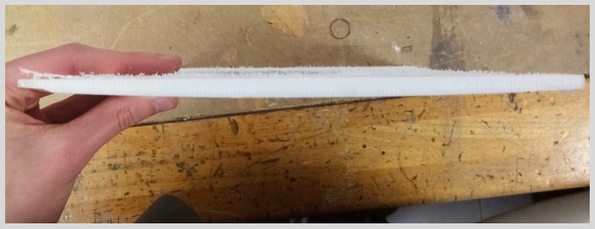


I started by milling one of the sideboard (as seen on the left image).

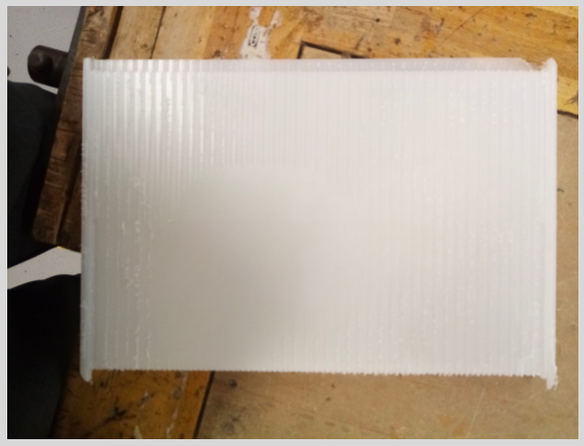
The piece came out okay as its top surface was not the flattest and the edges in the indent weren't smooth meaning that I would have to file them.
Before milling the other side, I cut away the excess using a bandsaw.

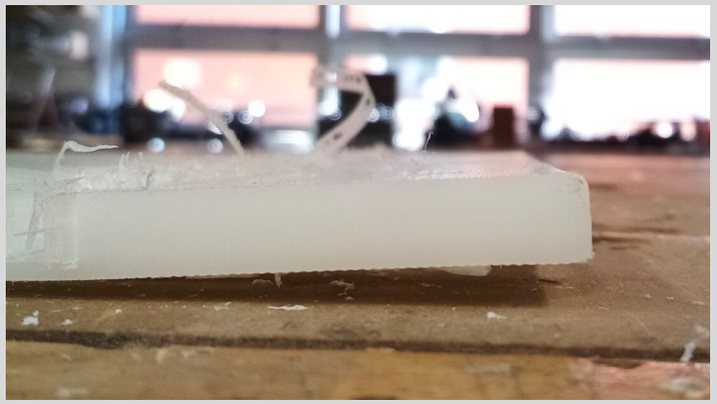




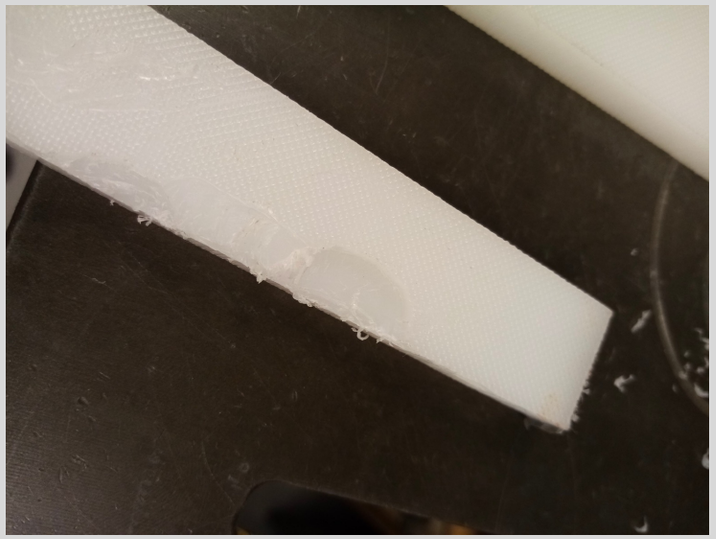


The milling of the other side did not come very well as the nylon warped cause it to be uneven and not smooth. The bottom left image shows how much of a curve the bottom had.





I tried to mill the nylon down from 10mm to 6, in a way which would allow me to cut the top side without having to worry about aligning the cams. However, When doing this the board warped so much due to the heat that the 4mm cut turned into 8mms at its extremities.



Therefore, this meant that I would need to route the indent by hand, so to try and do that I got a 20mm wide routing bit and put it on a pillar drill to try and pass the acrylic piece below it. However, that did not work very well as it was too hard to keep the nylon in place. Therefore, I went to go and use the handheld router, however, in the hour between setting it up and being able to use it, I had to come back to the plug being completely broken and unusable meaning that I was quickly running out of options of how I could build this.

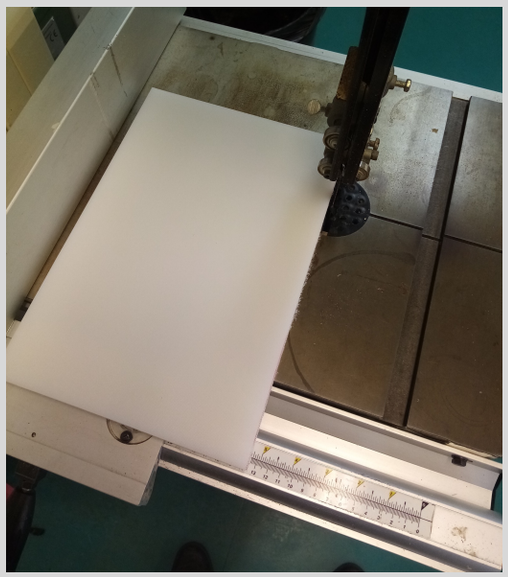
Diary Of Production 2:

I began by testing the handheld router after having fixed the plug by testing how I would produce the mid piece.

This started well until I started doing the second pass as I did not realise that the router would not have any support meaning that it dug out more than I needed.

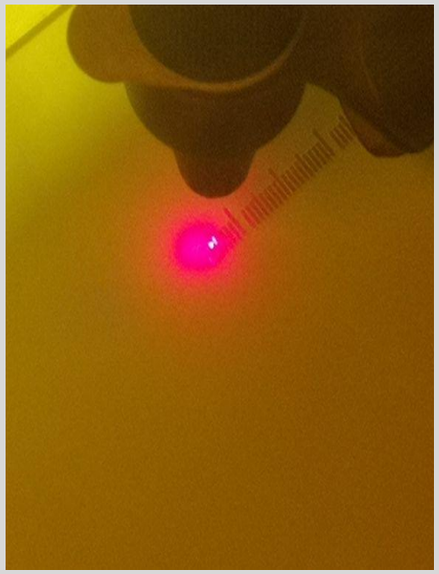
To fix this, I clamped two other pieces of the nylon to the side of it to support it in order to stop it from cutting as much.

Although this helped a lot, It was still a bit uneven so to fix it I used the orbital sander to smoothen its surface.

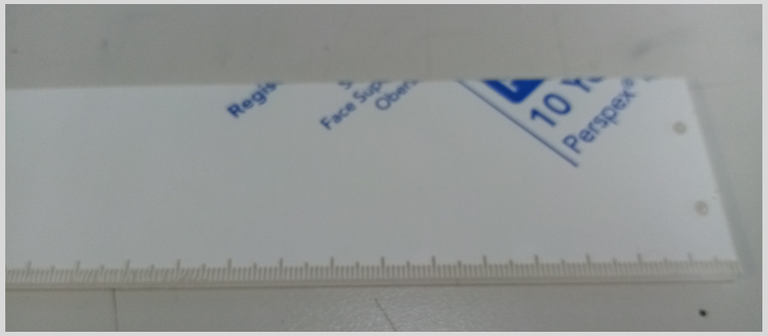


I then cut the first sideboard on the Bandsaw without the indent so that I could route it out.

I also built a jig for the routers flat edge to follow so that the board would have a 7mm border to it.

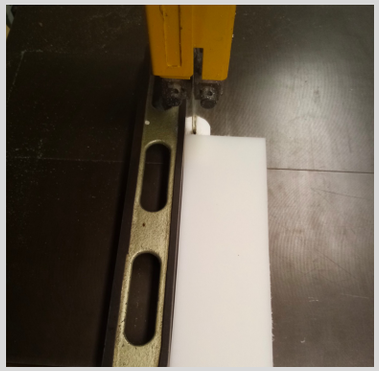


I then engraved and cut out the ruler using a Laser Cutter as I did not have access to the ruler at this point. It came out fairly well except for the screw holes which would need to be drilled out to the correct dimension, but the rest of the ruler was very accurate.

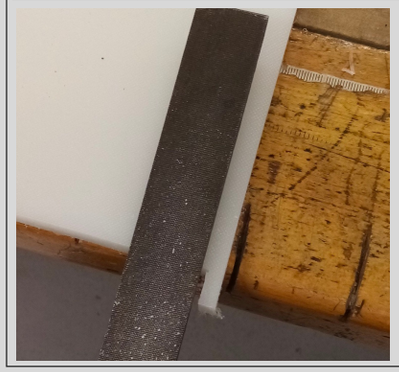


After this, started routing the side board's indent using the jig that I had built previously.

Diary Of Production 3:



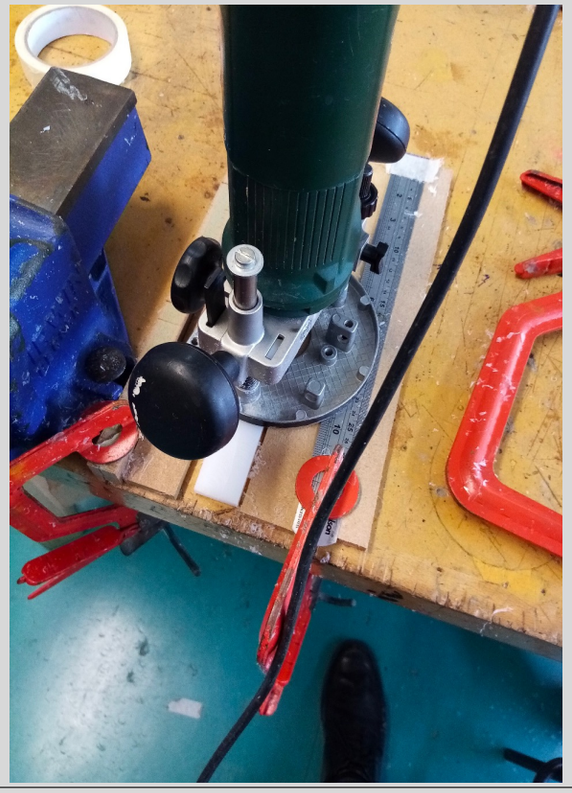
The next step was to finish the routing of the flat piece and to use an orbital sander to smooth the little bumps and imperfections that the routing had left.
After this, I cut a 7mm piece to be stuck at the end of the sideboard in order to have the indent on that side. I tried to use contact glue but ended up having to use hot glue as it did not hold the nylon together well.



After this, I filed the mechanism's Inverse CAM using a higher grit file to round it and to try and get it as close as I could on each side to make it smoother to use.

I decided to redo the middle piece as I was unhappy with how it had come out.

To start this, I began by cutting out the mechanism on a scroll saw and then I filed it down so that it was horizontal and could be attached to the inside piece.



I then cut the inside piece using a bandsaw and routed it to be a mm lower for the indentation and then used an orbital sander to smooth its surface as much as possible. I then attached to it the two T shapes using glue for the mechanism.

Diary Of Production 4:

The Mid Board with the first completed sideboard:

As the other Nylon still had not arrived, I was unable to build the other sideboard. Therefore, I used the available time to build the handles for the ruler.

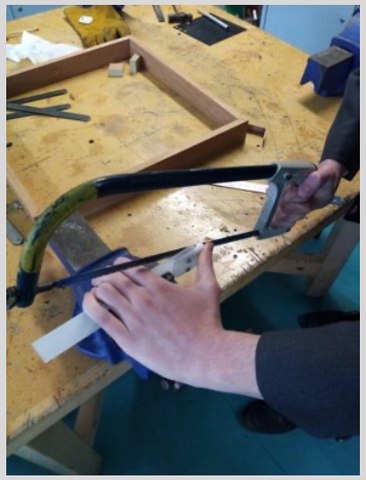
I began by cutting out the 50mm by 40mm rectangle for its frame and the 3mm indent on its longer side using a bandsaw.

I then started rounding the piece by passing it on the belt sander to round the edges and give it the rounder finish. Then, I used a craft Kinfe to cut off the excess flaking produced by the heat from the belt sander and then used sandpaper to smooth it and so make it much more comfortable to hold.

The next step was to drill the holes in the nylon for the screws using a pillar drill (as can be seen on the picture on the left).

I also had to widen the holes in the acrylic to the 3mm required so that the screws could fit in it.

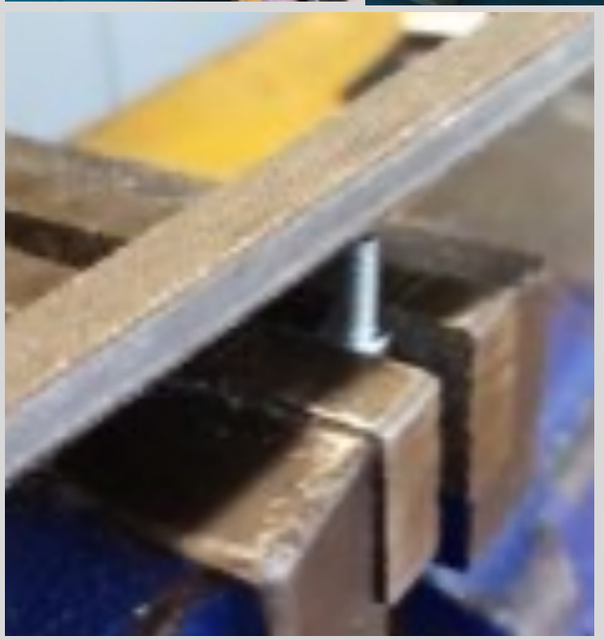
Diary Of Production 5:

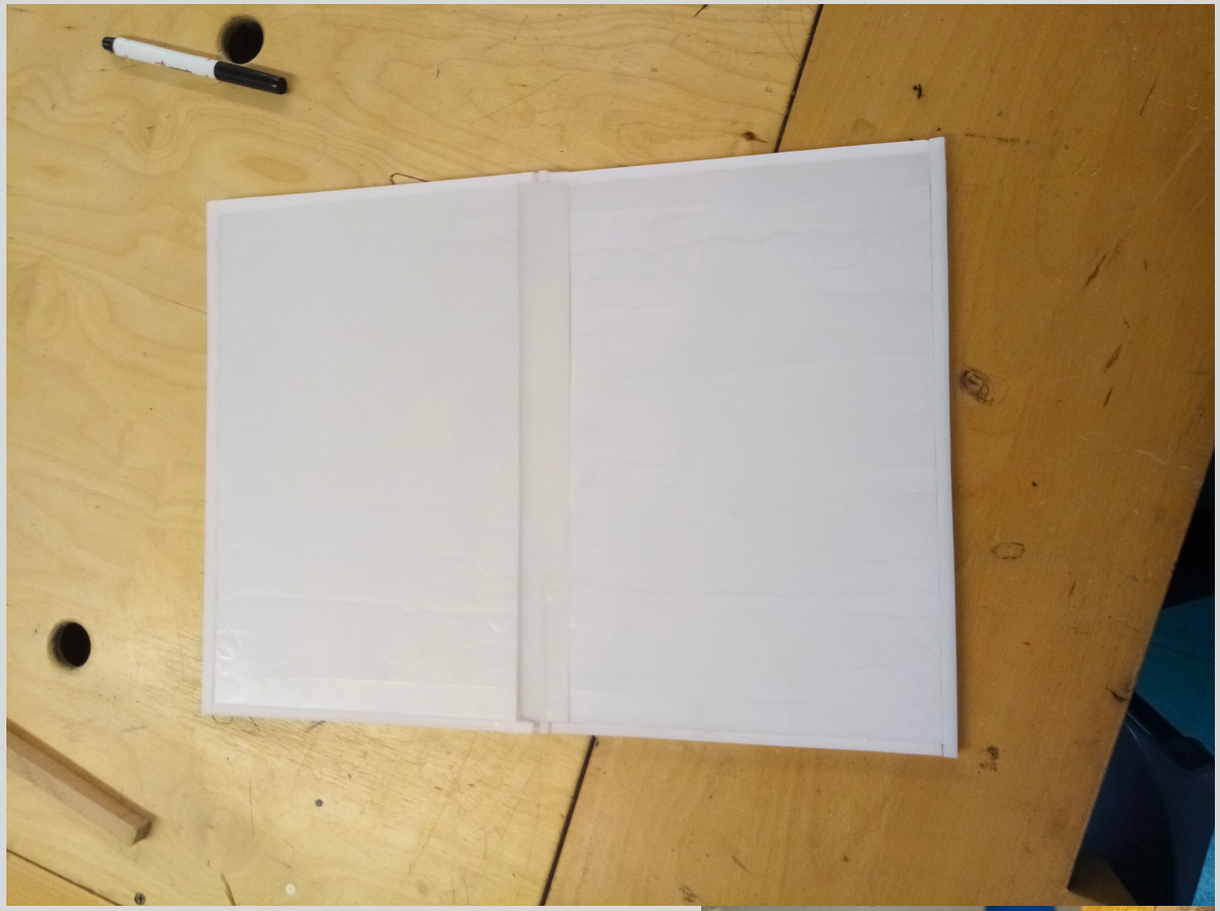


To screw the handles on, I had to use a tap and had to clamp the pieces to the table to hold them still.

The screw holes came out fairly well. However, as the shortest crews I could find were 16mm, I ended up having to cut the ends off using a hacksaw and then had to file them all by hand to make the ends smoother.

The bottom picture on the right shows how the handle ended looking on the acrylic (without the covering material having been ripped off yet). I was mostly happy with the finish even though the screws stuck out as I had not drilled a divot yet.





After this, the material for the second board had arrived, so I was able to build it using very similar methods to the first one, except that the sides were attached later in order to make the mechanism easier to build.

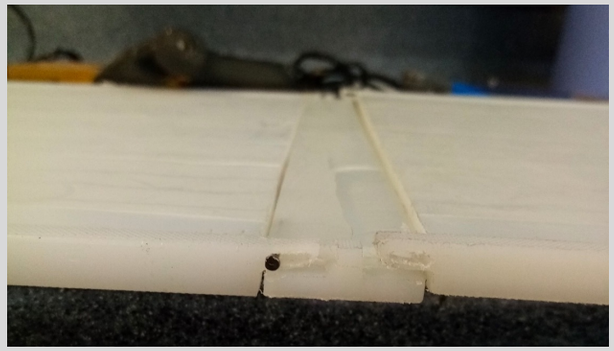
After this had been built, the next thing to do was to make the slots for the mechanism.

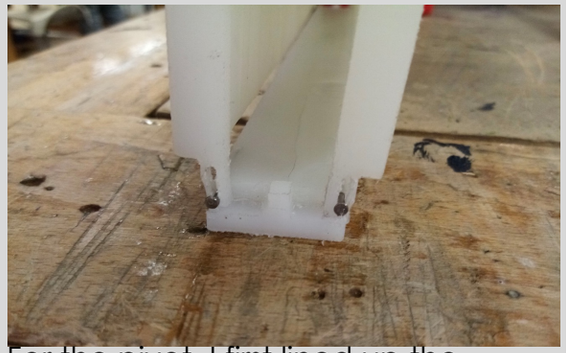
At first, I tried to use the pillar drill to drill these (using a 1mm drill bit). However, as the piece was too tall, it was too hard to hold the board straight. To fix this, I used a hand drill to drill every hole for each slot individually and then used a small saw blade to smooth the edges as much as possible by having it act like a file.

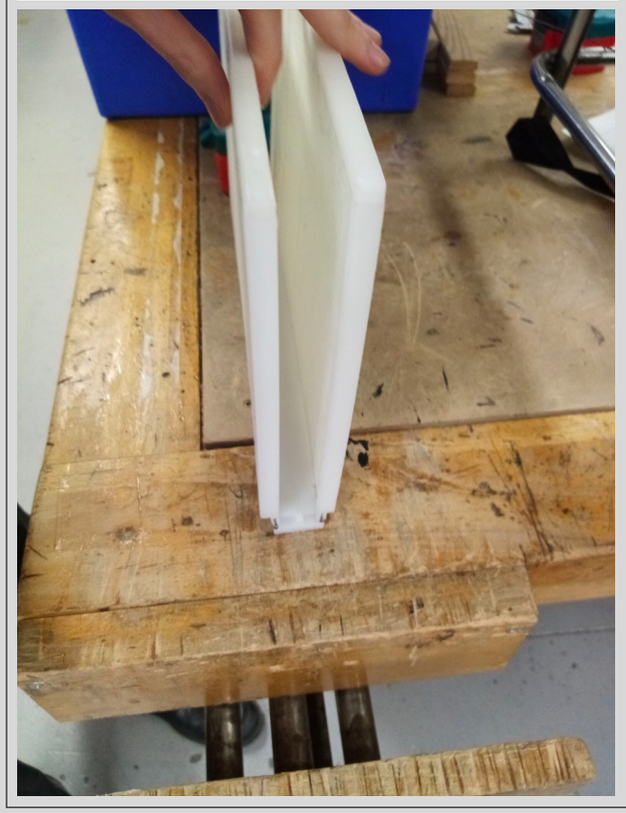
Diary Of Production 6:

As there were no 1mm rods, the next best option was to use 1mm thick nails, but before using them on the actual board, I wanted to test what they would actually work like and what was the best way of putting them in.

The best solution I found was to drill a small hole first and then put the nail inside it using a hammer to force it the rest of the way.



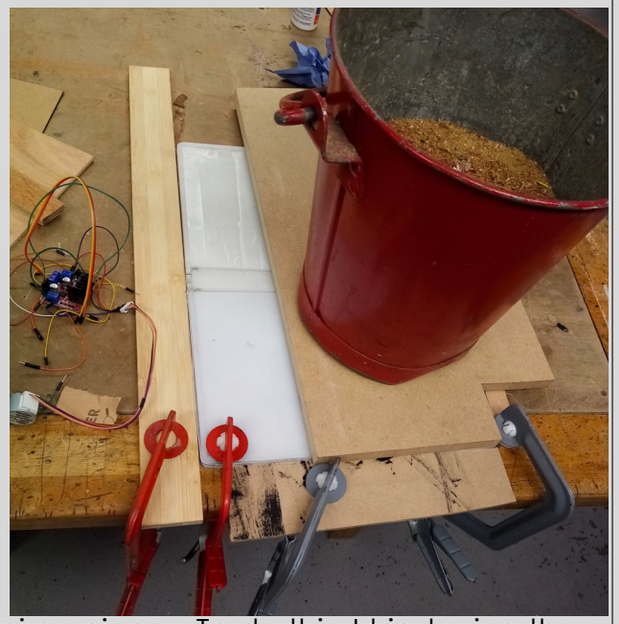




For the pivot, I first lined up the sideboard and middle board and clamped them together. Then, I placed the hand drill on the far side of the slot and drilled straight down for about 10 to 15mm. This gave me plenty of space in which to insert the nail in the nylon without it warping or cracking it weirdly.

This worked fairly well for the mechanism at first, as they held fairly well, however, as I got towards the end of the project, the nylon started to severely weaken and by the end most of them had snapped(as can be seen on the next page).

I had then realised that I had not rounded the corners yet, so I passed them on the belt sander.

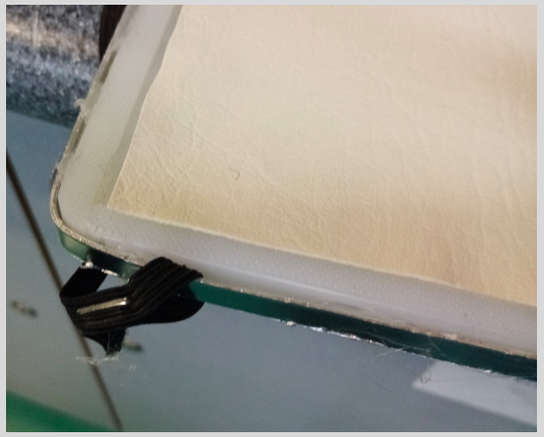


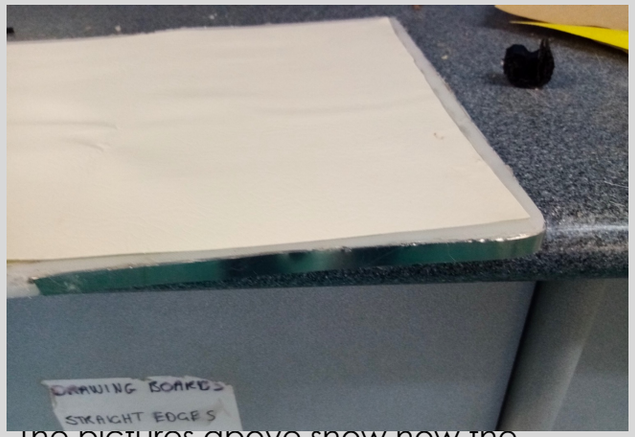
The next step was to make the side aluminium pieces. To do this, I tried using the guillotine to cut a 1mm strip following the blue line in the top left picture. However, this caused the piece to warp quite a bit and to not be too straight so to fix it I clamped it along the board and placed a heavy bucket on to try and shape it which worked relatively well and then using glue to attach it.

When I tried to use the same method for the 3mm steel strip, the guillotine cause it to warp and be cut really weird as can be seen in the bottom right picture. To fix this, I ended up using a bandsaw with a metal blade in it to try and cut a straight line. Although this worked much better than the guillotine, the pieces were cut were still not perfectly straight (going from 2mm to 5mm)

Diary Of Production 7:

I had found an elastic fabric that could be used as the holder to both keep the board closed (by going around the whole thing) and as the paper holder for the corners. For the large around bit, I stapled the fabric together so that it gave it more rigidity.

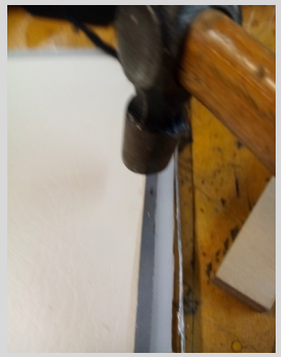




The pictures above show how the mechanism had snapped and what the black band that was supposed to go around the whole board. It also shows how the aluminium strip fits around the edge of the board.

This aluminium strip was at first glued on as it seemed to be the best way to bond a metal and nylon together. However, it did not end up holding very well, meaning that I ended up nailing it (placing 4 nails for each side) using the same method as for the steel strips, as can be seen on the right.

The leather was cut using a scissor by marking the underneath side first using a pencil and then cutting along that line. It was then glued on top of the board due to time constraints and as I was unable to make the indent that was supposed to clamp it.

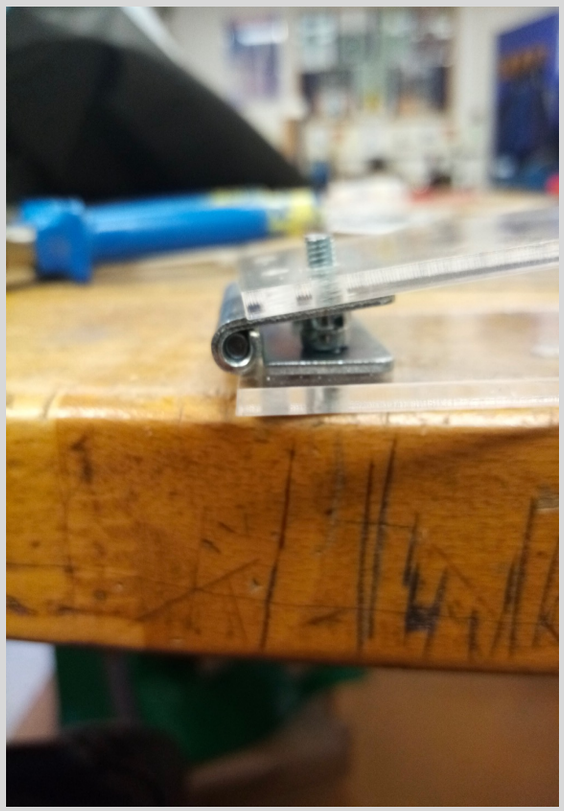


The steel strips were attached by first using a hand drill to drill the indents for the nails to go in. Those were then hammered in which in turned clamped the strips in place and the fabric as well.

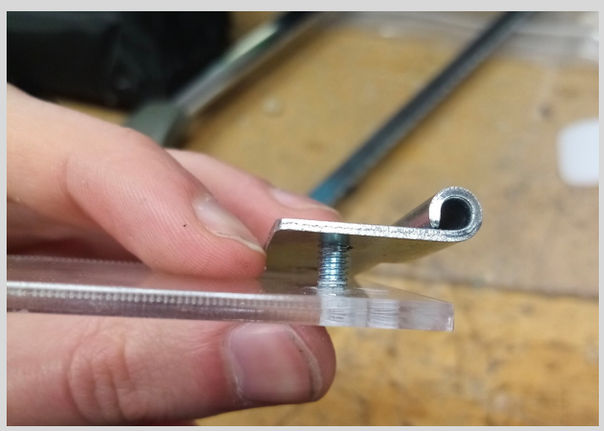
The picture below shows how the board looks with the steel strips and fabric underneath it before having it be sewn together to make the elastic corners to hold the paper.

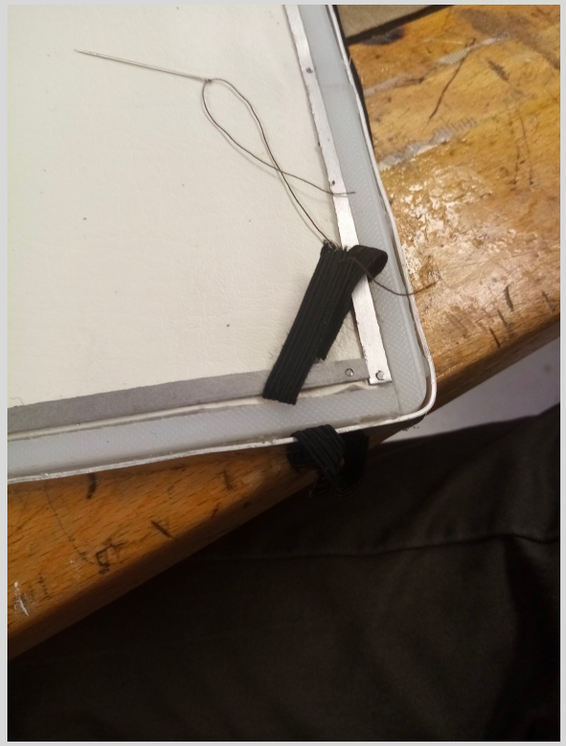
Diary Of Production 8:

As the main part of the board was finished, I decided to finish the ruler as well. Firstly, I added a chamfered hole for the top of the screws to go into the nylon using a 10mm drill bit and drilling around 3mm into the nylon.



After this, I screwed in the other screws and cut and filed the excess off to make the bottom smoother. However, I also noticed that when the ruler was closed the hinge screws would interfere and stop it from closing fully but as I did not have any thinner screws. I tried cutting the top of but the existing threads in the metal were not enough to hold it.

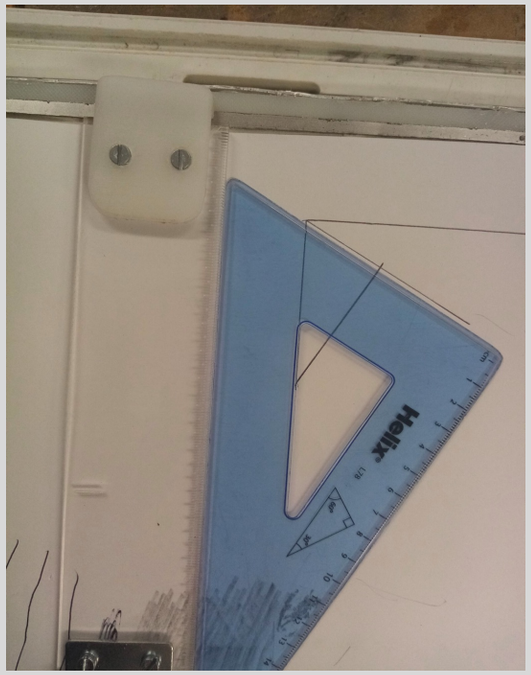


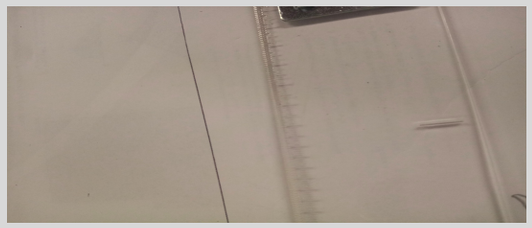


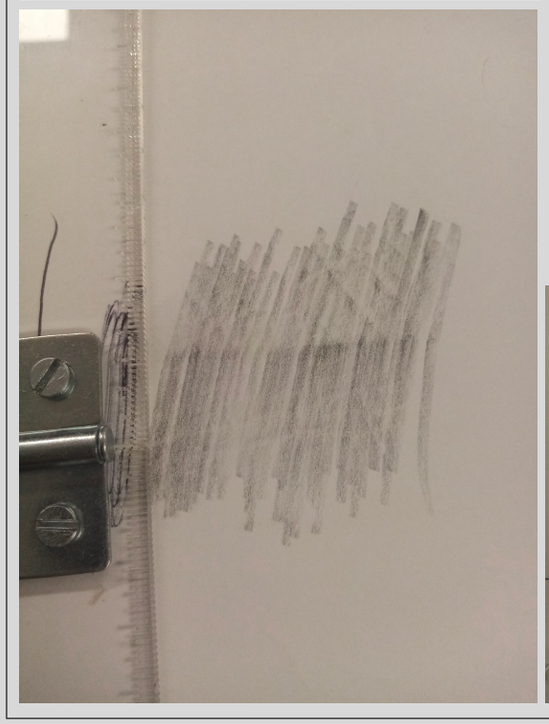
After this, I use needle and thread to saw the elastic bands on the corners to hold the paper.
I then cut and attached the magnetic strips to the bottom of the ruler using scissors.

Finished Product:

Testing And Evaluation 1 – Testing The Product:







Testing the drawing aspect:

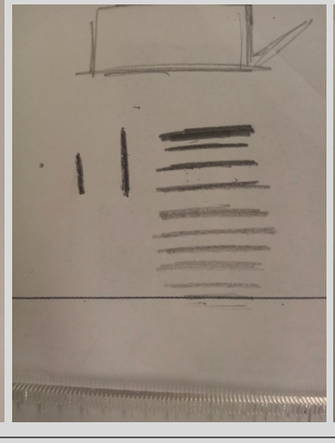
To do this, I got different people to test drawing different things on the board. These ranged from stuff as simple as straight lines and using a setsquare, to shading and trying to draw lines of different weights.

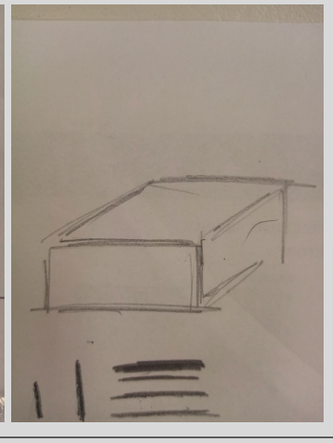
Although the drawing part worked relatively well in most circumstances, as the leather smoothed most of the imperfections of the board, there were some problems.

The main one was around the joint where there was a substantial drop between the sideboard and mid board, which was noticeable when drawing. This occurred due to the original warping of the nylon and so could not be fixed without replacing the whole piece.

Another issue, which I was expecting from the original leather tests, is that when you tried to draw with too much force, the paper would warp as the leather was not strong enough of a backing.

The general consensus for drawing on the overall board, however, was that the board was nice to draw on without too much struggle as long as you did not try to use too sharp of a pencil or too much force.



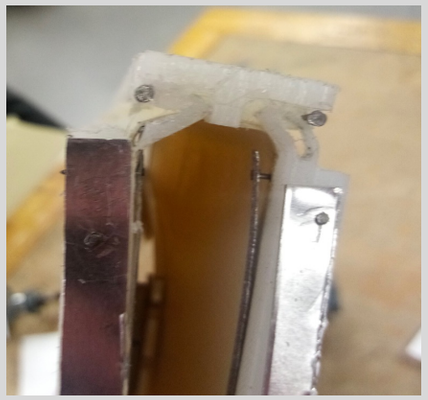


Testing the folding aspect:

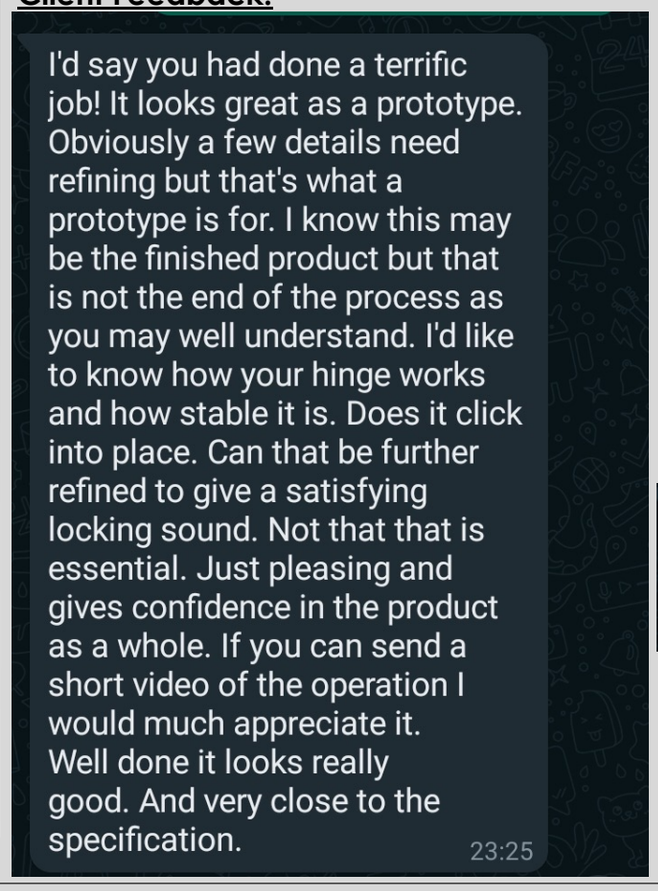
The board does not bend too well after a few uses as the nylon is just too weak and so deformed and snapped, as can be seen on the picture on the left.

However, it does fit into a backpack so at least that aspect of it works, even if it would probably break with the current prototype.

To fix these issues, I would both change the design of the mechanism and use metal for it, probably aluminium, to ensure that it would not snap.

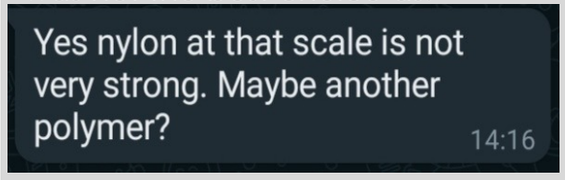


Client Feedback:



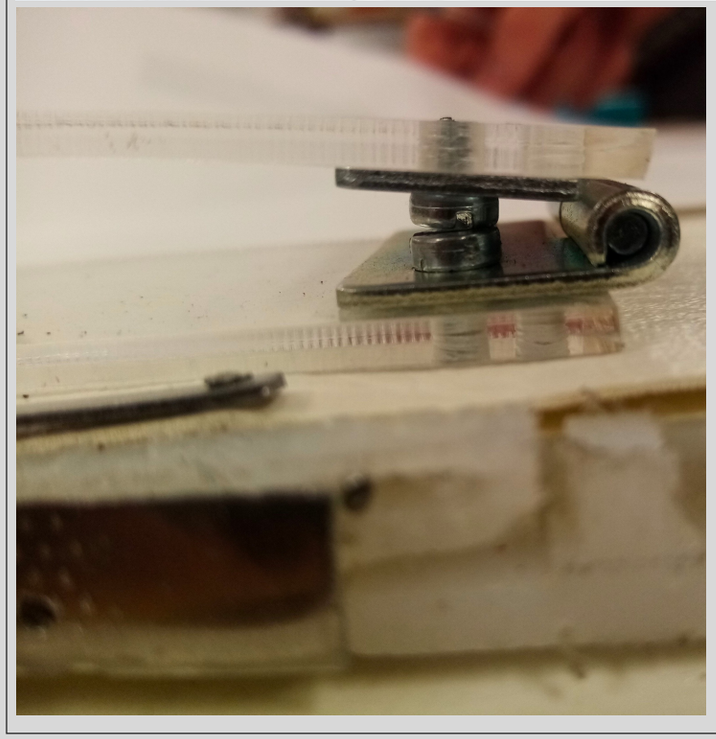
The client feedback was overall very good. Although he was unable to test the prototype, his rection to the various images (being all the ones from the Finished Product, Testing The Product, and Quality Control pages), was very good.

He did not have too many complaints at the start, however, after explaining the issues with the mechanism, he agreed that nylon was not the best choice as:



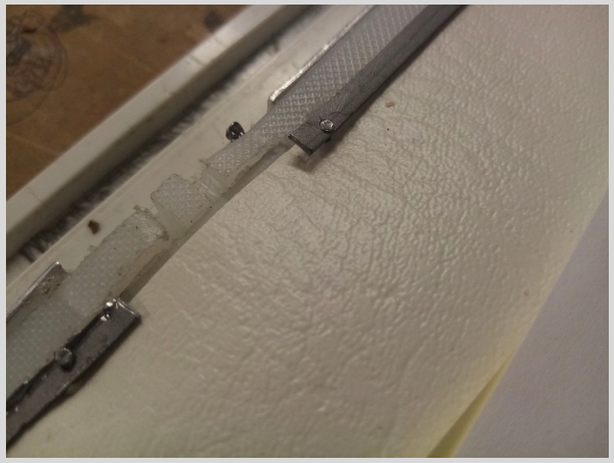
However, after explaining the redesign of the product to him he did not have any other issues to comment on and was happy with it. This can be seen in the last pages.

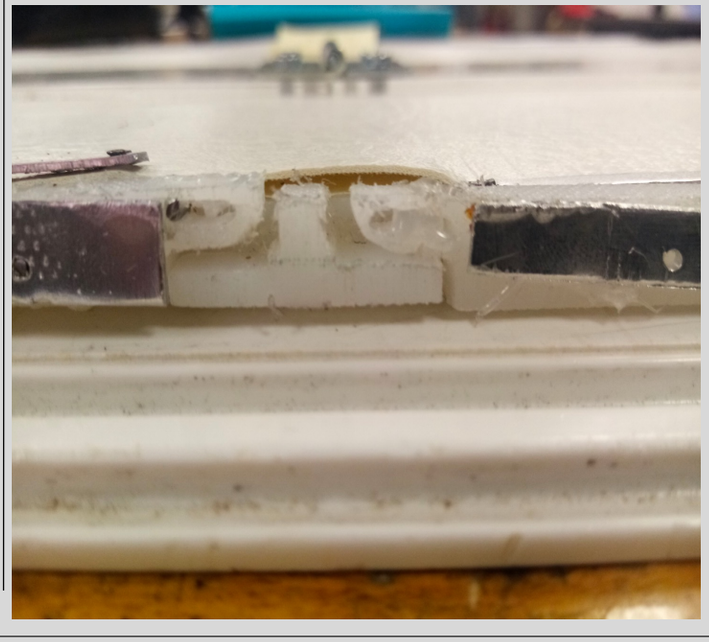
Testing And Evaluation 2 – Quality Control:



The top picture shows how an A3 sized paper fits inside the board. Although it is not a perfect fit, it still mostly fits and is held in place well.

The picture on the left shows how the screws are stopping the ruler from closing fully. The reason for this is that the screws are too tall, so the only way to fix this would be to use shorter screws, which the school did not have available and due to time constraints, I was unable to order any.





Another problem was that the steel strips would not sit perfectly flat and stick outwards and due to their sharp edges potentially injure someone. This warping mainly occurred due to the bandsaw, so buying pre-cut strips might help solve this as they would be flat.

I also was unable to put ther steel strips in the middle part of the board as it was too weak to hold them.

Other issues with the quality were the steel and aluminium strips which were not perfectly straight, with the steel strips ranging from 3mm to 8mm of width (as shown on the pictures on the left). This is due to the bandsaw not cutting perfectly straight so a more aligned bandsaw might work better for that. The nylon's thickness also varied from 8mm to 12mm along its length and it was seriously warped which greatly impacted the quality, however, some higher quality nylon might help remedy this.

Testing And Evaluation 3 – Comparing To Brief And Specification:

Design Brief:

A foldable A3 Drawing Board which can be stored inside a backpack. Its design must follow the 10 principles of good design meaning that it must be: minimalistic, unobtrusive, easy to use, useful, innovative, and aesthetically pleasing (The prototype fits most of these, however, due to the nylon snapping, it can be argued that it has made it not easy to use. However, replacing the nylon mechanism with aluminum should fix this.).

To make it unobtrusive, the design must be simplistic and thus minimalistic (The design is very simple overall, even if some aspects, such as the mechanism, are complicated,).

To make it easy to use, the mechanism must not be too complicated and if anything in it needs to be interacted with by the user (The mechanism is not too hard for the user to use as it should work is by having them twist the board), it should be easy to access and grasp or push (The overall design isn't hard to grasp, but the sharp edges of the metal hinder this in the prototype).

To make it useful, the board should remain rigid when opened but should also stay shut when it is closed (The Board does stay closed due to the elastic band and opened and flat when on a solid surface).

The design should aim to use materials which are eco-friendly and so can be recycled, reused, or are compostable (All materials used are recyclable).

The design should be able to fit into most backpacks when closed, so it should aim to be around A4 sized (It fits inside the backpack when tested).

Lastly, the design's materials should be as long lasting as possible and should attempt to not use any fixing methods such as glue to make it easier to recycle (The Nylon snapped, and some of the nylon was glued together, however, the rest of the pieces do not use permanent fixings).

Specification:

Size and weight:

1. The A3 foldable drawing board must be light enough so that a user can easily carry it without having to exert too much force or putting too much effort into it. ✓
2. The A3 foldable drawing board must be able to fit inside a backpack neatly and easily when closed, so it must aim to be slightly larger than an A4 paper at most and must not be thicker than 3cm. **X – Thicker then 3cms due to the nylon breaking.**

Function:

1. The A3 foldable drawing board must fold neatly and easily so that it is easier to use and will not occupy too much space in the consumer's backpack. **X – Due to nylon breaking, it is now too thick.**
2. The A3 foldable drawing board must be rigid when opened so that it is easier to draw on. ✓
3. The A3 foldable drawing board must not open by itself accidentally so that it does not open in a backpack and damage anything inside it. ✓
4. The A3 foldable drawing board must be strong and resistant so that it's lifetime increases. **X – Nylon Broke**
1. The A3 foldable drawing board must have all the normal functions of an A3 drawing board so that it can be used like a normal A3 drawing board. ✓
2. The A3 foldable drawing board must be able to store some accessories as the clientele have asked for that. **X – The final design didn't include that.**
3. The A3 foldable drawing board should be comfortable to use so that the user enjoys using it. **X – Although comfortable to draw on, the metal pieces have some sharp edges.**

Cost:

1. The A3 foldable drawing board must be affordable to the user. ✓
2. The A3 foldable drawing board should aim to be easy to disassemble so that broken parts can be replaced in it more easily. ✓

Materials:

1. The A3 foldable drawing board should aim use materials which are recyclable or reusable. ✓
2. The A3 foldable drawing board must use materials which are comfortable for the user to interact with in both their haptic response and visual response. **X – Metal pieces have some sharp edges.**
3. The A3 foldable drawing board should aim to use the least number of different materials so that at its end of life, it is much easier to recycle without creating a lot of different types of waste. ✓
4. The A3 foldable drawing board should aim to not use glues to connect pieces to it. **X – The nylon has some glue holding it together, but the metal ended up being nailed to the board.**
5. The A3 foldable drawing board must use lighter materials in order to reduce the weight of the board to make it more comfortable to use. ✓

Aesthetics:

1. The A3 foldable drawing board must have a minimalistic design. ✓
2. The A3 foldable drawing board must have a simple colour pallet. ✓
3. The A3 foldable drawing board must have an unobtrusive design. ✓
4. The A3 foldable drawing board must have an eye pleasing design. ✓
5. Parts the user interact with should have contrasting colours. **X – All lighter and similar colours except for the elastic bands.**

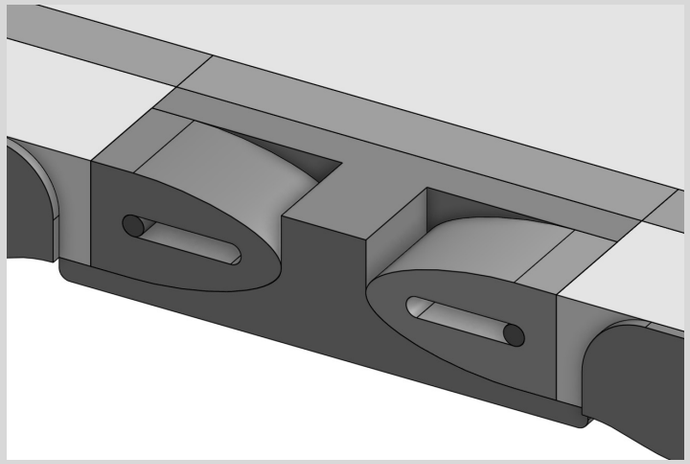
Maintenance:

1. The A3 foldable drawing board must be easy to disassemble so that pieces can be replaced by the user. ✓
2. The A3 foldable drawing board should aim to not use glues to attach pieces to it so that it is easier to replace pieces within it. **X – The nylon has some glue holding it together, but the metal ended up being nailed to the board.**

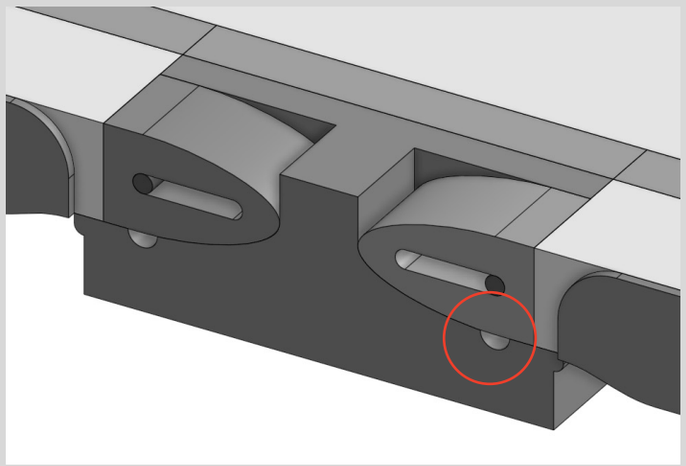
End of life:

1. The A3 foldable drawing board should aim to not use any materials which cannot be reused or recycled. ✓
2. The A3 foldable drawing board must be easy to disassemble so that different pieces can be recycled more easily. ✓
3. The A3 foldable drawing board should aim to not use glues to connect pieces to it so that when it is disassembled to be recycled, it is much easier to do so. **X – The nylon has some glue holding it together, but the metal ended up being nailed to the board.**

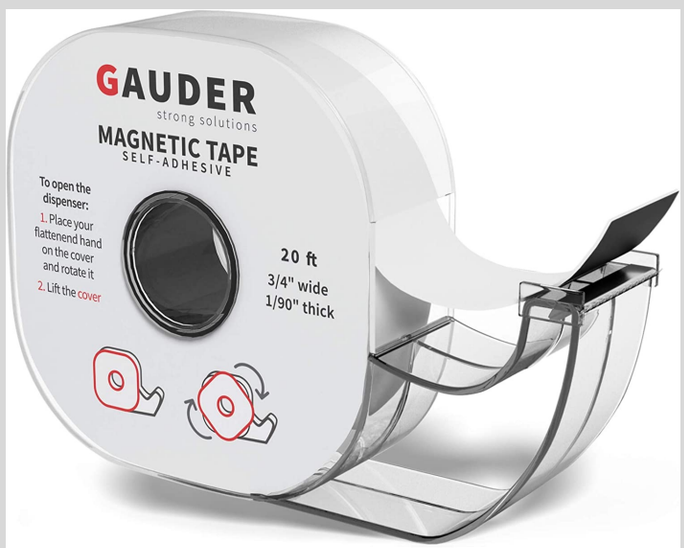
Testing And Evaluation 4 – Redesign:



Joint:
The main issue with the product was the joint being too weak due to the nylon snapping. To fix this, I decided to change the material at the joints of both the Larger CAM and Inverse CAM to Aluminium which can then be fixed on using screws on slots, much like the original wooden design.



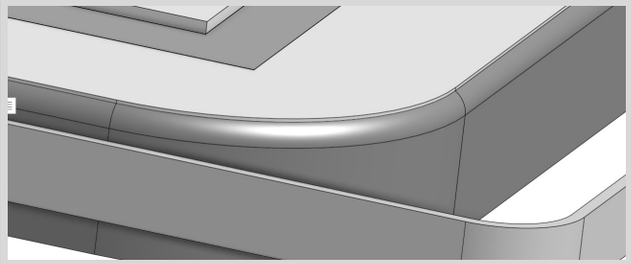
This design was then developed by increasing its height so that I could include a small slot into the bottom of the curve so that the aluminium would slot into it making a slight clicking sound, as the client requested. The increase of thickness was so that the design would be much stronger.



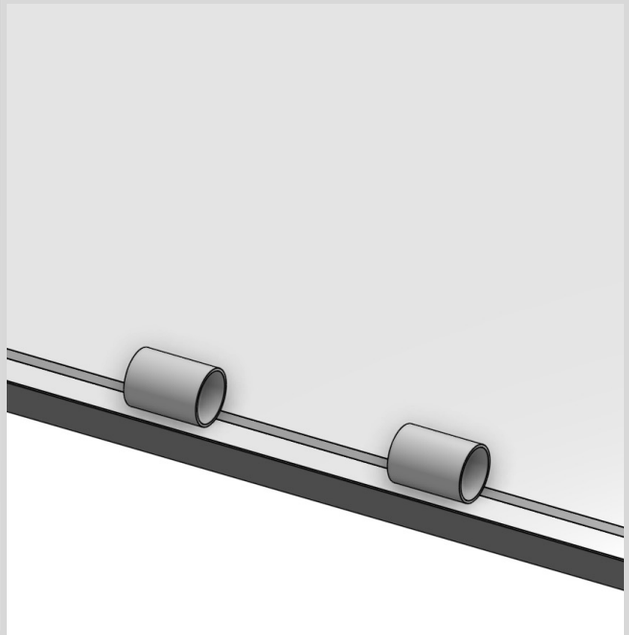
Changing the Steel Strips:
The steel strips did not work very well when attached on the bord. They are too rigid and cannot be fixed with enough force meaning that they come out really easily when the board is bent due to the leather pulling on them. To fix this, I could attach a flexible and thin magnetic tape which will also work better with the ruler or use steel foil strips (between 0.05mm and 0.8mm) which are flexible and so can be bent easily.

Material of Board:
The material of the board should definitely change from nylon. Although it was supposed to be a good material for this project, it has too many flaws, ranging from its quick yellowing to how easily it snapped.

I decided that I had to change it, originally, I was thinking of going back to the wooden design, however, I enjoyed the colour customizability of the nylon design, so I decided to use another plastic which was still recyclable. That being Acrylic.



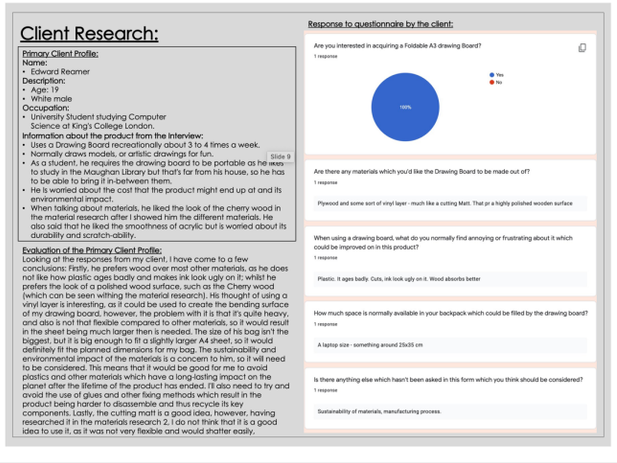
Rounding Of Edges:
To help with having a nicer feel on the edges, I decided that it would be good to give it a small chamfer in order to make them smoother and stop them from having as many sharp edges as before.



Pen Holder:
I decided that it would be a good idea to re-introduce the pen holder idea.

Originally, I had to remove it in my wooden design as I did not have anywhere where it would nicely fit as I had much smaller dimensions. However, the nylon uses a much thicker and larger frame meaning that there would be space to include two simple hoops of elastic or another similar method that would be able to store two pens as required by the client.

Testing And Evaluation 5 – Evaluation of the whole project:



Client Research:

Primary Client Profile:
Name:
- Edward Reamer

Description:
- Age: 19
- White male

Occupation:
- University Student studying Computer Science at King's College London.

Information about the product from the interview:
- Uses a Drawing Board recreationally about 3 to 4 times a week.
- Normally draws models, or artistic drawings for fun.
- As a student, he requires the drawing board to be portable as he needs to study in the Maughan Library but that's far from his house, so he has to be able to bring it in-between them.
- He is worried about the cost that the product might end up at and its environmental impact.
- When talking about materials, he liked the look of the cherry wood in the material research after I showed him the different materials. He also said that he liked the smoothness of acrylic but is worried about its durability and scratch-ability.

Evaluation of the Primary Client Profile:
Looking at the responses from my client, I have come to a few conclusions: Firstly, he prefers wood over most other materials, as he does not like how plastic ages badly and makes ink look ugly on it; whilst he prefers the look of a polished wood surface, such as the Cherry wood (which can be seen withing the material research). His thought of using a vinyl layer is interesting, as it could be used to create the bending surface of my drawing board, however, the problem with it is that it's quite heavy, and also is not that flexible compared to other materials, so it would result in the sheet being much larger then is needed. The size of his bag isn't the biggest, but it is big enough to fit a slightly larger A4 sheet, so it would definitely fit the planned dimensions for my bag. The sustainability and environmental impact of the materials is a concern to him, so it will need to be considered. This means that it would be good for me to avoid plastics and other materials which have a long-lasting impact on the planet after the lifetime of the product has ended. I'll also need to try and avoid the use of glues and other fixing methods which result in the product being harder to disassemble and thus recycle its key components. Lastly, the cutting matt is a good idea, however, having researched it in the materials research 2, I do not think that it is a good idea to use it, as it was not very flexible and would shatter easily.

Response to questionnaire by the client:

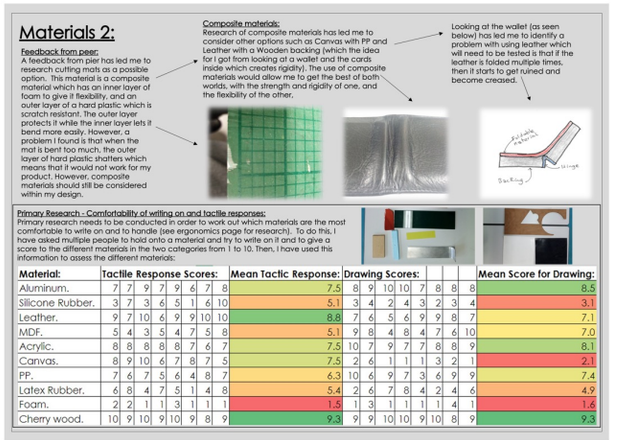
Are you interested in acquiring a Foldable A3 drawing Board? — 1 response — Yes 100%

Are there any materials which you'd like the Drawing Board to be made out of? — 1 response — Plywood and some sort of vinyl layer - much like a cutting Matt. That pr a highly polished wooden surface

When using a drawing board, what do you normally find annoying or frustrating about it which could be improved on in this product? — 1 response — Plastic. It ages badly. Cuts, ink look ugly on it. Wood absorbs better

How much space is normally available in your backpack which could be filled by the drawing board? — 1 response — A laptop size - something around 25x35 cm

Is there anything else which hasn't been asked in this form which you think should be considered? — 1 response — Sustainability of materials, manufacturing process.



Materials 2:

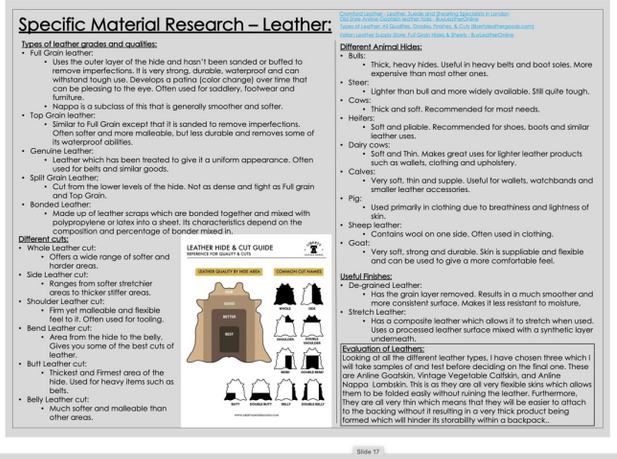
Feedback from peer:
A feedback from pier has led me to research cutting mats as a possible option. This material is a composite material which has an inner layer of foam to give it flexibility, and an outer layer of a hard plastic which is scratch resistant. The outer layer protects it while the inner layer lets it bend more easily. However, a problem I found is that when the mat is bent too much, the outer layer of hard plastic shatters which means that it would not work for my product. However, composite materials should still be considered within my design.

Composite materials:
Research of composite materials has led me to consider other options such as Canvas with PP and Leather with a Wooden backing (which the idea for I got from looking at a wallet and the cards inside which creates rigidity). The use of composite materials would allow me to get the best of both worlds, with the strength and rigidity of one, and the flexibility of the other.

Looking at the wallet (as seen below) has led me to identify a problem with using leather which will need to be tested is that if the leather is folded multiple times, then it starts to get ruined and become creased.

Primary Research - Comfortability of writing on and tactile responses:
Primary research needs to be conducted in order to work out which materials are the most comfortable to write on and to handle (see ergonomics page for research). To do this, I have asked multiple people to hold onto a material and try to write on it and to give a score to the different materials in the two categories from 1 to 10. Then, I have used this information to assess the different materials:

Material:	Tactile Response Scores:	Mean Tactic Response:	Drawing Scores:	Mean Score for Drawing:
Aluminum.	7 7 9 7 9 6 7 8	7.5	8 9 10 10 7 8 8 8	8.5
Silicone Rubber.	3 7 3 6 5 1 6 10	5.1	3 4 2 4 3 2 3 4	3.1
Leather.	9 7 10 6 9 9 10 10	8.8	7 6 5 6 9 9 8 7	7.1
MDF.	5 4 3 5 4 7 5 8	5.1	9 8 4 8 4 7 6 10	7.0
Acrylic.	8 8 8 8 8 7 6 7	7.5	10 7 9 7 7 8 8 9	8.1
Canvas.	8 9 10 6 7 8 7 5	7.5	2 6 1 1 1 3 2 1	2.1
PP.	7 6 7 5 6 4 8 7	6.3	10 6 9 7 3 6 9 9	7.4
Latex Rubber.	6 8 4 7 5 1 4 8	5.4	2 6 7 8 4 2 4 6	4.9
Foam.	2 2 1 1 3 1 1 1	1.5	1 3 1 1 1 1 4 1	1.6
Cherry wood.	10 9 10 9 10 9 8 9	9.3	9 9 10 10 9 10 8 9	9.3



Specific Material Research – Leather:

Cromford Leather - Leather, Suede and Sheeting Specialists in London
Dry Style Aniline Goatskin leather Side - BuyLeatherOnline
Types of Leather: All Qualities, Grades, Finishes, & Cuts (ilovestyleorganic.com)
Italian Leather Supply Store: Full Grain Hides & Sheets - BuyLeatherOnline

Types of leather grades and qualities:
- Full Grain leather:
  - Uses the outer layer of the hide and hasn't been sanded or buffed to remove imperfections. It is very strong, durable, waterproof and can withstand tough use. Develops a patina (color change) over time that can be pleasing to the eye. Often used for saddlery, footwear and furniture.
  - Nappa is a subclass of this that is generally smoother and softer.
- Top Grain leather:
  - Similar to Full Grain except that it is sanded to remove imperfections. Often softer and more malleable, but less durable and removes some of its waterproof abilities.
- Genuine Leather:
  - Leather which has been treated to give it a uniform appearance. Often used for belts and similar goods.
- Split Grain Leather:
  - Cut from the lower levels of the hide. Not as dense and tight as Full grain and Top Grain.
- Bonded Leather:
  - Made up of leather scraps which are bonded together and mixed with polypropylene or latex into a sheet. Its characteristics depend on the composition and percentage of bonder mixed in.

Different cuts:
- Whole Leather cut:
  - Offers a wide range of softer and harder areas.
- Side Leather cut:
  - Ranges from softer stretchier areas to thicker stiffer areas.
- Shoulder Leather cut:
  - Firm yet malleable and flexible feel to it. Often used for tooling.
- Bend Leather cut:
  - Area from the hide to the belly. Gives you some of the best cuts of leather.
- Butt Leather cut:
  - Thickest and Firmest area of the hide. Used for heavy items such as belts.
- Belly Leather cut:
  - Much softer and malleable than other areas.

Different Animal Hides:
- Bulls:
  - Thick, heavy hides. Useful in heavy belts and boot soles. More expensive than most other ones.
- Steer:
  - Lighter than bull and more widely available. Still quite tough.
- Cows:
  - Thick and soft. Recommended for most needs.
- Heifers:
  - Soft and pliable. Recommended for shoes, boots and similar leather uses.
- Dairy cows:
  - Soft and Thin. Makes great uses for lighter leather products such as wallets, clothing and upholstery.
- Calves:
  - Very soft, thin and supple. Useful for wallets, watchbands and smaller leather accessories.
- Pig:
  - Used primarily in clothing due to breathiness and lightness of skin.
- Sheep leather:
  - Contains wool on one side. Often used in clothing.
- Goat:
  - Very soft, strong and durable. Skin is suppliable and flexible and can be used to give a more comfortable feel.

Useful Finishes:
- De-grained Leather:
  - Has the grain layer removed. Results in a much smoother and more consistent surface. Makes it less resistant to moisture,
- Stretch Leather:
  - Has a composite leather which allows it to stretch when used. Uses a processed leather surface mixed with a synthetic layer underneath.

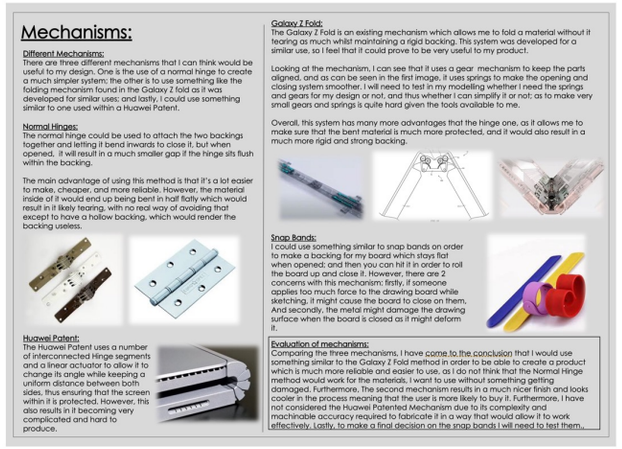
Evaluation of Leathers:
Looking at all the different leather types, I have chosen three which I will take samples of and test before deciding on the final one. These are Aniline Goatskin, Vintage Vegetable Calfskin, and Aniline Nappa Lambskin. This is as they are all very flexible skins which allows them to be folded easily without ruining the leather. Furthermore, They are all very thin which means that they will be easier to attach to the backing without it resulting in a very thick product being formed which will hinder its storability within a backpack..

Slide 17

The beginning of the project's brief went fairly well up until the Client section. Although at the time they seemed fine as clients, contacting them became harder as I went through the project as one of them moved too far away for me to see them while the other stopped replying to me. If I was to redo this project, I would definetely try and find someone who was more likely to keep in touch and was actually vested into the product enough so that I could have them test it.

The research section went fairly well, the only thing I would have changed there is that I should have tested my materials more thoroughly throughout it. Although the nylon wasn't chosen in the original research phase, I still feel like having looked at it then and tested it would have helped me avoid it in my prototype as I'd have probably seen how weak it was, or at least would have made my design change in order to make it stronger.

Something else I should have researched, although this may fall more under modelling, is how I would have realistically produced my board using the tools at school. This would definitely have helped me with the material choice as I would have focused on using an easier to work with material that wasn't nylon.



Mechanisms:

Different Mechanisms:
There are three different mechanisms that I can think would be useful to my design. One is the use of a normal hinge to create a much simpler system; the other is to use something like the folding mechanism found in the Galaxy Z fold as it was developed for similar uses; and lastly, I could use something similar to one used within a Huawei Patent.

Normal Hinges:
The normal hinge could be used to attach the two backings together and letting it bend inwards to close it, but when opened, it will result in a much smaller gap if the hinge sits flush within the backing.

The main advantage of using this method is that it's a lot easier to make, cheaper, and more reliable. However, the material inside of it would end up being bent in half flatly which would result in it likely tearing, with no real way of avoiding that except to have a hollow backing, which would render the backing useless.

Huawei Patent:
The Huawei Patent uses a number of interconnected Hinge segments and a linear actuator to allow it to change its angle while keeping a uniform distance between both sides, thus ensuring that the screen within it is protected. However, this also results in it becoming very complicated and hard to produce.

Galaxy Z Fold:
The Galaxy Z Fold is an existing mechanism which allows me to fold a material without it tearing as much whilst maintaining a rigid backing. This system was developed for a similar use, so I feel that it could prove to be very useful to my product.

Looking at the mechanism, I can see that it uses a gear mechanism to keep the parts aligned, and as can be seen in the first image, it uses springs to make the opening and closing system smoother. I will need to test in my modelling whether I need the springs and gears for my design or not, and thus whether I can simplify it or not; as to make very small gears and springs is quite hard given the tools available to me.

Overall, this system has many more advantages that the hinge one, as it allows me to make sure that the bent material is much more protected, and it would also result in a much more rigid and strong backing.

Snap Bands:
I could use something similar to snap bands on order to make a backing for my board which stays flat when opened; and then you can hit it in order to roll the board up and close it. However, there are 2 concerns with this mechanism: firstly, if someone applies too much force to the drawing board while sketching, it might cause the board to close on them. And secondly, the metal might damage the drawing surface when the board is closed as it might deform it.

Evaluation of mechanisms:
Comparing the three mechanisms, I have come to the conclusion that I would use something similar to the Galaxy Z Fold method in order to be able to create a product which is much more reliable and easier to use, as I do not think that the Normal Hinge method would work for the materials, I want to use without something getting damaged. Furthermore, The second mechanism results in a much nicer finish and looks cooler in the process meaning that the user is more likely to buy it. Furthermore, I have not considered the Huawei Patented Mechanism due to its complexity and machinable accuracy required to fabricate it in a way that would allow it to work effectively. Lastly, to make a final decision on the snap bands I will need to test them..

Possible Color Combinations And Client Reviews

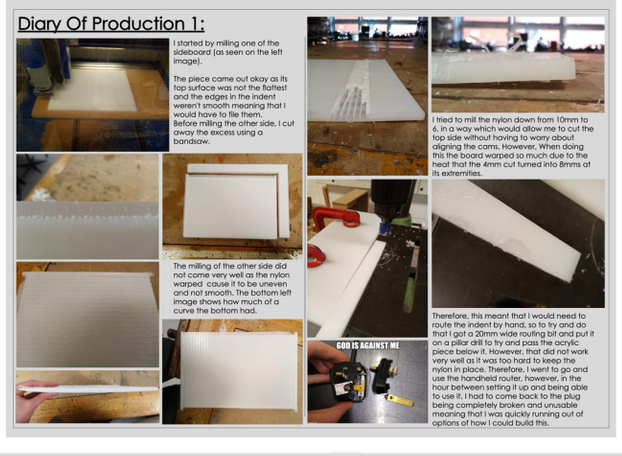
Client Feedback:
First client:
Very cool. Very Swag. I like it.
I especially like the variation in color options. Very simple.
Could easily use it for drawing.

Second client.

I like it. Sleek and not bulky.
Looks like it could easily fold and be carried with no effort.
I like the grey with black border(no4)
Overall I think it meets with the initial criteria my only query is the hinge on the ruler looks a bit like an afterthought. Would be nicer incorporated better of flush. And will it engage with the edges of the board somehow? 13:48

Thank you sir. It will use magnets to engage the ruler with the edges of the board. 13:51

Excellent. I think it's developed very well. Hope the prototypes performed as expected. 13:54



Diary Of Production 1:

I started by milling one of the sideboard (as seen on the left image).

The piece came out okay as its top surface was not the flattest and the edges in the indent weren't smooth meaning that I would have to file them. Before milling the other side, I cut away the excess using a bandsaw.

I tried to mill the nylon down from 10mm to 6, in a way which would allow me to cut the top side without having to worry about aligning the cams. However, When doing this the board warped so much due to the heat that the 4mm cut turned into 8mms at its extremities.

The milling of the other side did not come very well as the nylon warped cause it to be uneven and not smooth. The bottom left image shows how much of a curve the bottom had.

Therefore, this meant that I would need to route the indent by hand, so to try and do that I got a 20mm wide routing bit and put it on a pillar drill to try and pass the acrylic piece below it. However, that did not work very well as it was too hard to keep the nylon in place. Therefore, I went to go and use the handheld router, however, in the hour between setting it up and being able to use it, I had to come back to the plug being completely broken and unusable meaning that I was quickly running out of options of how I could build this.

The designing section went fairly well. I think the mechanism was optimised as much as it could in order to make it simple as possible, but, if I had the chance to do it again, I would probably completely change it and use something more similar to a layered hinge, similar to the Huawei One that can be seen on this page on the left, as even though it might be a lot harder to build, it would probably end up looking a lot better and hopefully working better.

With the overall design, I think that brainstorming a little bit more at the start could have helped me come up with more ideas, however, the final design still looked very good. I am especially happy with the colour customization at the end which I feel is something that aids in the user with expressing themselves.

In the making section, the first thing that I know I should have changed is, once I realised that the nylon could not be milled, I should not have used it as my main material. I wasted too much time trying to get it to work and, in the end, it did not even work that well. The other big issue during the making was that a lot of the pieces did not cut very straight as all of the bandsaws were not tensioned and straightened properly so they cut askew, messing up the steel and aluminium strips, and the boards themselves.